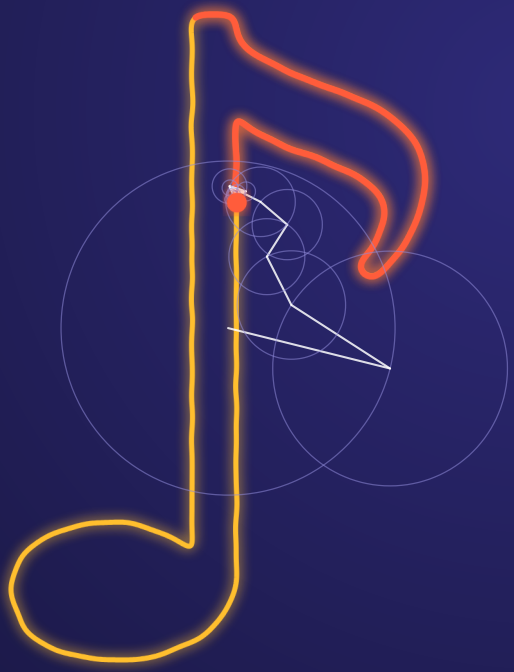
Over the Fence

Big ideas in mathematics, one neighbourly conversation at a time



CHAPTER 1
Fourier series

CHAPTER 2
Vector spaces

CHAPTER 3
Metric

CHAPTER 4
Eigenvalues and eigenvectors

CHAPTER 5
Groups

CHAPTER 6
Bayes' theorem

CHAPTER 7
Regression to the mean

Guhan Purushothaman

M.SC. MATHEMATICS · IIT MADRAS

from the blog
guhanpurushothaman.wordpress.com

BEFORE WE START

How to read this

This isn't a textbook. It's a season of conversations between two neighbours, one of whom can't stop explaining things. You don't need any maths beyond school. You just need to be willing to picture things.

Me

Me, Guhan: the narrator

Your neighbour. A mathematician from IIT Madras who owns an old analogue watch, an expensive robot and far too many analogies. My lines are set in plain type.

J

Jim

Asks the questions you're too polite to ask. Rents superbikes. Makes the big leaps himself. His lines sit in the blue bubbles.

B

Bruno

Jim's German Shepherd. Scared of the vacuum cleaner. Has opinions during voice notes.

R

The robot

Imported from Germany. Memorises everything and understands nothing. A warning to us all.

Signposts along the way

- **Pause and picture it** — Stop reading and imagine it first. It's where the learning happens.
- **Side quest** — A detour worth taking: history, surprises, trivia.
- **Tool unlocked** — The mental model you walk away with at the end of each part.
- **Try this** — Something to do with your hands, a guitar or a phone.
- **Hello, my name is** — Names arrive *after* the ideas. When you see one of these, you've earned it.
- **Scribbles in the margin** — Small asides you can skip, or not.

One rule runs through the whole book: no formula appears until you've already understood what it says.

THE ROUTE

What's inside

CHAPTER 1 Fourier series 5

1.1 Circles all the way down 6
Introduction to Fourier series · 8 min

1.2 A song is one wiggly line 14
From circles to sound · 8 min

1.3 The resonance test 21
Developing the mathematical architecture for Fourier analysis · 10 min

1.4 How many circles? 29
Convergence, corners and a 159-year argument · 6 min

CHAPTER 2 Vector spaces 35

2.1 Anything you can add and scale 36
Introduction to vector spaces · 7 min

2.2 The minus-two-onions problem 43
Developing the mathematical architecture for vector spaces · 8 min

2.3 Same song, different language 50
Changing the basis · 6 min

CHAPTER 3 Metric 58

3.1 How far is the airport? 59
Introduction to metrics · 5 min

3.2 Distances between things that aren't places 64
Developing the mathematical architecture for metrics · 4 min

3.3 Getting closer 68
Convergence, horns and Stigler's law · 4 min

CHAPTER 4 Eigenvalues and eigenvectors 72

4.1 The directions that don't turn 73
Introduction to eigenvectors and eigenvalues · 4 min

4.2 Hunting eigenvectors 77
Developing the mathematical architecture for eigenvectors · 4 min

4.3 Where the future goes 81
Where eigenvectors take us · 4 min

CHAPTER 5 **Groups** 85

5.1 **The mattress problem** 86
Introduction to groups · 4 min

5.2 **The grammar of moves** 90
Developing the mathematical architecture for groups · 4 min

5.3 **The power of impossibility** 95
Parity, puzzles and proofs · 4 min

CHAPTER 6 **Bayes' theorem** 99

6.1 **Car alarms and rare diseases** 100
Introduction to Bayes' theorem · 3 min

6.2 **Updating, one clue at a time** 104
Developing the mathematical architecture for Bayes' theorem · 4 min

6.3 **The costliest mistake in probability** 108
When "if" gets turned around · 4 min

CHAPTER 7 **Regression to the mean** 112

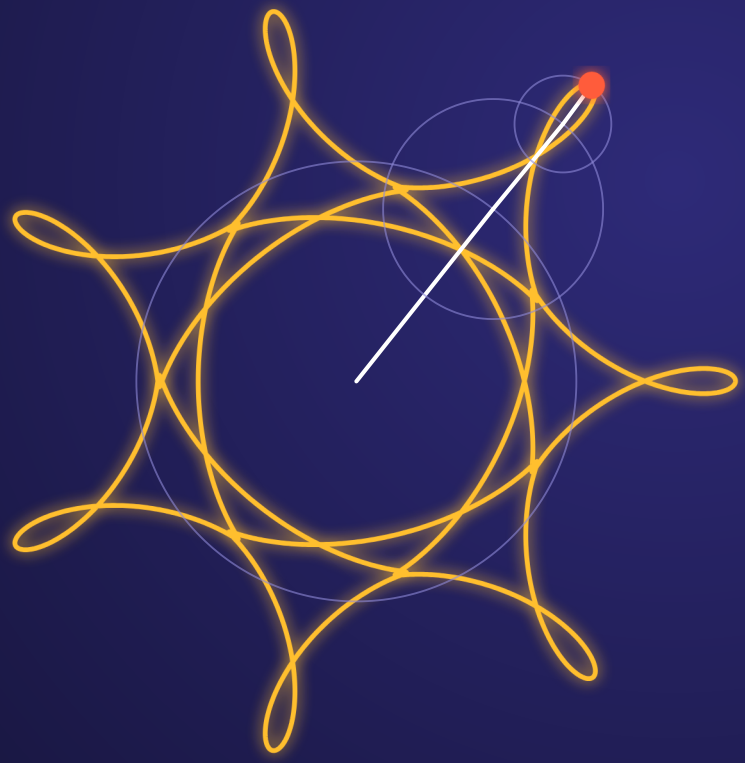
7.1 **Why the yelling seems to work** 113
Introduction to regression to the mean · 4 min

7.2 **How far do things fall back?** 117
Developing the mathematical architecture for regression · 4 min

CHAPTER

1 Fourier series

Every repeating thing is a sum of circles



How do you build a song out of circles?

YOUR ROUTE THROUGH THIS CHAPTER

1.1 **Circles all the way down**
Introduction to Fourier series

1.2 **A song is one wiggly line**
From circles to sound

1.3 **The resonance test**
Developing the mathematical architecture for Fourier analysis

1.4 **How many circles?**
Convergence, corners and a 159-year argument

1.1 Circles all the way down

Introduction to Fourier series

BY THE END
You'll draw a musical note with nothing but spinning circles.

8 min read · A conversation over the fence

SCENE *Tuesday evening. The bins are out. So, it seems, is a robot.*

J "Howdy, neighbour! Taking out the trash, eh?"

"That's not trash, Jim. That's an imported robot from Germany. It cost more than my first car."

J "Then why is it sitting next to the recycling?"

"Because this morning your German Shepherd walked past the gate and it called him a wolf. Then it called the police."

J "Bruno? A wolf? Bruno is scared of the vacuum cleaner."

"I know. I spent a fortnight teaching that machine what a dog is. I showed it five thousand photos of dogs, and every one of them, it turns out, was small, white and fluffy."

J "So it learned 'small, white and fluffy'."

"It learned five thousand photographs. That's the problem. It stored every example perfectly and understood none of them. The moment something new walked past, a large black-and-brown animal with pointy ears, it had nothing to go on."

J "Isn't that how we all learn, though? From examples?"

"Examples are the raw material. What matters is what you do with them. You can store them, like the robot, or you can squeeze them down into a small working machine in your head, one that makes predictions about things you've never seen. When you were three you'd met perhaps a dozen dogs, and you could still recognise a breed you'd never laid eyes on."

J "I squeezed."

"You squeezed. It's the difference between memorising a thousand sentences of a language and learning its grammar. Learn the grammar and you can say things nobody has ever said to you. The psychologist Kenneth Craik had a name for these little machines back in 1943: mental models."

J "All right, professor. What's the best mental model you own?"

Craik's book The Nature of Explanation (1943) argued that the mind builds "small-scale models" of reality so it can run ahead of it.

"Easy. It's the one I'd put in every head if I could: every repeating thing, whether a song, a heartbeat, the tides or the hum of your fridge, can be built out of circles spinning at steady speeds."

J "Circles. You're going to make a song out of circles."

"Give me twenty minutes and my watch."

J "All ears."

> *I unstrap an old analogue watch and hold it over the fence.*

"Three hands. The hour hand goes round once every twelve hours, the minute hand once an hour, the second hand once a minute. Each is just a stick, spinning at a steady speed around the centre."

J "Riveting."

"Now for the trick. Imagine I unscrew the second hand and bolt it onto the tip of the minute hand, so it whirls around the end of the minute hand instead of around the centre. And let's speed the whole thing up, so we're not standing here for an hour."

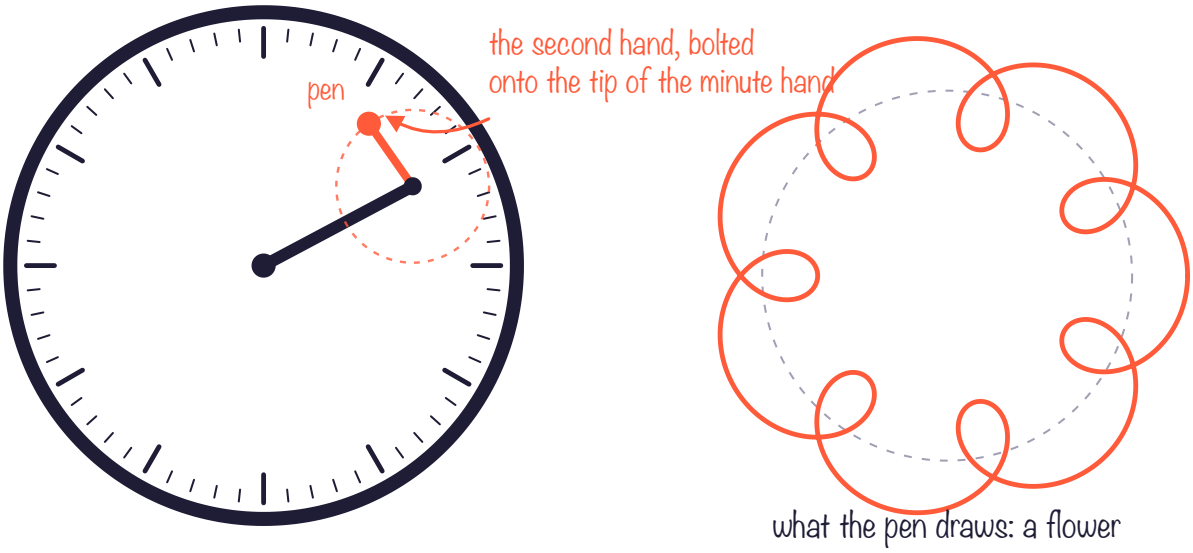


FIGURE 1.1 The second hand, unscrewed and bolted onto the tip of the minute hand. Tape a pen to the end of the little hand and let it draw.

"Tape a pen to the tip of the little hand. What does it draw?"

PAUSE AND PICTURE IT

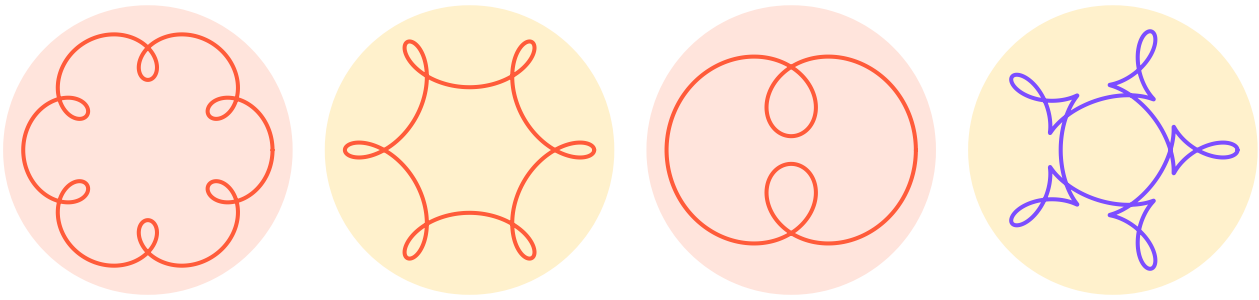
Before you read Jim's answer, picture it. The big hand sweeps slowly round; the little hand whirls around its tip. What shape does the pen leave behind?

J "Well... the big hand drags the little one round in a big circle. But the little one is whirling at the same time, so the pen keeps looping as it goes. A big circle with little loops all the way round. A flower-ish pattern?"

"A flower-ish pattern. Exactly."

J "Like a Spirograph!"

"A Spirograph is exactly that: one circle rolling on another. The pen doesn't know it's doing mathematics. Now watch what happens when we change just two things about the little hand: how long it is, and how fast it spins. Or even which way."



short & fast
petals

spinning backwards
a star

long & fast
loops through the middle

three hands
a crown

FIGURE 1.2 Two hands are already a zoo. Change the little hand's length, speed or direction, and the same machine draws petals, stars or loops. Add a third hand and things get stranger still.

J "The backwards one makes a star! And the long one loops right through the middle."

"Two or three spinning sticks, and already we're drawing shapes you'd struggle to draw by hand. Before we stack more of them, though, I want to be sure we agree on what a 'hand' really is. Let's play a game. I've buried treasure somewhere in your garden."

J "Please don't dig up my garden."

"Hypothetically. You may ask me two questions, and only two. What do you ask?"

J "'Which way?' and 'How far?'"

"Why not 'What colour is the box?'"

J "Because it doesn't get me there. Which way and how far is all I need. Stand here, face north-east, walk forty paces, dig."

"Then you've just invented the most important object in this book. Something that carries a 'which way' and a 'how far' is called a vector. You draw it as an arrow: it points the way, and its length is the distance."

J "So a watch hand is a vector. Its length is 'how far' and where it points is 'which way'."

"A vector that happens to be spinning. Now, say my treasure map has two steps: forty paces north-east, then twenty-five paces south. Where do you end up?"

J "I walk the first arrow, then start the second one from wherever the first one left me."

"Tip to tail. That's all 'adding vectors' means: follow one set of directions, then the next."

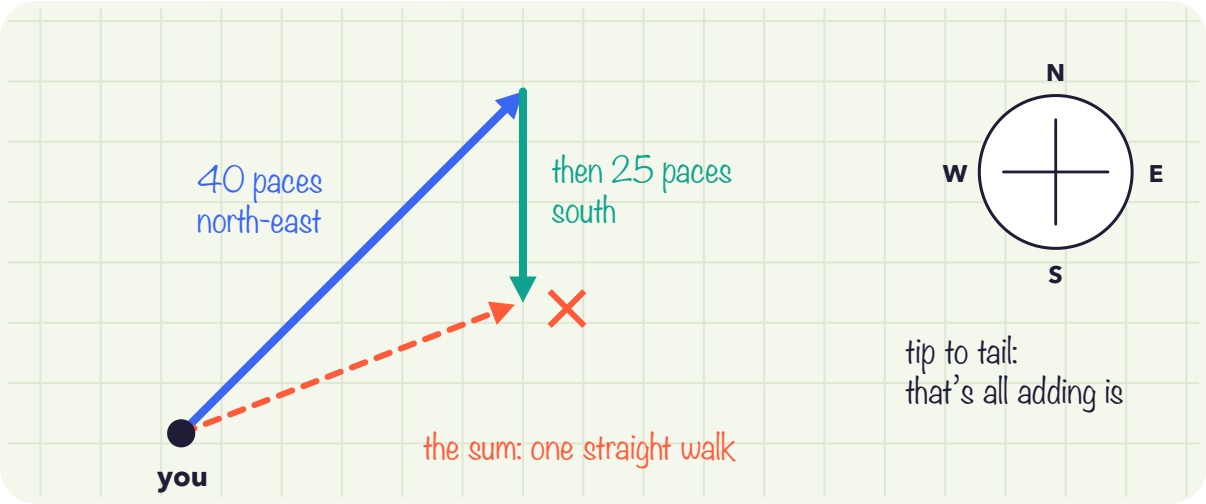


FIGURE 1.3 Adding vectors is just following directions: walk the first arrow, then start the second from its tip. The dashed arrow, the shortcut, is their sum.

J "Oh. Oh! That's what the watch was doing. The little hand starts from the tip of the big one. So the pen is at the end of the big arrow plus the little arrow."

"You've hit the nail on its conceptual head. At every instant, the pen is the sum of all the hands. Stack a hundred hands, each riding on the tip of the one before, and the pen is the sum of a hundred spinning vectors."

J "And with a hundred hands you can draw...?"

"Name something."

J "Something musical. An eighth note, the little flag thing."

"Done."

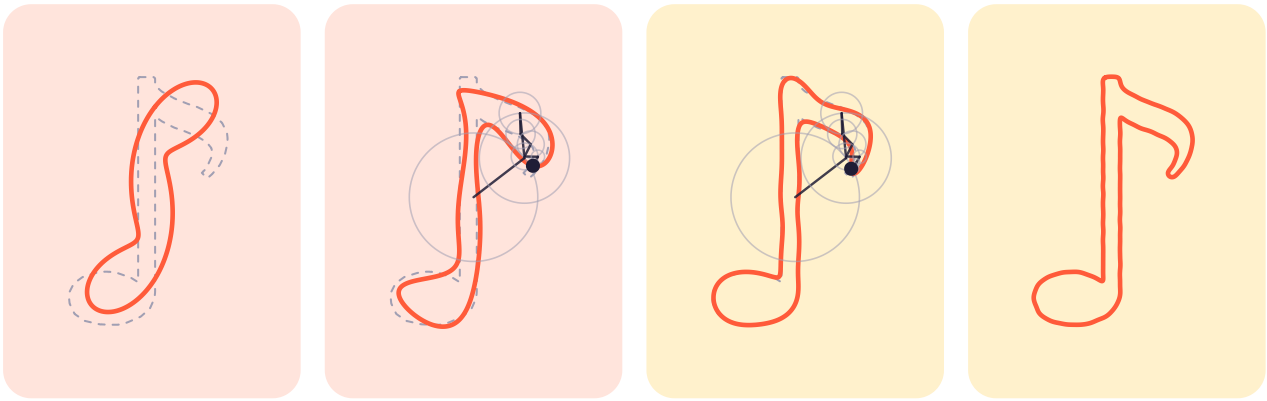


FIGURE 1.4 A musical note drawn by nothing but spinning circles. With 3 hands it's a blob, with 8 a suggestion, with 20 you'd recognise it, and with 80 it's crisp. The faint circles show the chain of hands at one moment, each riding on the last.

J "That's absurd. There's no note in any of those circles."

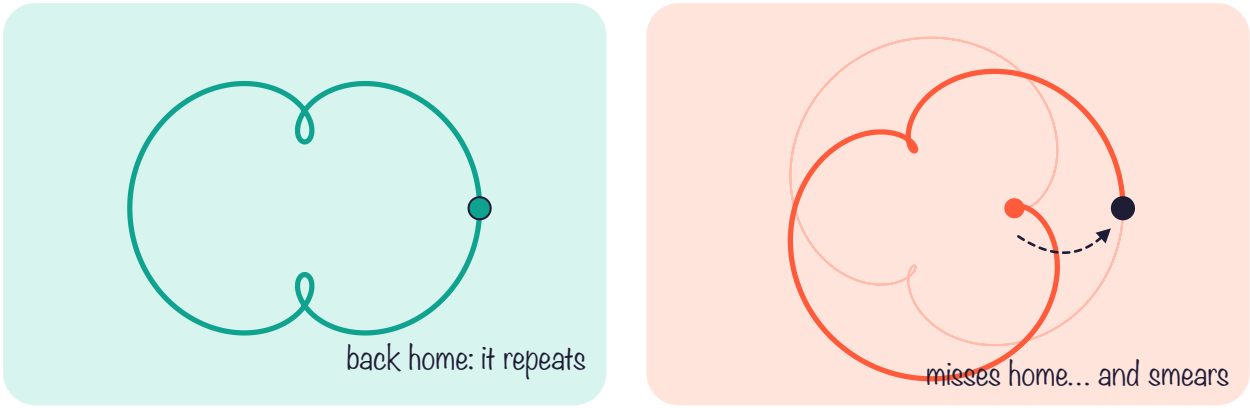
"There's no note in any single one of them. The note lives in how they're combined: their lengths, their speeds and where each one starts. Change those numbers and the same machine draws your signature, the outline of Bruno, the coastline of India: anything you can draw in one loop without lifting your pen."

J "Why 'in one loop'?"

"Because spinning hands always come back. Think about what that forces on us. Say the drawing takes one second to go round once, and then it repeats. Every hand has to be back where it started when that second is up. Otherwise the next loop starts somewhere else and the drawing smears."

J "So each hand has to go round... a whole number of times?"

"Once, twice, three times, a hundred times. Never two and a half."



3 turns per loop **2½ turns per loop**

FIGURE 1.5 Why the speeds must be whole numbers. Left: the little hand turns exactly 3 times per loop, so the pen comes home and the drawing repeats perfectly. Right: at 2½ turns it misses home, and every loop smears a little further.

J "And backwards?"

"Backwards is fine: minus once, minus twice. About half the hands drawing that note are spinning backwards. What isn't allowed is a hand that doesn't come home."

J "So the speeds aren't really a choice at all. The only choices are how long each hand is and where it starts."

"And that's why this idea is so powerful. The menu of speeds is fixed: 1, 2, 3, 4 turns per loop, forwards or backwards. Every loop-shaped drawing in the universe is just a different set of lengths and starting angles on that one menu."

SIDE QUEST: WHY THIS WORKS AT ALL

You might wonder why circles, and not squares or triangles, are the building blocks. Circles are the only shapes that spin at a perfectly steady speed and come back to where they started. That steadiness is what makes them easy to add, to measure and, as you'll see in 1.3, to detect.

"Now I can tell you what it's called. Writing a repeating thing as a sum of spinning circles, each turning a whole number of times per loop, is called a Fourier series."



"It's named after Joseph Fourier, who in 1807 used it to predict how heat spreads through a metal bar. The grandest mathematicians in Paris didn't believe him. We'll get to that fight."

J "Heat? We started with a watch and ended with a note. Where does heat come in?"

"Heat, sound, light, radio, the tides, your heartbeat, the wobble in the orbit of Mars. Anything that repeats. But you're right to be suspicious: so far we've only drawn pictures. Tomorrow I'll show you why these circles are hiding inside every sound you've ever heard."

J "And the robot?"

"The robot needed five thousand dogs and learned nothing. You needed one watch."

TOOL UNLOCKED

The spinning-circles model

Anything that repeats can be built from circles that each turn a whole number of times per loop. At any moment, the pen is simply the sum of all the hands.

1.2

A song is one wiggly line

From circles to sound

BY THE END
You'll see a whole song as one wiggly line, and hear it as an orchestra of circles.

8 min read A conversation over the fence

SCENE *Wednesday. Jim is on his porch with a guitar, playing the same four chords he always plays.*

"Evening, Jim. Still those four chords?"

J "They're good chords. And you owe me something: you said circles are hiding in sound. I've been staring at this guitar for an hour. No circles."

"Then let's go and find them. Remember the watch hand? Shine a torch at it from the side, so the hand's shadow falls on the wall."

J "The shadow's just a line on the wall."

"Watch the tip of the shadow as the hand turns."

J "It goes up… slows down at the top… comes down… slows at the bottom… up again. Up and down, smoothly, forever."

"Now pin a strip of paper to the wall and pull it sideways at a steady speed while a pen follows the shadow's tip."

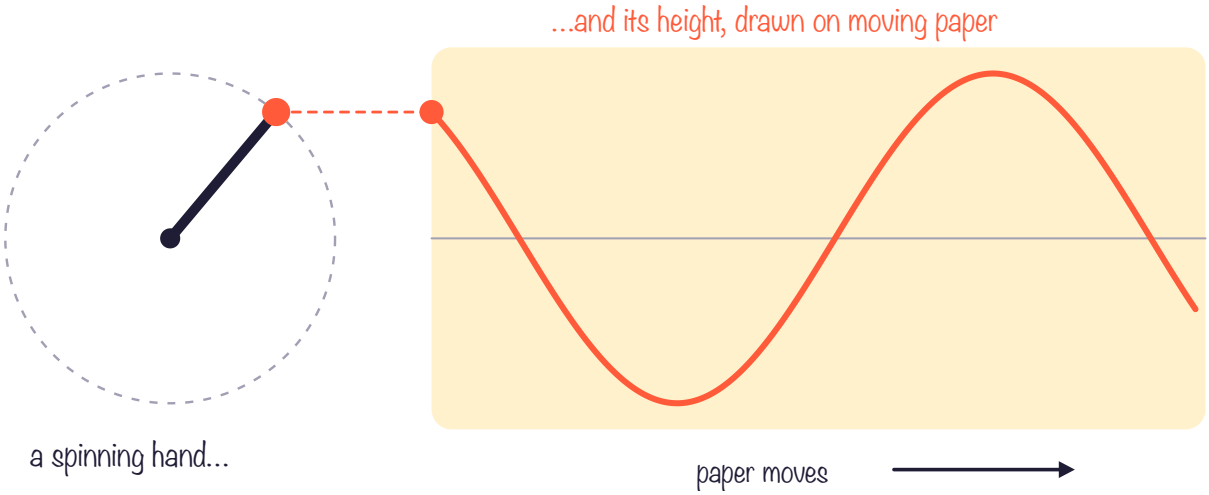


FIGURE 1.6 A spinning hand, seen edge-on, is a wave. Pull paper past its shadow at a steady speed and the up-and-down motion unrolls into the smoothest wave there is.

J "A wave. The circle turned into a wave."

"The circle is the wave, seen from the side. Whenever you see that smooth up-and-down curve, there's a spinning hand behind it, whether you can see it or not. Mathematicians call it a sine wave; I'll call it a pure wave."

J "Then two stacked hands would give a wobblier wave. The flower's shadow."

Heart monitors and seismographs work this way: a pen that only moves up and down, over paper that scrolls at a steady speed.

"Exactly. Now, your guitar. When you pluck a string, what actually reaches my ear?"

J "Sound."

"Which is?"

J "...air wobbling?"

"Air being pushed and pulled, very fast. Your eardrum moves in when the pressure rises and out when it falls. A loudspeaker does the same in reverse: a cone that moves in and out. And here's the thing I'd like you to sit with. At every instant, the only thing a loudspeaker needs to know is how far out its cone should be. One number. Then the next instant, another number."

J "One number at a time... so a song is just a list of positions over time."

"Draw them on scrolling paper and you get one wiggly line. Your favourite song, with every instrument, the singer's breath and the crowd, is a single wiggly line."

J "No. The whole thing? The drums and the bass and the vocals, all in one line?"

"All in one line. An old vinyl record is exactly that: one long spiral groove whose wiggles are the song. The needle rides the wiggles and the speaker copies them."

J "Then how do I hear a drum and a voice as separate things, if it's one line?"

Stereo records cut two wiggly lines, one into each wall of the same groove: one for each ear.

"That's the question this whole chapter is really about. Let's build a chord, and you'll see."

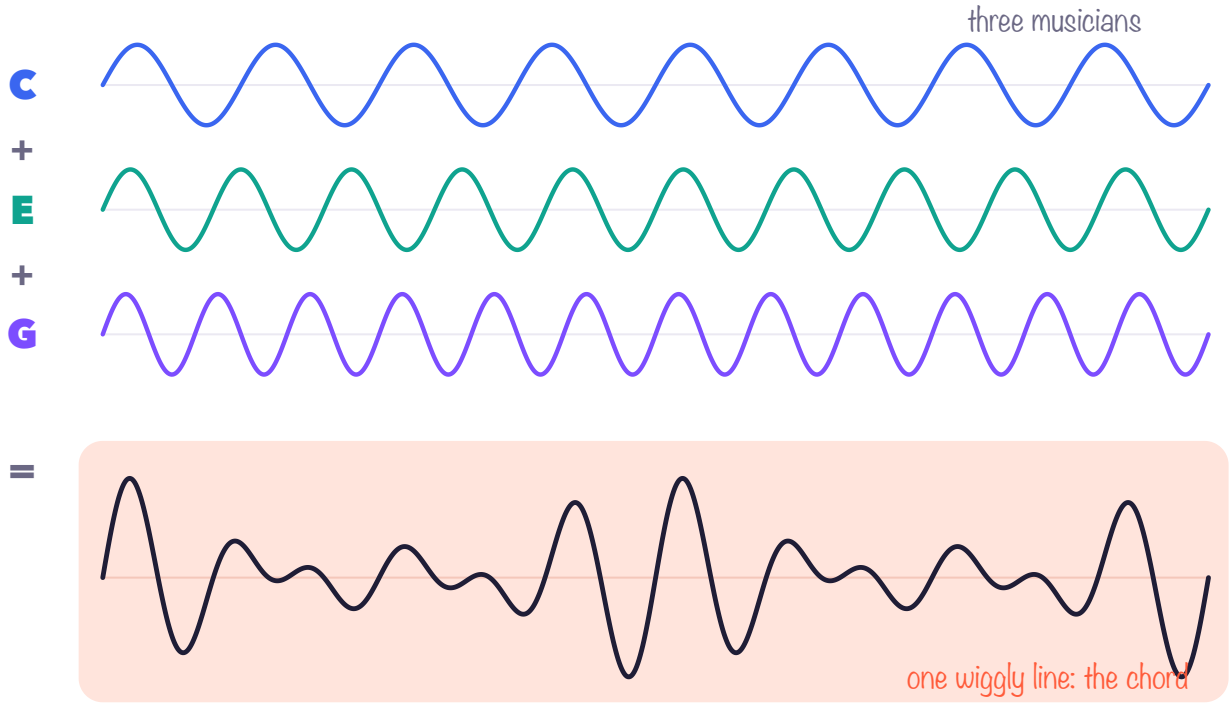


FIGURE 1.7 Three musicians, each playing one pure wave, add up to a single wiggly line: a major chord. Nothing in the bottom line tells you it's made of three parts, and yet it is.

"Three pure waves, spinning at speeds in the ratio 4 to 5 to 6. Add them up moment by moment, which is exactly what the air does when three strings sound at once, and you get the bottom line."

J "It doesn't look like anything. It looks like a heart monitor having a bad day."

"And yet your ear hears three notes in it, instantly. Now turn it around. Suppose I hand you any wiggly line, say a recording of a full orchestra. Is there a set of pure waves that adds up to it?"

J "Well, a pure wave is a spinning hand seen from the side, and yesterday you said stacked hands can draw anything that repeats..."

“Keep going.”

J “So yes: any wiggly line is a pile of pure waves. Like an orchestra where each musician can only play one pure note, at one loudness, the whole time. Give each one the right note and the right loudness, put enough of them together and... even though none of them understands what they’re playing, the collective result is the song.”

> Even though none of them understands what they’re playing, the collective result is the song.
> JIM, FIGURING IT OUT

“That’s exactly it. You’ve just described a Fourier series better than most textbooks do. Each of your musicians has three dials. How fast their hand spins sets the pitch: physicists call that the frequency. How long their hand is sets the loudness: the amplitude. And where their hand starts is called the phase.”

J “Does ‘where it starts’ matter? It’s the same note.”

“For a steady note your ear is famously relaxed about it. Shuffle the starting points and the shape of the wiggly line changes completely, but it sounds almost the same. For drawings, it matters enormously: start the hands of that musical note at different angles and you’d get a scribble.”

Jim plays an A. Badly, then less badly.

“That string goes back and forth 110 times a second, so the wiggly line repeats 110 times a second. What does yesterday’s rule tell us about the hands?”

J “Every hand has to come home at the end of each repeat. So they have to spin a whole number of times per repeat: 110 times a second, 220, 330, 440...”

“Those are the harmonics. The slowest one, the 110, sets the pitch you hear. The others, and how long each one is, set the character of the sound. So why do you think a flute and a violin playing the same A sound nothing alike?”

J “Different recipes! Same slowest hand, different amounts of the others.”

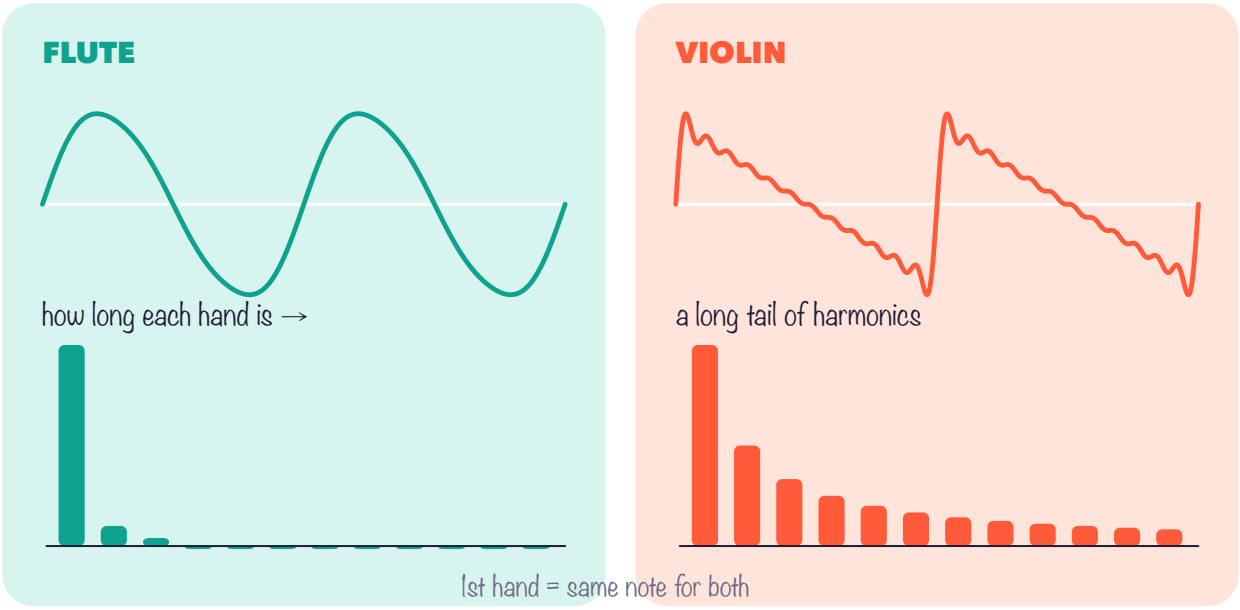


FIGURE 1.8 Same note, different recipe. A flute's wave is nearly one pure hand. A violin's sharp, buzzy wave hides a long tail of strong harmonics. The bars show how long each hand is.

"Musicians call that recipe the timbre. You can even hear the hands on your own guitar."

TRY THIS: HEAR A HIDDEN HARMONIC

Rest a fingertip lightly on a guitar string, without pressing, exactly above the 12th fret (the halfway point), and pluck. The slowest hand needs the middle of the string to swing, so your finger silences it. What's left is the next hand: a clear, bell-like note an octave higher.

J "Oh, that's pretty. A little bell."

"You just silenced one musician with a fingertip and heard the one standing behind him. Now let me show you why any of this is useful. Last night I recorded a voice note for my sister. Halfway through, Bruno had opinions."

J "He does that."

"The recording is one wiggly line: my voice and Bruno, added together. You can't cut Bruno out with scissors; he's woven through every moment of it. But break the line into its hands, and my voice and his bark mostly live in different places. My voice is a low hand plus its tidy harmonics. A bark is a sudden, sharp burst of higher hands."

J "So you find Bruno's musicians and tell them to stop playing. Then play the rest."

"And the bark is gone. In practice we do it slice by slice, a twentieth of a second at a time, because a bark comes and goes. Line up all the slices and

you get a picture like this.”

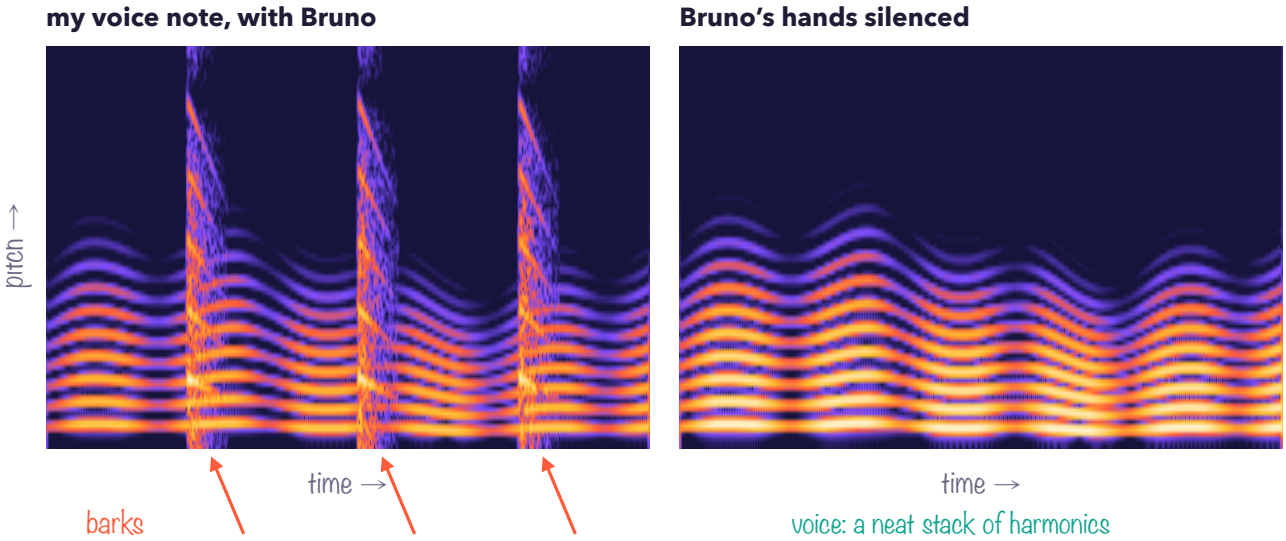


FIGURE 1.9 A spectrogram: time runs left to right, pitch runs up, and brightness shows how long each hand is. My voice is the neat stack of harmonics; Bruno’s barks are the bright vertical smears. Silence those, and the recording comes back clean. (A made-up recording, built to show the idea.)

J “That looks like sheet music.”

“It is sheet music, written by the sound itself. Noise-reducing microphones, hearing aids and every music-editing app are built on pictures like this.”

J “Does light have hands too?”

Shazam recognises a song from the pattern of the brightest points in exactly this kind of picture: a fingerprint made of hands.

“Newton showed it in the 1660s with a prism. White light goes in, a rainbow comes out. A prism is a glass machine for sorting light’s hands by speed: the slow ones come out red, the fast ones violet. He called the rainbow a spectrum, and we still use his word for any list of hands and their lengths, sound included.”



J “So the chord, the violin, my voice note, a rainbow: they’re all orchestras of spinning hands.”

“All of them. But we’ve been cheating. I’ve been telling you which hands are inside. If I gave you only the wiggly line, how would you find them?”

J “Can’t you just look at it?”

“Can you look at a smoothie and tell me the recipe?”

J “…No.”

“Then tomorrow we’ll build a machine that can. And, spoiler: you already own one.”

TOOL UNLOCKED

Waves are circles seen side-on

Any sound is one wiggly line, and any repeating wiggly line is an orchestra of pure waves. Each has a speed (pitch), a length (loudness) and a start (phase). The slowest hand sets the note; the rest set its character.

1.3

The resonance test

Developing the mathematical architecture for Fourier analysis

BY THE END

You'll find every hidden hand in a wiggly line, and discover your ear already does it.

10 min read A conversation over the fence

SCENE *Thursday. A deep burble outside. Jim is sitting on something red.*

"Is that a Ducati Monster I am looking at?"

J "Ha, yes! I'm renting superbikes for a few weeks: the Ducati, then a Harley, then a Kawasaki. For science."

"And the verdict?"

J "It lives up to its name. But something's off. Around four thousand rpm the mirrors blur and the handlebars buzz like they're angry. At five thousand it's smooth as butter."

"That's not the bike being faulty. That's the bike being an orchestra. The engine shakes the frame at a speed set by the revs. And the frame, like every object, has favourite speeds of its own, speeds it loves to wobble at: its natural frequencies. When the engine's shaking lines up with one of them, the frame joins in and the wobble grows. That's resonance."

J "So at four thousand, the engine is singing the frame's favourite song."

"Beautifully put. Your body has natural frequencies too. Your belly likes to wobble at roughly 4 to 8 times a second, your head at 20 to 30. That's partly why hours on a buzzing bike wear you out, and why lorry seats sit on springs."

"Have you ever pushed a child on a swing?"

J "My nephew. Endlessly."

"What happens if you push exactly in time with the swing?"

J "Higher and higher. Every push adds a bit."

"And if you push at your own rhythm, a little faster than the swing?"

J "Some pushes help and some land while it's coming back at me. It goes a bit higher, then drops, then higher again. It never really gets going."

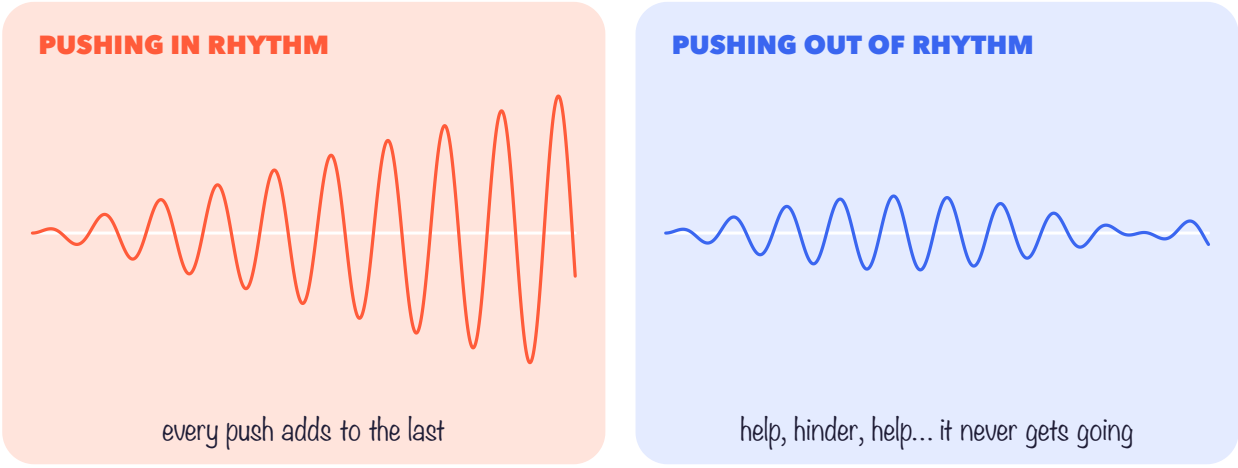


FIGURE 1.10 Push in rhythm and every push adds to the last: the swing climbs and climbs. Push out of rhythm and the helps and hinders take turns, and over time they cancel out.

"That might be the most important picture in this chapter, because it means a swing is a detector. It ignores every rhythm except its own. Feed it a jumble of pushes and it responds only to the part of the jumble that matches its rhythm."

J "So if I wanted to know whether a wiggly line has, say, a 220-times-a-second hand hiding in it..."

"...build a swing that loves 220, push it with your wiggly line, and see whether it swings."

J "That's clever. That's actually clever."

"Nature thought of it first. Let's turn it into arithmetic. Here's our job: we have the wiggly line, call it F. We want a little calculation that we can point at any speed, say 'three turns per loop', and it hands back how long F's three-turns hand is. If there's no such hand, it should hand back zero."

J "Then we need a test wave: a pure wave at the speed we're testing. And I'm naming it after the engine I'm trying to catch: `ducati(x)`."

“Of course you are. So we have F and `ducati(x)`. What do we do with them?”

J “Add them?”

“Adding just piles one on top of the other. You’d get a new wiggly line, not an answer.”

J “Subtract?”

“Same problem. Go back to the swing. When does a push help?”

J “When I push the same way the swing is already moving.”

“And when does it hurt?”

J “When I push against it.”

“So at each instant, the effect is push times motion: positive when they agree, negative when they fight. And the swing’s total gain is all those little effects added up over time.”

J “So: multiply F by `ducati(x)` at every moment, and add it all up.”

This is physics, not just an analogy: the energy a push gives a swing is exactly the push times the swing’s speed, added up over time.

“That’s the whole test. Multiply the two lines point by point, then add up the result over one full loop. And adding up a line over time is just finding the area under it, where area above the axis counts as plus and area below counts as minus.”

J “Why area?”

“Same reason as on your bike. Ride at 90 km/h for two hours. How far did you go?”

J “180 kilometres.”

“Draw your speed against time: a flat line at 90, two hours long. What’s the area under it?”

J “90 times 2 is… 180. The distance is the area!”

“Adding up something that changes over time is always finding an area. Mathematicians call it integration; I’ll keep calling it ‘adding up’. Let’s try the test. I’ll make an F with two hands inside: a long one spinning twice per loop and a shorter one spinning five times. Then we’ll test it for ‘twice’ and for ‘three times’.”

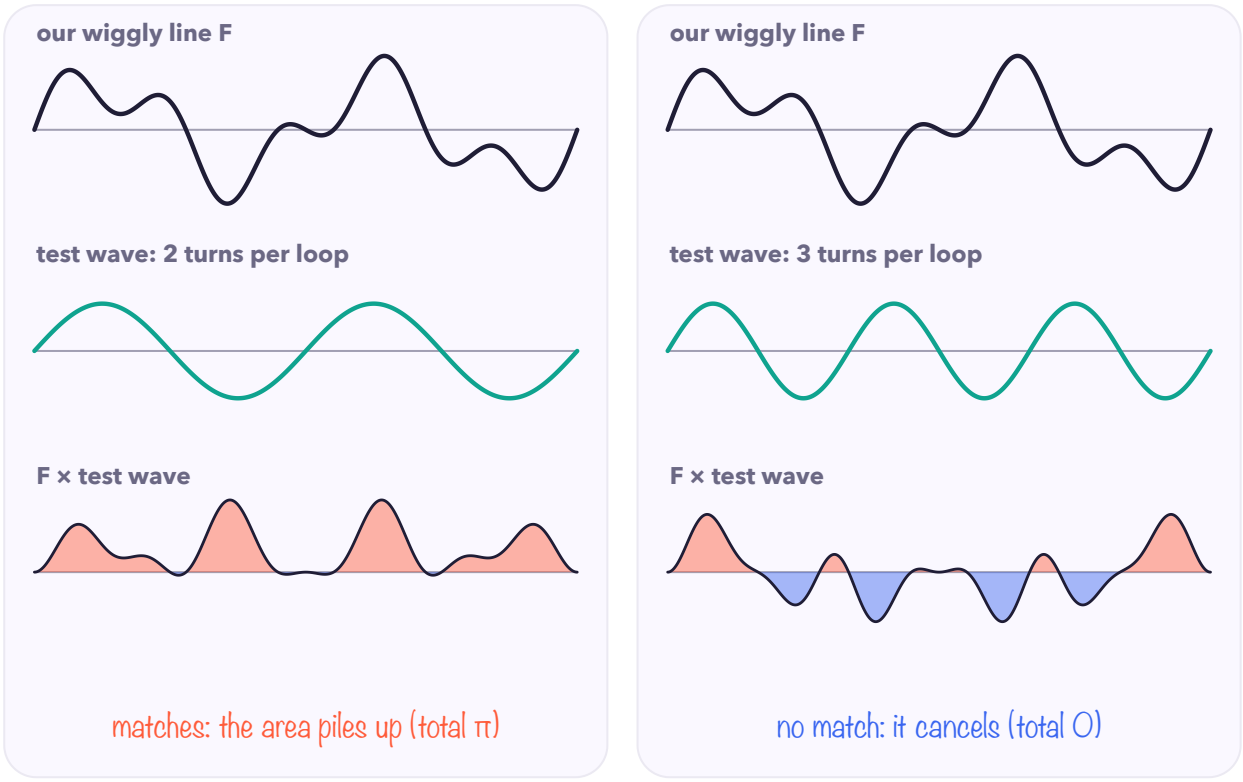


FIGURE 1.11 The resonance test. Top: our wiggly line F. Middle: two test waves. Bottom: F multiplied by each test wave. When the test speed is inside F (left), the product sits mostly above the line and the area piles up. When it isn't (right), every bit above is matched by a bit below, and the total is exactly zero.

J "Look at the right-hand one! The coral bits and the blue bits cancel perfectly."

"Like pushing the swing out of rhythm: help, hinder, help, hinder, and over a full loop, nothing. And that's true for any two different whole-number speeds. Two hands that don't match always cancel out over a loop."

J "And the left one doesn't cancel because..."

"Because the matching part of F, times `ducati(x)`, is the same wave multiplied by itself. What happens when you multiply a number by itself?"

J "It's never negative. Minus times minus is plus."

"So the product can't dip below the line. There's nothing to cancel, and the area just grows."

J "So the test says 'zero' for speeds that aren't there, and 'something big' for ones that are. How big?"

"Let's work it out. Suppose the hand inside F has length A, and our test wave has length 1."

J "Why 1? Can I have 7? It's my lucky number."

"You can, but then every answer comes out seven times too big and you'd have to divide the seven back out. A ruler should be marked in ones. Length 1 keeps it honest."

J "Fine. Length 1."

"Then the matching part of the product is A times a pure wave multiplied by itself. So we need to know what a pure wave times itself looks like. Here's a lovely fact."

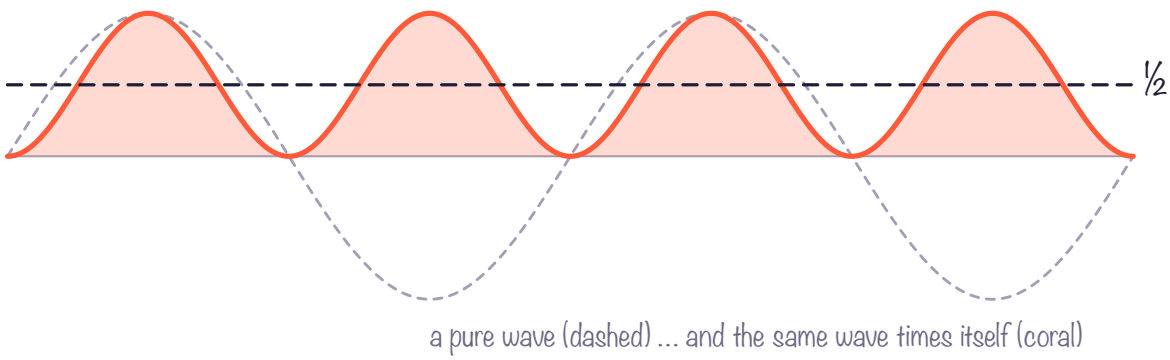


FIGURE 1.12 A pure wave times itself is another wave: twice as fast, never negative, and floating exactly halfway up, at one half. So over a loop, it averages exactly ½.

J "It's just a wave sitting on the line at a half."

"So over a full loop, it averages exactly one half, which means F times `ducati(x)` averages A times a half."

J "So to get A back, I double the average."

"And that's the formula. In words:"

THE RESONANCE TEST, IN WORDS

The length of the hand spinning n times per loop = 2 × the average, over one loop, of (F × a pure wave spinning n times per loop)

J "That's it? That's the whole machinery?"

"Almost. You're about to find the hole in it yourself. Try this F: a single hand of length 1, spinning twice per loop, that started a quarter-turn late."

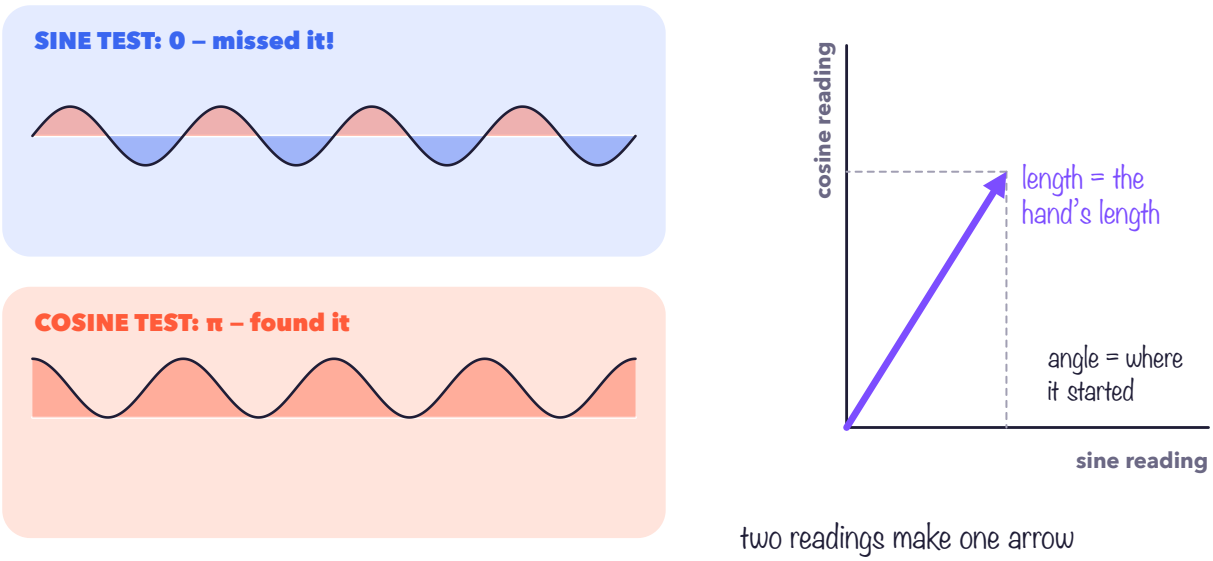


FIGURE 1.13 The hole in the test, and the fix. A hand that starts a quarter-turn late is invisible to the sine test: the product cancels to zero. A second test wave, a quarter-turn ahead, catches it. Together the two readings form an arrow: its length is the hand's length and its angle is where the hand started.

J "It cancels! There IS a twice-per-loop hand in there, and the test says zero."

"The swing again. Push in the right rhythm but at the wrong moment, when the swing is at the very top and not moving. What do your pushes do?"

J "Nothing. It isn't moving, so push times motion is zero."

"So timing matters, not just rhythm. The fix is to test twice: once with `ducati(x)`, and once with a copy shifted by a quarter-turn. Mathematicians call the pair sine and cosine. Any hand, whatever its starting point, shows up in one or the other or both."

J "And then I combine them... the two readings are like 'how far east' and 'how far north'..."

"Go on."

J "So they make an arrow! Its length, by Pythagoras, is the hand's length. And the direction it points is... where the hand started!"

"That's the phase. And you've just rediscovered why engineers love spinning hands so much: a spinning test hand does both tests at once and gives you the arrow directly. Mathematicians write spinning hands with complex numbers. For our purposes, that's all a complex number is: a neat way to write an arrow that turns."

J "What about a hand that doesn't spin at all?"

“Speed zero. It just lifts the whole line up or down. The test for it is the plainest one there is: the average height of F.”

“Now you’ve earned the names. The lengths we’re measuring are the Fourier coefficients. The test itself, multiply and add up, is called an inner product. And because different whole-number speeds always cancel each other out in the test, mathematicians call them orthogonal, which is the Greek-flavoured word for perpendicular.”

J “Perpendicular? They’re wiggly lines, not arrows at right angles.”

“Hold that thought. In the next chapter I’ll show you that they really are at right angles, just in a space with more directions than you can picture.”



IN TEXTBOOK LANGUAGE (FOR THE CURIOUS)

For a wiggly line F that repeats every L seconds, textbooks write the length of the n-times-per-loop sine hand as $b_n = (2/L) \int F(t) \sin(2\pi nt/L)\, dt$, taken over one loop, and the cosine hand as $a_n = (2/L) \int F(t) \cos(2\pi nt/L)\, dt$. The “2/L” is our “double the average”. That’s all it is.

“Now for the thing I promised. You own one of these machines. Two, actually.”

J “I do?”

“Your ears. Deep inside each one is a snail-shaped tube called the cochlea, about 35 millimetres long if you uncoiled it. Running along it is a membrane that’s stiff and narrow at the entrance and wide and floppy at the far end.”

J “Let me guess. Different parts have different natural frequencies.”

“The entrance responds to high notes, the far end to low ones, and everything in between in order. When a sound arrives, each spot along the membrane shakes hardest for its own hands. About three and a half thousand hair cells sit in a row along it and report how hard their spot is shaking.”

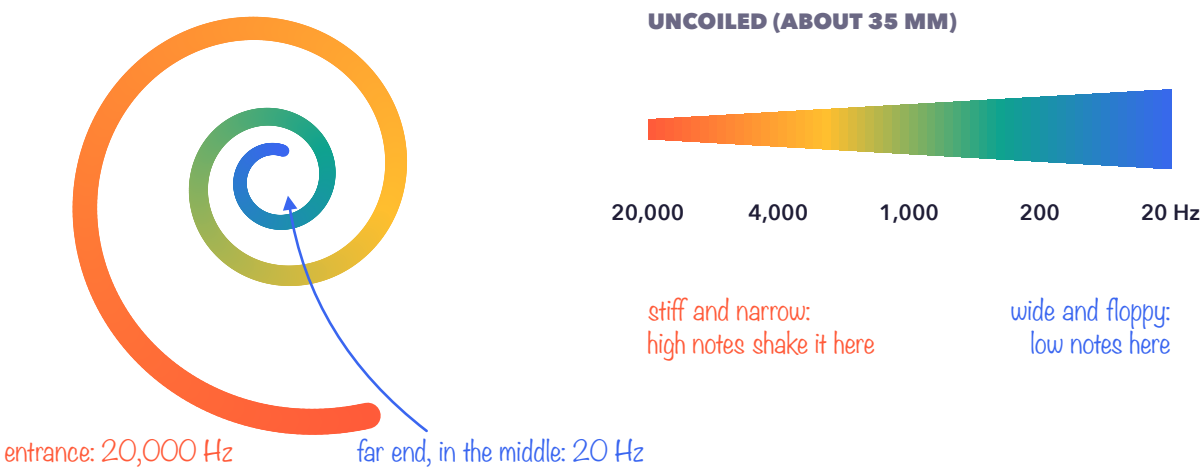


FIGURE 1.14 Your built-in resonance test. The membrane in the cochlea responds to high pitches near its entrance and low pitches deep in the coil, so each spot along it reports how long one hand is. Position along the spiral is pitch.

J "So my ear is doing the resonance test. On every sound. All the time."

"Something very like it. Hermann von Helmholtz guessed in 1863 that the ear held rows of tiny resonators, like the strings inside a piano. The truth, which won Georg von Békésy a Nobel Prize in 1961, is subtler: a wave rolls along the membrane and peaks at a spot that depends on the pitch. But the upshot is Helmholtz's: position is pitch. That's how you hear a drum and a voice as separate things in one wiggly line. Your ear splits the sound into hands before your brain ever sees it."

J "And I thought I was just listening."

TOOL UNLOCKED

The resonance test

To find the hand spinning n times per loop, multiply the wiggly line by a pure test wave of that speed and add up over one loop. Different speeds cancel to zero and the matching one survives. Double the average to get its length, and test with both sine and cosine to catch where it started.

1.4

How many circles?

Convergence, corners and a 159-year argument

BY THE END
You'll learn why sharp corners are expensive, and why mathematicians fought about it for 159 years.

6 min read A conversation over the fence

SCENE *Friday morning. Jim is waiting at the fence with a coffee and a suspicious expression.*

J "I couldn't sleep. You keep saying 'infinitely many hands'. If I add up infinitely many things, don't I get infinity?"

"Not necessarily. Walk towards that wall, but at each step only cover half the distance that's left."

J "Half, then a quarter, then an eighth... I never quite get there."

"But you never go past it, either. One plus a half plus a quarter plus an eighth, on forever, adds up to exactly two. Infinitely many pieces, finite total."

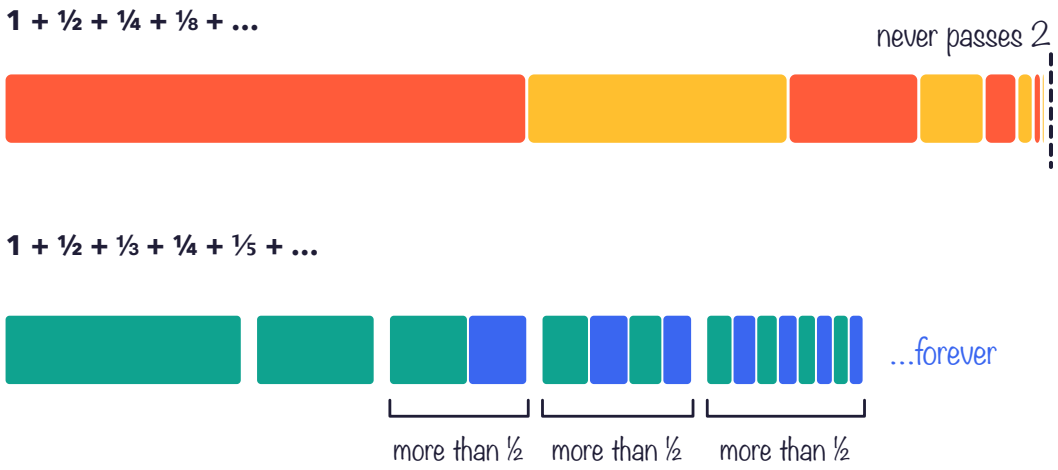


FIGURE 1.15 Top: halve the gap each time and the pieces pile up to exactly 2, never beyond. Bottom: the harmonic series looks just as tame, but group its pieces cleverly and every group is bigger than ½, so it grows past any number you name.

J "Right, so as long as the hands keep getting smaller, the total settles."

"Careful. Try one, plus a half, plus a third, plus a quarter, plus a fifth..."

J "They're getting smaller. So it settles."

"It never settles. It grows forever."

J "It can't!"

"A French bishop called Nicole Oresme proved it around 1350, with a trick you can do in your head. Group the pieces. A third plus a quarter is more than a half. The next four pieces, a fifth to an eighth, add up to more than a half. The next eight, more than a half again. You get an endless supply of halves, and endless halves add up to anything you like."

J "That's outrageous."

"It's slow, mind. You need more than twelve thousand pieces to pass 10, and about 272 million to pass 20. But it never stops. And here's the delicious part: it's called the harmonic series, after music, because a string's harmonics have wavelengths of one, a half, a third and a quarter of the string's length."

J "So the one sum named after music is the one that runs away."

"Which tells you that 'getting smaller' isn't enough. What matters is how fast the hands shrink, and that depends on the shape you're drawing. Let me show you two wiggly lines. A triangle wave: straight slopes and pointed tops. And a square wave: flat, then a sudden jump, then flat again."

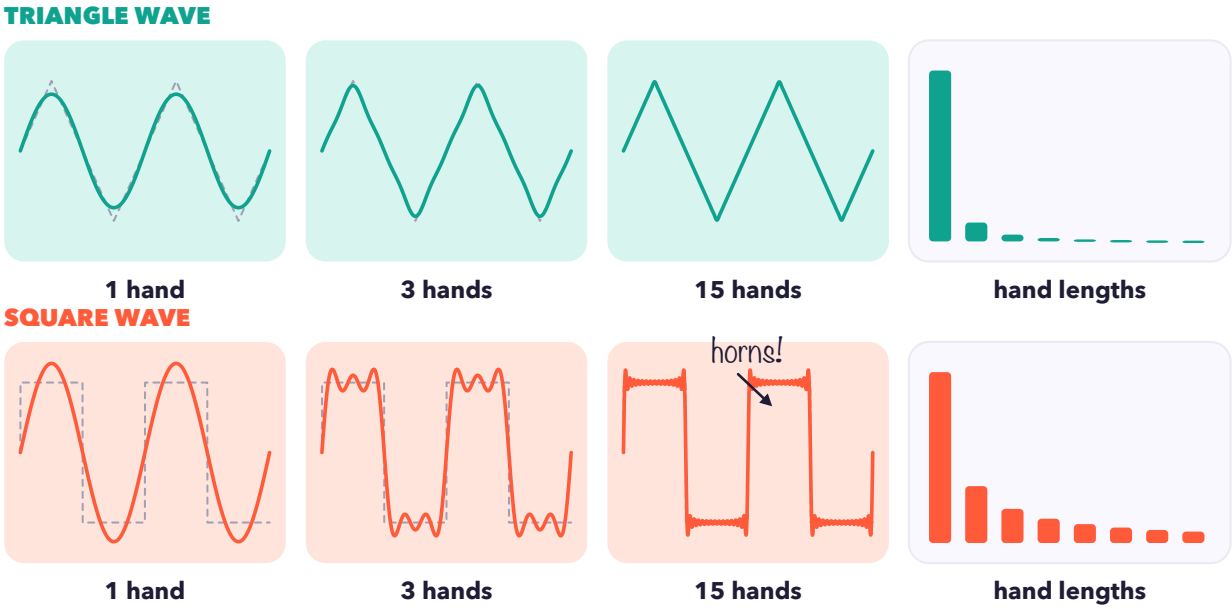


FIGURE 1.16 Drawing with more and more hands. The triangle wave (top) is nearly perfect with just three, because its hands shrink fast: the second is 1/9 the length of the first, the third 1/25. The square wave (bottom) needs far more, because its hands shrink slowly: 1/3, 1/5, 1/7... And look at the little horns beside each jump.

J "The triangle's practically there with three hands. The square is still wobbling with fifteen."

"Why, do you think?"

J "The square has a jump. It has to get from the bottom to the top instantly, and slow, lazy circles can't turn that sharply. You need fast hands for sharp corners, and lots of them."

"That's the whole thing. Smooth things need only slow hands. Sharp things need fast ones, and plenty of them. Corners are expensive."

J "Is that why a clap sounds so bright? It's a sudden jump."

"Exactly. A clap, a snare drum, a snapped twig: all sharp, all full of fast hands. A hum is smooth and has almost none. And you know what a square wave sounds like, by the way."

J "Do I?"

"The Super Mario theme. The old Nintendo played the melody on square-wave voices and the bass line on a triangle wave."

J "So that buzz is all those slowly shrinking hands?"

"That buzz is precisely the slowly shrinking hands. The triangle bass sounds soft and round because its hands vanish so fast."

SIDE QUEST: YOUR EAR CAN HEAR HANDS SHRINKING

A square wave's hands shrink like 1/3, 1/5, 1/7, so the hand spinning five times per loop is still a fifth as long as the first. That long tail of loud, fast hands is the buzz in retro game music. In a triangle wave the same hand is only 1/25 as long, which is why it sounds almost as pure as a flute.

J "What are those horns on the square, next to each jump?"

"Ah. Add more hands and they squeeze closer to the jump, but they never get any shorter. They stay at about nine percent of the jump, forever."

J "Forever? But you said the hands could draw anything!"

"Hold that thought. It has a name, a strange history and a whole section of Chapter 3."

J "So did everyone just believe Fourier?"

"Nobody believed Fourier. In 1807 he told the Paris Academy that any function could be built from pure waves, even ones with corners. Joseph-Louis Lagrange, perhaps the greatest mathematician alive, objected. How could smooth waves possibly add up to a sharp corner?"

J "I'm with Lagrange. That's exactly what I just said."

"And you were both right to be suspicious. The argument lasted 159 years."

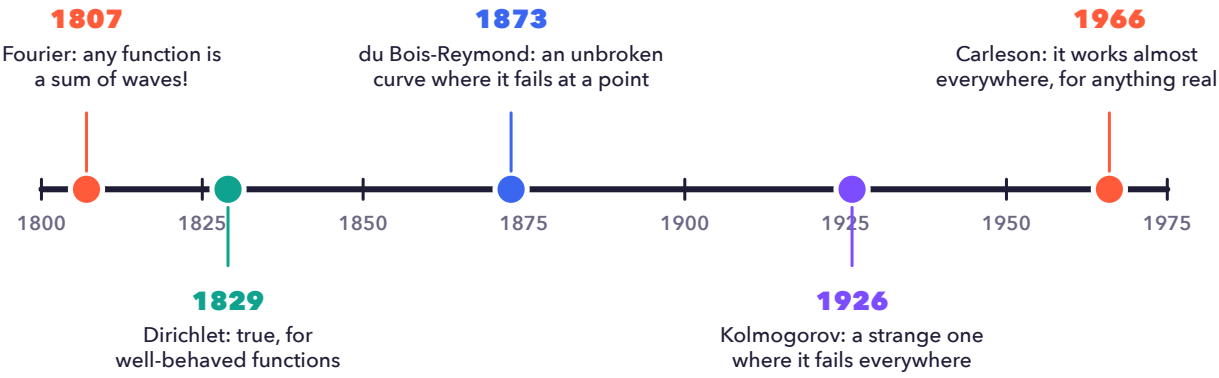


FIGURE 1.17 The 159-year argument. Engineers used Fourier series happily the whole time; mathematicians took until 1966 to settle exactly when they work.

J "Kolmogorov found one where it fails everywhere?"

“A function so strange that its Fourier series doesn’t settle at a single point. Then in 1966 Lennart Carleson proved that for every function you’d meet in real life, and plenty you wouldn’t, it settles almost everywhere. Physicists and engineers didn’t wait for permission. They used it the whole time, to build telegraphs, radios and radar.”

“One last thing, and it’s the one in your pocket. Your phone stores songs as MP3s, or something like them. Guess how.”

“Almost everywhere” is a precise idea: the places where it fails take up no length at all.

J “It breaks the song into hands, slice by slice…”

“And then?”

J “…throws away the hands I can’t hear?”

“Exactly. A loud hand drowns out quieter ones close to it in pitch, the way a shout drowns out a whisper. So the file keeps the hands your ear will notice and throws most of the rest away. That’s how a song shrinks to about a tenth of its size and you barely notice.”

J “And photos? JPEGs?”

“The same trick. But for that you need the idea in the next chapter.”

J “One complaint. All week, you never once said ‘Fourier coefficient’ or ‘sine wave’ until the very end. Why not tell me the names up front?”

“Because a name up front is a lid on a box you haven’t looked inside yet. Richard Feynman liked to say there’s a difference between knowing the name of something and knowing something. You can know the name of a bird in every language on earth and still know nothing about the bird.”

J “Well. I know something now.”

“You know a great deal. You know that every repeating thing is made of circles, that circles seen side-on are waves, and that a song is an orchestra of them. You know how to find each musician with a resonance test, and why corners cost so much. Plenty of people with a maths degree couldn’t tell you half of that over a garden fence.”

TOOL UNLOCKED

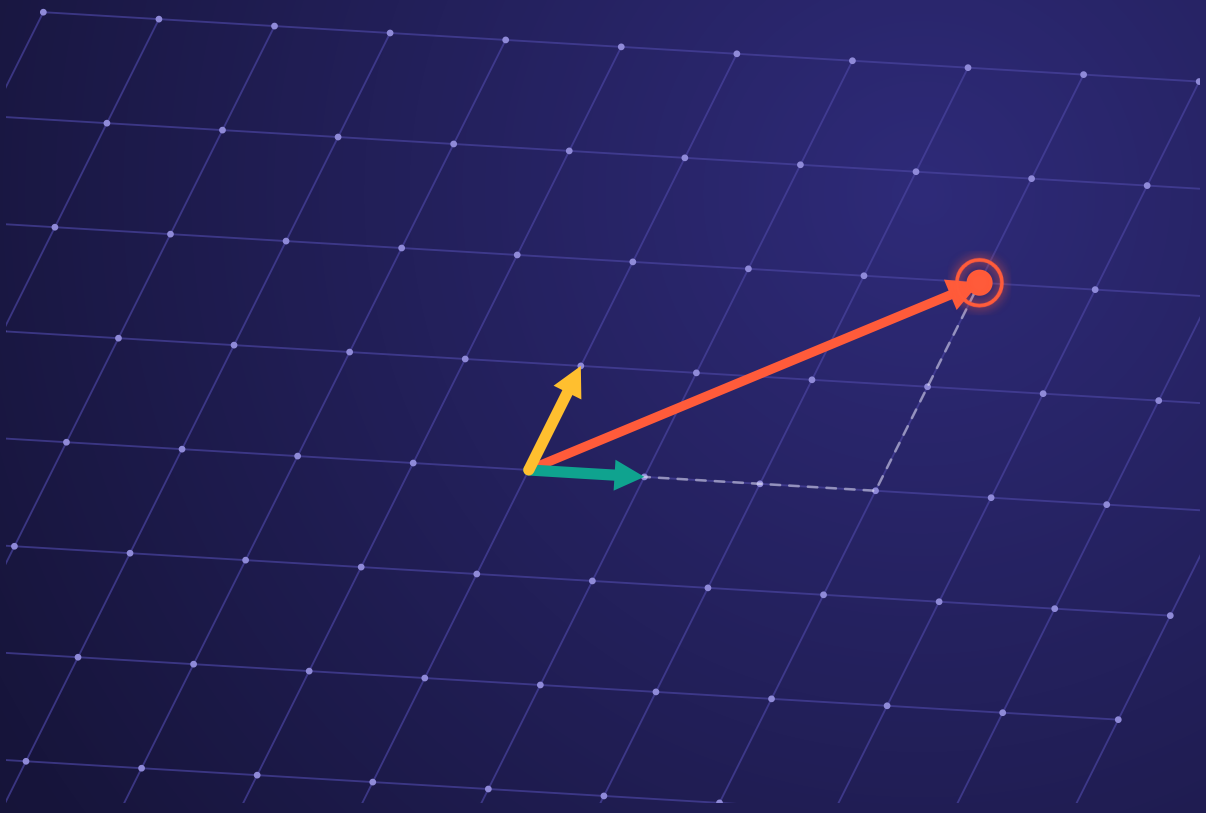
How many circles?

Infinitely many hands can still add up to something finite, but only if they shrink fast enough. Smooth shapes need few, slow hands; sharp corners need many fast ones. That one fact runs MP3s, JPEGs and the sound of Super Mario.

CHAPTER

2 Vector spaces

Anything you can add and scale



What do a biryani, a rainbow and a pop song have in common?

YOUR ROUTE THROUGH THIS CHAPTER

2.1 **Anything you can add and scale**
Introduction to vector spaces

2.2 **The minus-two-onions problem**
Developing the mathematical architecture for vector spaces

2.3 **Same song, different language**
Changing the basis

2.1 Anything you can add and scale

Introduction to vector spaces

BY THE END
You'll see recipes, arrows, songs and colours as the same kind of thing.

7 min read · A conversation over the fence

SCENE *Saturday afternoon. A smell of frying onions, and a cry for help from Jim's kitchen.*

J "Neighbour! Emergency!"

"Is Bruno all right?"

J "Bruno's fine. I'm not. My mother-in-law's biryani recipe serves four, and twenty people are coming tonight. It says 'a few cardamom pods'. What's five times a few?"

"A handful. Show me the recipe."

> *A stained index card: 2 cups rice, 500 g chicken, 2 onions, 1 tsp cumin, 4 cardamom pods, 1 bay leaf, "salt to taste".*

"For twenty, you multiply every line by five."

J "Obviously."

"And your brother's bringing vegetarian friends, so you're cooking his paneer biryani as well. What goes on the shopping list?"

J "Both recipes added together, line by line. Rice plus rice, onions plus onions…"

"Then congratulations: you're doing vector arithmetic."

J "I'm doing groceries."

"Same thing. Remember the treasure hunt? A vector was 'which way and how far'. But look at how you'd actually write one down: 'so far east, so far north'. Two amounts."

J "Two... ingredients. East and north."

"And your recipe is seven amounts. Seven ingredients."

J "So an arrow is a recipe with two ingredients?"

"Let's check. Take a tiny recipe, just chilli and salt, and draw it as an arrow: so many teaspoons of chilli across, so many of salt up."

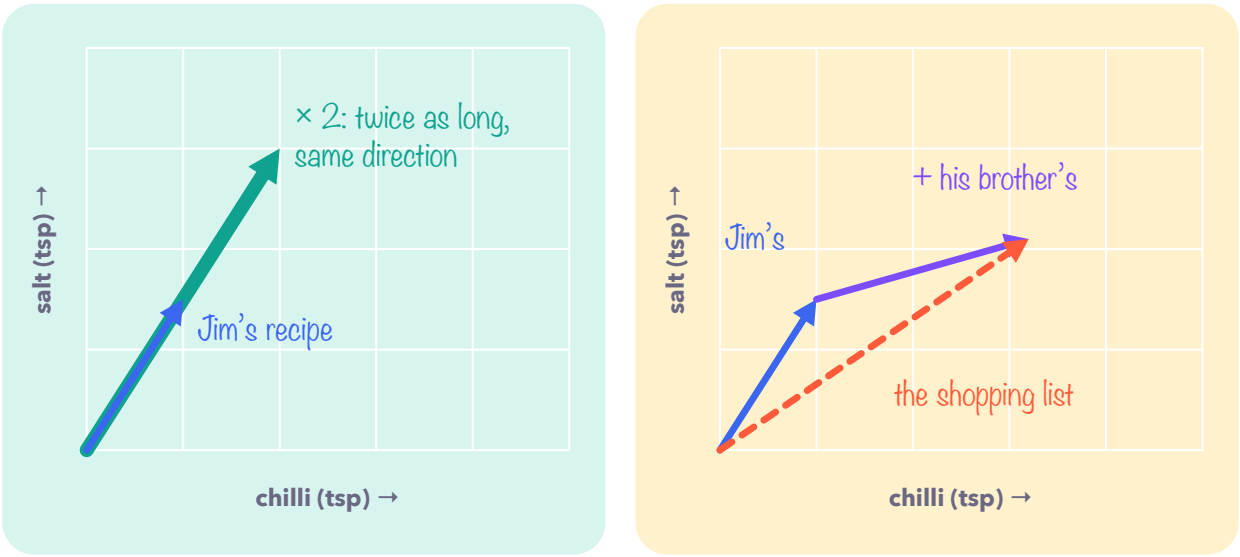


FIGURE 2.1 Two-ingredient recipes as arrows. Scaling a recipe stretches its arrow; combining two recipes is tip-to-tail, exactly like the treasure hunt.

J "Doubling the recipe doubles the arrow. And adding my brother's is walking one arrow, then the other."

"Tip to tail. The kitchen and the treasure map follow exactly the same rules. And that's the entire idea of this chapter: a vector isn't an arrow. A vector is anything you can add and scale, where adding and scaling behave the way you'd expect. The arrow is just one costume it wears."

J "What other costumes?"

"Recipes. Shopping lists. Forces. Colours of light. Sounds. Pictures. You'll meet them all. In every costume, each ingredient is a direction, and the amount of that ingredient is how far you go in that direction."

Jim holds up a jar of garam masala.

J "Then what's this? A new direction?"

"What's in it?"

J "Cumin, coriander, cardamom, cloves, cinnamon, pepper..."

"And do you have all of those in your cupboard, separately?"

J "Yes."

"Then garam masala isn't a new direction. Anything you can cook with it, you could cook without it, just by using the spices inside it. It's convenient, but it adds nothing new."

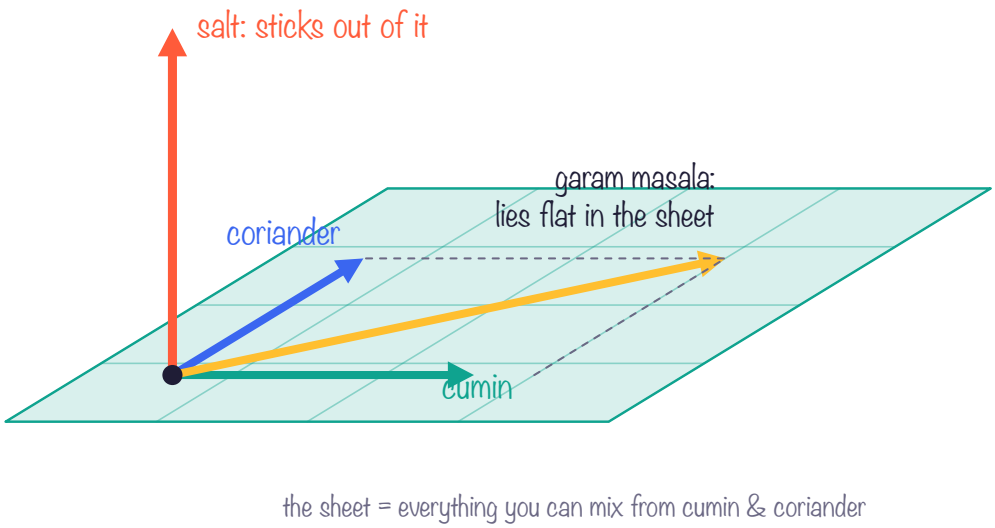


FIGURE 2.2 Cumin and coriander, and every mix of them, fill a flat sheet. Garam masala (simplified here to a mix of just those two) lies flat in that sheet, so it's not a new direction. Salt sticks straight out of it: no mix of the two spices makes salt.

J "And salt?"

"Can you make salt from any amount of cumin and coriander?"

J "Of course not."

"So salt is a genuinely new direction: it sticks out of the sheet of everything you can mix from those two. Mathematicians have names for all of this. Everything you can reach by mixing some ingredients is their span. A set of ingredients where none is a mix of the others is linearly independent. And an independent set that can reach everything you need is a basis."

J "And how many there are?"

"That's the dimension: how many truly independent ingredients the space needs."

J "So everything I could cook from my seven lines is a seven-dimensional space."

“As long as none of them is secretly a mix of the others. Can you make a chocolate cake in it?”

J “No chocolate in the span.”



“Now tell me what’s been bothering you since Wednesday.”

J “Wednesday? ...Oh. Oh no. The orchestra. Each musician plays one pure wave at some loudness. That’s a recipe! The pure waves are the ingredients and the loudnesses are the amounts.”

“So a song is...”

J “A vector. A song is a vector.”

“And a song is a wiggly line, which mathematicians call a function...”

J “Functions are vectors?”

“Add two wiggly lines moment by moment and you get a wiggly line. Turn up the volume on one and it’s still a wiggly line. It passes the test. The pure waves are a basis, and the Fourier coefficients you found with the resonance test are the amounts. They’re the recipe.”

J “With infinitely many ingredients.”

“One for every whole-number speed. It’s an infinite-dimensional space, and you’ve been cooking in it all week.”

“Now for the costume that surprises everyone. What’s the dimension of colour?”

J “Colour? There are infinitely many colours.”

“Infinitely many colours, but how many ingredients?”

J “Red, green and blue? Like on a TV?”

“And why does a TV have three?”

J "Because… that's what TVs have?"

"TVs have three because you have three. Light itself has infinitely many ingredients: Newton's prism showed you every wavelength in a sunbeam. But the back of your eye has only three kinds of colour sensor, called cones, and each is most sensitive to a different part of the rainbow."

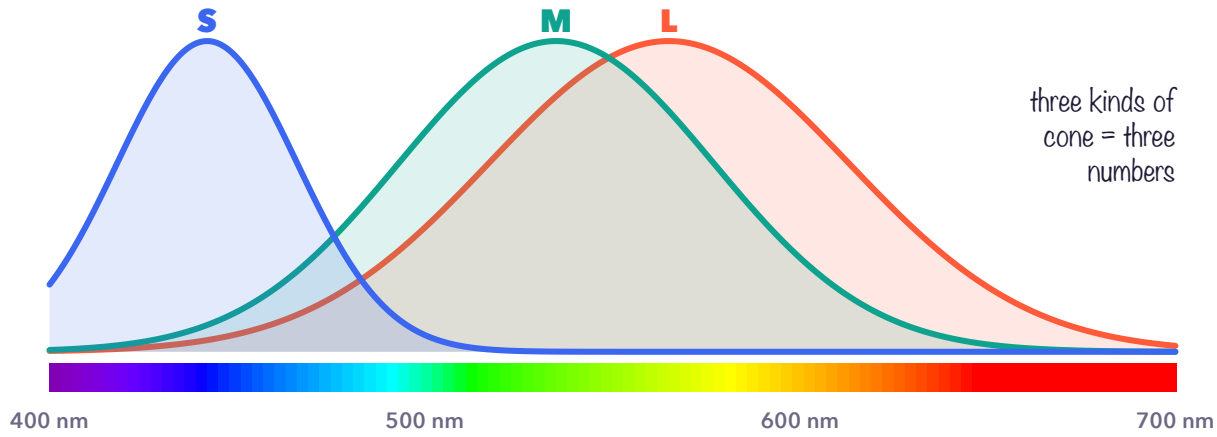


FIGURE 2.3 Your three kinds of cone and the parts of the rainbow they respond to. However complicated a light is, your brain receives just three numbers: how strongly it tickles each kind. (Curves simplified.)

J "So whatever light comes in…"

"…your eye squeezes it down to three numbers. Of all the infinitely many directions light can vary in, you see a three-dimensional shadow. And that has a startling consequence: two completely different lights that happen to produce the same three numbers look identical to you."

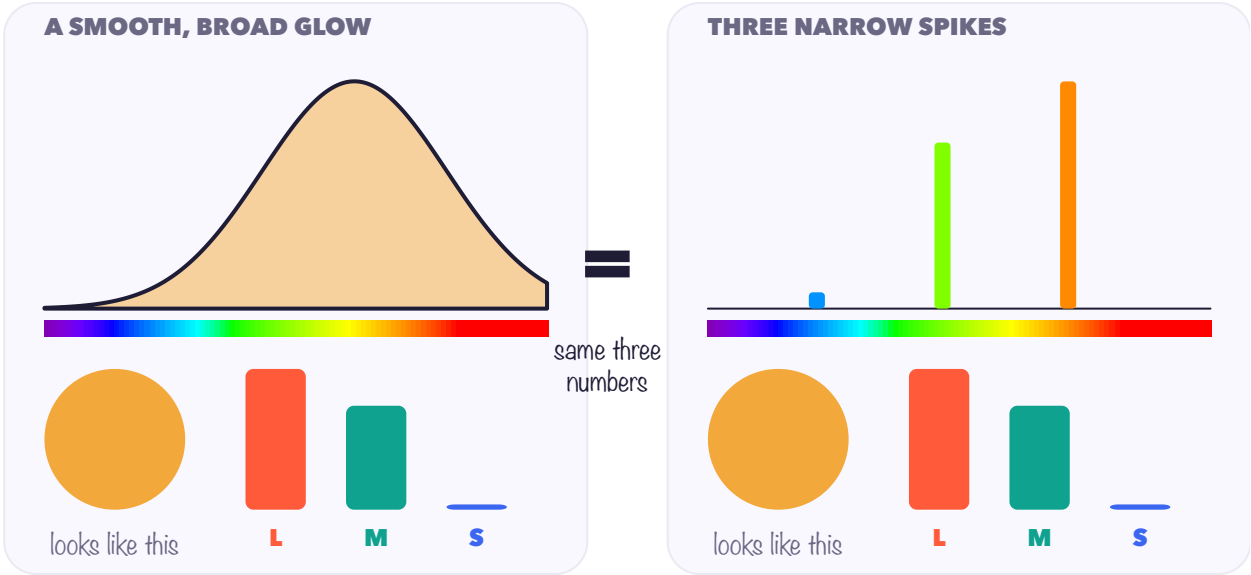


FIGURE 2.4 Two very different lights, one colour. Left: a smooth, broad glow. Right: three narrow spikes, like a screen's red, green and blue dots. Both tickle your three cones by exactly the same amounts, so you can't tell them apart. (Worked out with the simplified cones.)

J “So my TV is lying to me.”

“Beautifully, and only because you’re a three-ingredient animal. Colour scientists call such look-alikes metamers. Your screen builds a sunset out of three narrow spikes of light, and you’re none the wiser.”

J “What about Bruno?”

“Dogs have two kinds of cone, so Bruno lives in a two-dimensional colour space. To him, a red ball on green grass is mostly a matter of brightness.”

J “Poor Bruno.”

“Don’t pity him. Think of the mantis shrimp instead: it has twelve kinds of colour sensor. And yet experiments suggest it’s worse than you at telling similar colours apart.”

J “Twelve ingredients and it can’t cook?”

“It seems to use them differently, for quick recognition rather than fine comparison. More dimensions aren’t automatically better. What matters is what you do with them.”

SIDE QUEST: THE TWO-DIMENSIONAL DOG

Dogs, like many people with red–green colour blindness, have two kinds of cone, so their colours live in a two-dimensional space. A small number of people seem to carry a fourth kind of cone, and a very few may actually see with it.

J “So recipes, arrows, songs, colours…”

“…are all the same kind of thing, because you can add and scale them. Once you see that, every tool you have for arrows works on all of them, and that’s why the idea is worth so much.”

J “The robot would have memorised a thousand arrows.”

“And you learned the one thing they all do. Go and cook. And Jim: in your mother-in-law’s presence, do not call her biryani a vector.”

TOOL UNLOCKED

Vectors wear costumes

A vector is anything you can add and scale. Pick independent ingredients (a basis) and every vector becomes a list of amounts. Arrows, recipes, songs and colours all live in vector spaces: of 2, 7, infinitely many and (for you) 3 dimensions.

2.2

The minus-two-onions problem

Developing the mathematical architecture for vector spaces

BY THE END

You'll learn the rules that make something a vector, why negative red exists, and how headphones erase noise.

8 min read A conversation over the fence

SCENE *Sunday morning. The party was a success. Jim's mother-in-law had notes.*

J "Morning. The biryani was a triumph, thank you. But my mother-in-law says next time it needs 'two fewer onions and one more chilli'. And that got me thinking about vectors, which is your fault."

"Dangerous. Go on."

J "I wrote her advice down as her recipe minus mine. And for onions I got minus two. What on earth is minus two onions? You can't put minus two onions in a pot."

'Jim, that's the best question anyone has asked me all week. You've just caught recipes failing to be vectors."

J "But you said..."

"I said a vector is anything you can add and scale, where adding and scaling 'behave the way you'd expect'. It's time to say exactly what we expect. Let's write the rules by testing them on recipes. First: does the order matter when you add? Your recipe plus your brother's, or your brother's plus yours?"

J "Same shopping list."

"Does the grouping matter? With three recipes, add the first two and then the third, or the last two and then the first?"

J "Same again."

"Is there a recipe that changes nothing when you add it?"

J "The empty recipe. Nothing at all."

"Scaling: double a combined list, or double each list and then combine?"

J "Same."

"Scale by 2 and then by 3, or scale by 6 in one go?"

J "Same."

"Now the hard one. For every recipe, is there another one you can add to it that brings you back to nothing?"

J "...No. That would need minus two cups of rice. That's where it breaks."

"So recipes pass every rule but one: they have no negatives, and you can't scale them by negative numbers either. That's why your subtraction produced nonsense."

SIDE QUEST: THE RULES, IN PLAIN WORDS

Adding: order doesn't matter; grouping doesn't matter; there's a "nothing" that changes nothing; and everything has a negative that cancels it.
Scaling: scaling by 1 changes nothing; scaling twice is scaling by the product; and scaling spreads over adding, both ways round.
Mathematicians call these the vector space axioms. Every one of them is obvious; the skill is noticing when one fails.

"But look at what your mother-in-law actually gave you. Not a recipe: a change to a recipe. Two fewer onions, one more chilli."

J "A tweak."

"Can you add tweaks?"

J "One tweak, then another? Sure. 'Two fewer onions' then 'one more onion' is 'one fewer onion'."

"Scale them?"

J "'Do it twice as much': four fewer onions, two more chillies."

"A tweak that changes nothing?"

J "The do-nothing tweak."

"And a negative?"

J "Undo it: two more onions, one less chilli. That makes perfect sense!"

"So tweaks pass every rule. Recipes are places. Tweaks are the journeys between them, and the journeys are the vectors."

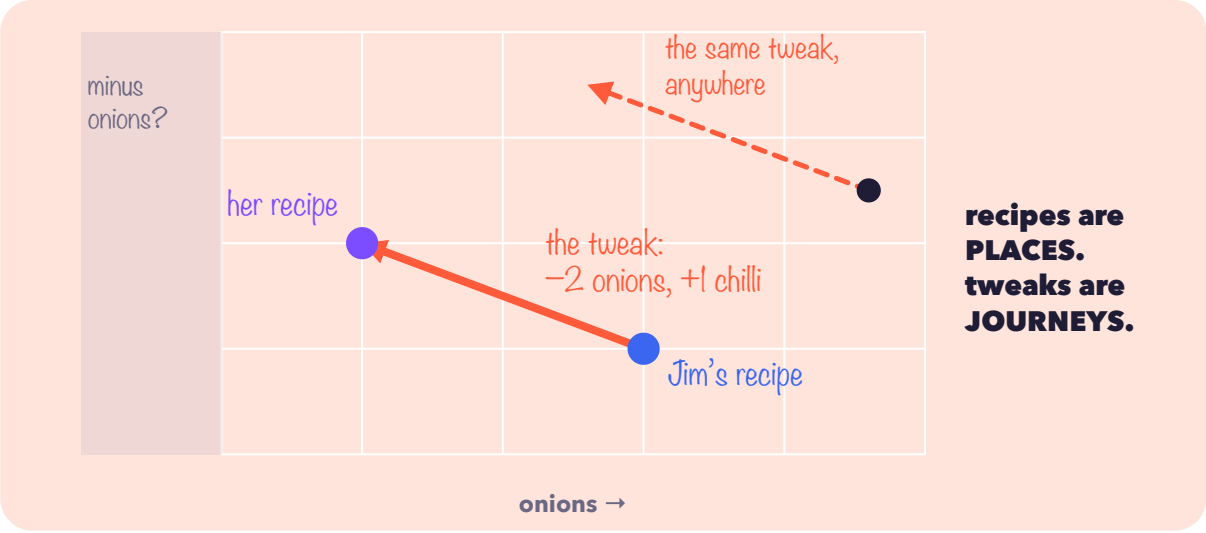


FIGURE 2.5 Recipes are points: two of them don't add up to anything meaningful, and there's no such thing as a negative one. Tweaks, the arrows between recipes, can be added, scaled and undone, and the same tweak works from any starting recipe. The tweaks are the vectors.

J "Like the treasure hunt! The spot where the treasure's buried isn't a vector. 'Forty paces north-east' is."

"Exactly. You can't add two cities, but you can add two journeys. Or take temperature: is a 30-degree day plus a 30-degree day a 60-degree day?"

J "Ha! No. But 'five degrees warmer' plus 'five degrees warmer' is 'ten degrees warmer'."

"Temperatures are places; changes in temperature are vectors. Once you see this, you'll see it everywhere. Let's play a game: 'vector or not'. I name a collection, and you tell me whether it's a vector space."

J "Go."

"All possible wiggly lines one second long."

J "Add two, you get a wiggly line. Scale one, still a wiggly line. Negative: flip it upside down. Nothing: a flat line. Vector space."

"All the sounds quieter than 100 decibels."

J "Scale one up by a thousand and it's too loud to be in the club. Not a vector space."

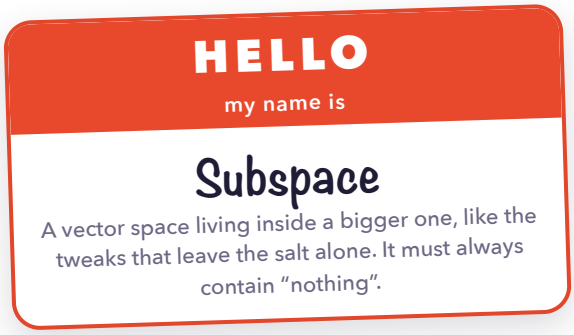
"All the tweaks that don't change the salt."

J "Add two of them and the salt still doesn't change. Scaling doesn't change it, undoing doesn't, and the do-nothing tweak doesn't. Vector space!"

"All recipes with exactly one teaspoon of salt."

J "Add two and you've got two teaspoons, so it's out. And the empty recipe isn't in it either."

"That's the quickest test there is. If a collection doesn't contain 'nothing', it can't be a vector space. A rule like 'the salt must be one' fails; 'the change in salt must be zero' works. And a vector space living inside another one, like your salt-free tweaks, is called a subspace."



J "So negative amounts only make sense for tweaks. Never for real stuff, like light."

"That's what everyone assumed. Then, in the late 1920s, two British scientists, William David Wright and John Guild, ran an experiment. Volunteers looked at a split circle: one half lit by a pure colour from the rainbow, the other by a mix of three lamps, red, green and blue. They turned the knobs until the two halves matched."

J "Three ingredients to match any colour. Just like the TV."

"Except that for some bright blue-greens, no setting worked. However they turned the knobs, the mix looked washed out next to the target."

J "So what did they do?"

"They moved the red lamp to the other side. They shone red onto the target instead, and then it could be matched."

J "Adding red to the target is like... taking red away from the mix."

"It's exactly that. Written as an equation, the mix needed negative red."

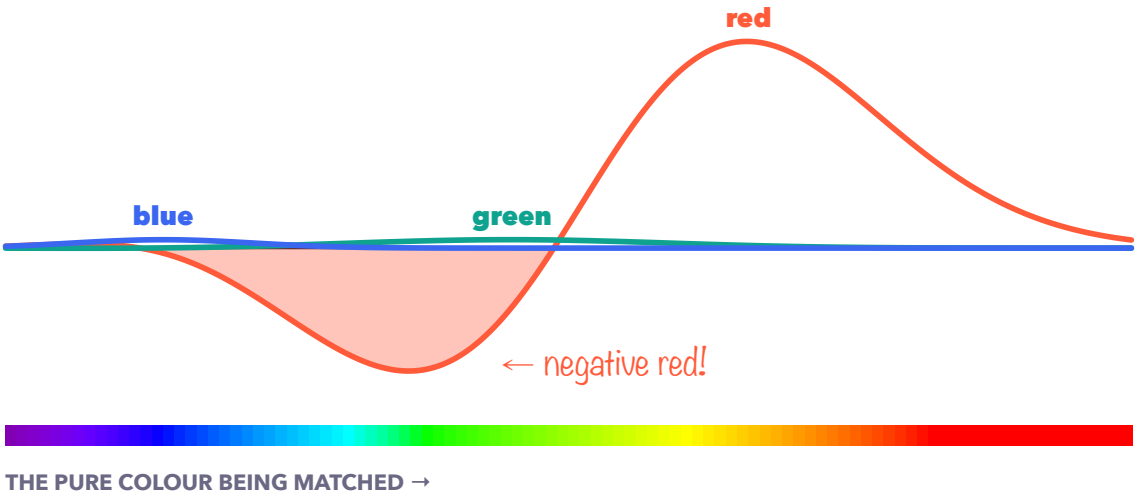


FIGURE 2.6 How much red, green and blue light you need to match each pure colour of the rainbow. Across the blue-greens, the red amount dips below zero. Nothing can shine negative light, so the experimenters shone it on the other side instead. (Worked out with the simplified cones, so trust the shape, not the exact numbers.)

J "Minus two onions, in light!"

"The mathematics insisted on negatives even though no lamp could make them, so the scientists simply moved the lamp across the equals sign. Their numbers became the 1931 international standard for colour, which still sits behind every screen and printer. And the negative red tells you something practical: no screen made of three coloured lights can show every colour you can see."

J "I didn't know my TV had limits."

"Look at a real rainbow, then at a photo of one. The photo's blue-greens are always a little duller. Now: does sound have negatives?"

J "Flip the wiggly line upside down. Push where it pulled."

"And what's a sound plus its negative?"

J "...Nothing. Silence."

"You've just invented noise-cancelling headphones."

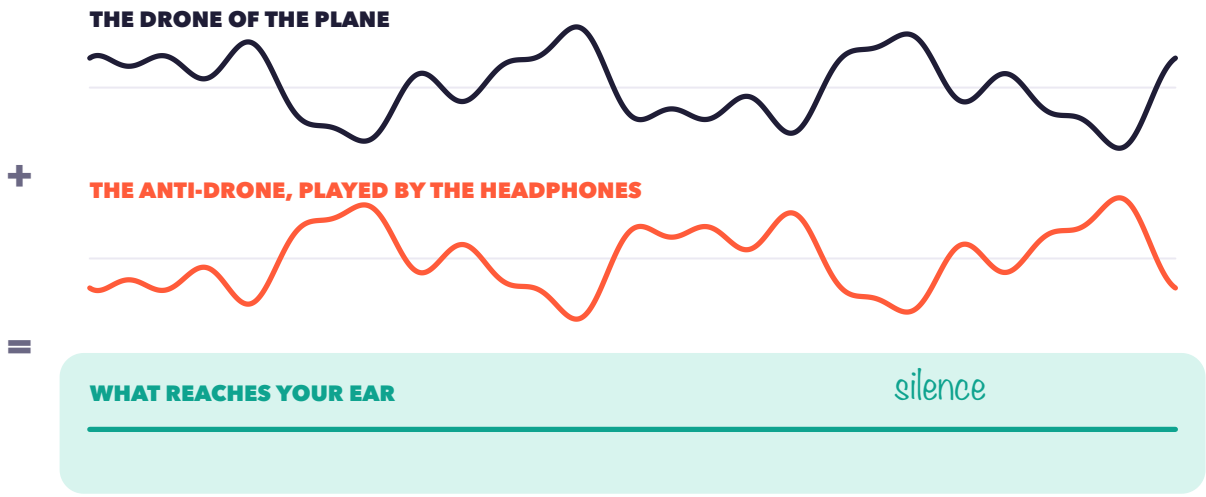


FIGURE 2.7 A noise, its exact negative and their sum: silence. Noise-cancelling headphones listen to the noise outside and play its negative inside, thousands of times a second.

J "They play the anti-noise?"

"A little microphone listens to the drone of the plane, a chip works out its negative, and the speaker plays that into your ear. Noise plus anti-noise is nothing. It works best on steady, low hums like engines. For sudden, high-pitched sounds the chip can't keep up, so the padding of the ear cups does that job instead."

J "Bruno's barks would get through."

"Bruno gets through everything. One more example, and it's the reason Chapter 1 worked at all. In 1807 Fourier was studying heat spreading through a metal bar. He found some very simple heat patterns, one for each pure wave, that he could solve exactly. But real bars start with messy temperatures."

J "So he added the simple ones up?"

"But why was he allowed to? Adding two ways that heat could spread doesn't obviously give you another way it could spread."

J "Unless... the ways heat can spread form a vector space?"

"They do. If pattern A is a possible way for heat to spread, and pattern B is too, then A plus B is also possible, and so is three times A. Physicists call equations that behave like this linear. So the possible heat patterns form a vector space, the simple patterns are a basis, and any messy start can be built from them."

J "So Fourier series only work because of this."

“It’s called the superposition principle, and an enormous amount of physics rests on it. Sound in air at ordinary loudness is linear, which is why my voice and Bruno’s barks simply added together, and could be pulled apart again.”

J “And when things aren’t linear?”

“Play your guitar through a distortion pedal. Double how hard you strum and the output doesn’t double; it gets squashed flat at the top. The scaling rule breaks, and that breakage is precisely the crunchy sound rock guitarists love.”

J “So distortion is a failure of the vector space rules.”

“A glorious one.”

TOOL UNLOCKED

The rules of the game

A vector space is a collection where adding and scaling behave: order and grouping don’t matter, there’s a “nothing”, and everything has a negative. Places aren’t vectors, but journeys are. When nature obeys the rules (superposition), you can build any solution out of simple ones.

2.3

Same song, different language

Changing the basis

BY THE END
You'll understand what a JPEG really stores, and why Chapter 1 was a translation all along.

6 min read · A conversation over the fence

SCENE *Monday evening. Jim is giving directions to a delivery driver over the phone. Badly.*

J "...no, not left at the river. Along the river. Then up the hill. No, the other hill."

He hangs up.

J "My phone says the takeaway is 300 metres east and 100 north. The man at the corner shop says 'follow the river for 200 metres, then 150 up the hill'. Which of them is right?"

"Both. It's the same place: the same vector, described in two different bases."

J "The river and the hill are ingredients?"

"They're directions to go in, just like east and north. Any two directions that aren't the same will do: go so far along one and so far along the other, and you can reach anywhere."

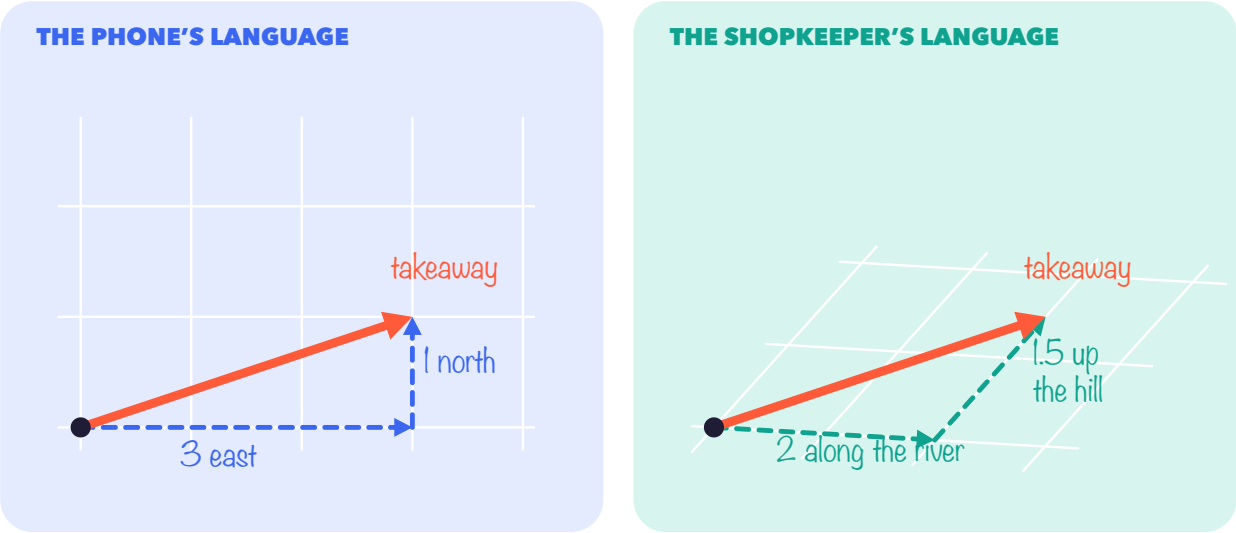


FIGURE 2.8 One journey, two languages. On the east-north grid it's "3 east, 1 north" (in hundreds of metres). On the river-hill grid it's "2 along the river, 1.5 up the hill". The arrow doesn't care which grid you use.

J "So the numbers change, but the journey doesn't."

"The numbers are the coordinates, and they belong to the basis, not to the vector. It's like language. Bruno is 'dog' in English, 'Hund' in German and 'naai' in Tamil. The robot would call him 'Wolf', but that's a different problem."

J "Same dog, different words."

"And here's the punchline of the book so far. Remember the wiggly line of a song? There are two natural ways to write it down. One: list where the speaker cone is at each instant. A CD does exactly that, with 44,100 numbers every second. Call that the 'moment by moment' basis. Two: list which pure waves are in it, and how long each hand is. That's the 'hand by hand' basis."

J "And those are two languages for the same song!"

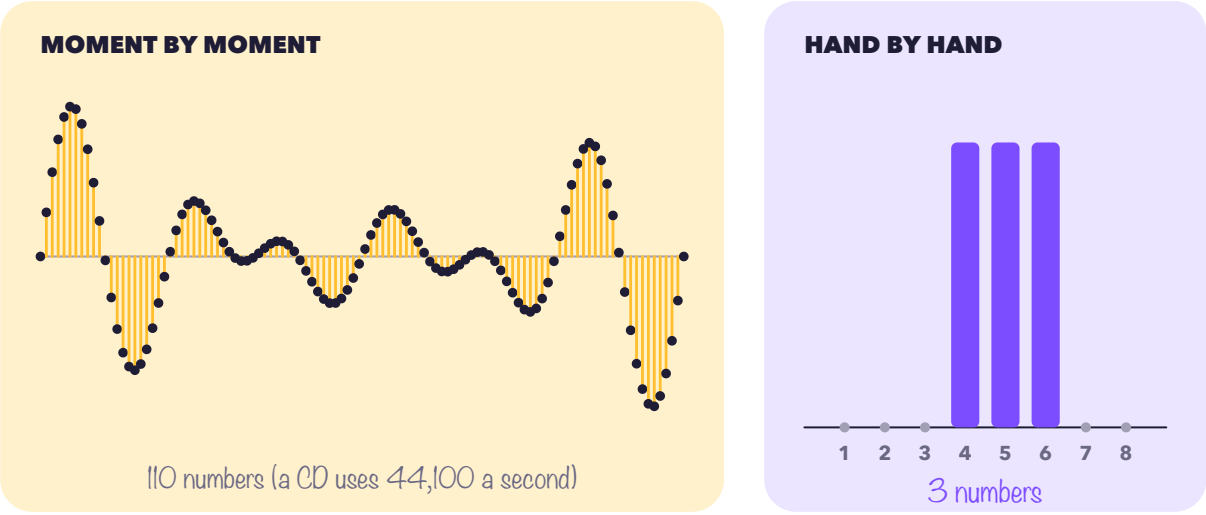


FIGURE 2.9 The same chord in two languages. Left: moment by moment, the height of the wiggly line at each instant. Right: hand by hand, how long each pure wave is. Three bars say everything the hundred-odd dots say.

"So what was Chapter 1?"

J "...A translation. A Fourier series is just a song translated from 'moment by moment' into 'hand by hand'."

"The most useful translation in science. Now, how do you translate? How did the man at the corner shop know it was '200 along the river'?"

J "He's lived here forty years."

"Suppose he hadn't. With the east–north grid there's an easy trick. Hold a torch straight above the arrow and look at its shadow on the east line. The length of the shadow is the east amount."

J "A shadow again."

"Now try the same trick on the river–hill grid. Shine the light straight down onto the river line."



FIGURE 2.10 Shadows only tell the truth when the directions are at right angles. Left: with perpendicular directions, each shadow is exactly the coordinate. Right: with slanted ones, the shadow on the river line picks up part of the hill and overstates how far along the river you went.

J "The shadow's too long! It's picking up some of the hill, because the hill leans towards the river."

"That's the trouble with slanted directions: they overlap, so a bit of 'hill' looks like 'river'. It's like a taste test. If there's garam masala and cumin in a dish, tasting for cumin also picks up the garam masala, because they overlap. You can't taste one without the other."

J "But with directions at right angles, each shadow measures just one thing."

"Right angles are what make translation easy. Each coordinate is one shadow, measured on its own, ignoring all the others. And the shadow has a formula: for arrows, multiply the matching coordinates and add them up. It's called the dot product."

J "Multiply and add up... that's the resonance test! Multiply F by the test wave at every moment and add it all up!"

"It's the same operation, just with a coordinate for every instant instead of two."

J "And the resonance test worked because different pure waves cancelled out... because they don't overlap... because they're..."

"Say it."

J "...at right angles. They're perpendicular! That's what you meant by 'orthogonal'!"

“Every pure wave is perpendicular to every other one, in a space with a direction for every instant. That’s why the test for one hand completely ignores all the others. With a slanted basis you’d have had to untangle all the overlaps at once; with a perpendicular one, you can measure each hand on its own.”



“Now for the payoff in your pocket. Every photo on your phone is a grid of dots, called pixels, each with a brightness. What’s the obvious basis?”

J “One ingredient per pixel: ‘this much brightness here, this much there’.”

“A JPEG chops the photo into blocks of eight by eight pixels, which makes 64 numbers per block. Then it translates each block into a different basis: 64 wave patterns, running from perfectly smooth to finely striped.”

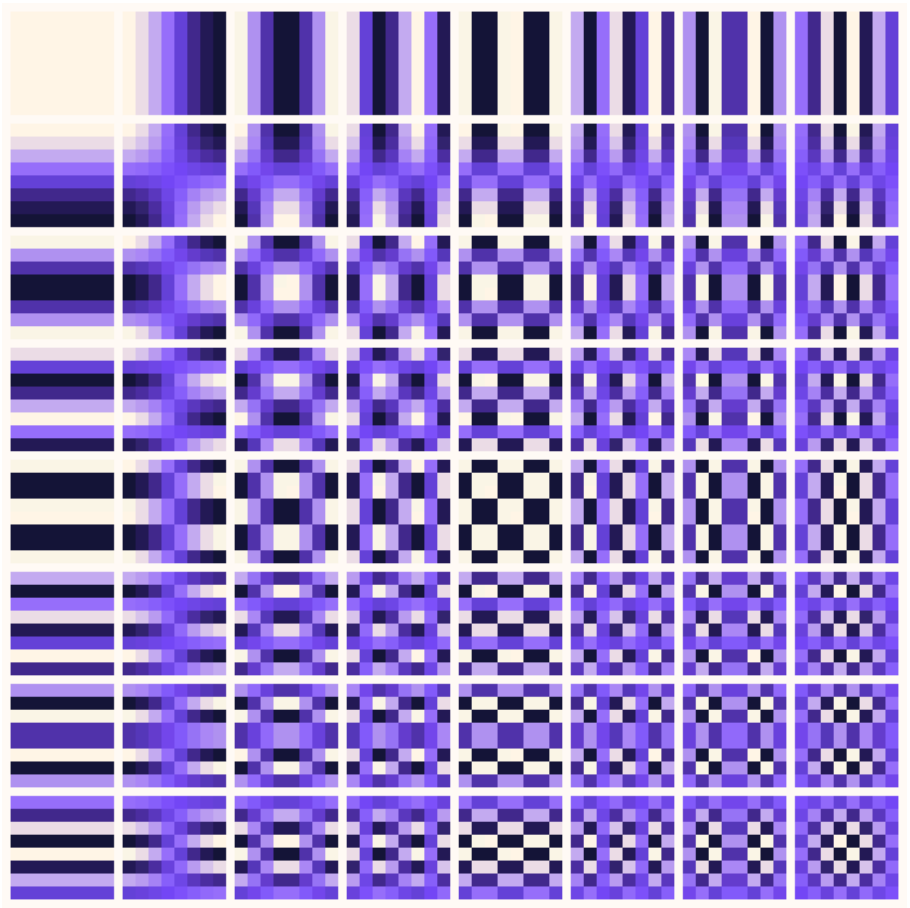


FIGURE 2.11 The 64 ingredients of a JPEG. Every 8×8 block of every JPEG is a mix of these wave patterns: slow, smooth ones at the top left, fast, fine stripes at the bottom right.

J "Waves again! Two-dimensional hands."

"Now think about a real photo. Sky, skin, walls: mostly smooth. What happens to them in the pattern basis?"

J "Smooth things only need slow hands... so almost all the weight lands on the top-left patterns, and the fast ones get tiny amounts."

"So?"

J "So throw away the tiny amounts!"

"That's JPEG. Translate each block into the pattern basis, keep the few big numbers, round the small ones to zero, and store what's left. Watch."



1 of 64 a mosaic | **3 of 64** recognisable | **10 of 64** nearly perfect | **the original: 64** every number

FIGURE 2.12 Rebuilding a photo from only the biggest pattern amounts in each block. With 1 number out of 64 it's a mosaic, with 3 it's recognisable, and with 10 it's hard to tell from the original. The photo is of Rear Admiral Grace Hopper, a pioneer of computer programming (U.S. Navy photo).

J "Ten numbers out of sixty-four, and it's nearly perfect. That's a sixth of the storage!"

"The real JPEG is a little cleverer, but that's the heart of it. Compression means choosing a basis in which your data is lopsided, so that a few co-ordinates carry nearly everything. And now you know why a JPEG of text looks smudgy around the letters."

J "Letters have sharp edges... sharp things need fast hands... and those are the ones JPEG throws away. Corners are expensive!"

"Chapter 1 and Chapter 2 in one smudge. JPEG even changes the colour basis before it starts, from red–green–blue to brightness plus two 'colour difference' coordinates, and then it keeps less detail in the colour ones."

J "Because my eye cares more about detail in brightness than in colour?"

"You're three steps ahead of me now."

J "Let me see if I've got it. A vector is anything you can add and scale. A basis is a language for describing it. When a basis is at right angles, translating is just taking shadows. And the whole art is choosing the language in which the thing you care about is simple."

"If you ever teach this, start with that paragraph."

J "And next? You said pure waves are perpendicular. If songs have directions, can they have distances too? How far apart are two songs?"

"Now that, Jim, is Chapter 3."

TOOL UNLOCKED

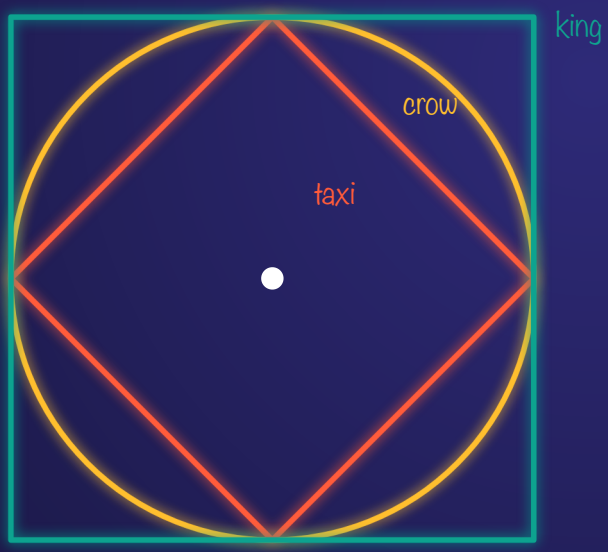
Changing the basis

The same vector has different coordinates in different bases. With a perpendicular basis, each coordinate is just a shadow (a dot product) measured on its own. A Fourier series is a change of basis, and so is a JPEG. Compression means choosing the basis where a few numbers say nearly everything.

CHAPTER

3 Metric

What "distance" really means



How far apart are two songs?

YOUR ROUTE THROUGH THIS CHAPTER

3.1 **How far is the airport?**
Introduction to metrics

3.2 **Distances between things that aren't places**
Developing the mathematical architecture for metrics

3.3 **Getting closer**
Convergence, horns and Stigler's law

3.1

How far is the airport?

Introduction to metrics

BY THE END

You'll discover that "distance" is a choice, and that a circle can be a square.

5 min read A conversation over the fence

SCENE *Tuesday, early. Jim is dragging a suitcase down the drive and arguing with his phone.*

J "Neighbour! Settle something. How far is the airport?"

"Good morning to you too. Why?"

J "My phone says 18 kilometres. My brother says 'it's only 11 as the crow flies'. And the taxi driver says 'forty minutes'. Three answers. Somebody's lying."

"Nobody's lying. They're measuring three different distances."

J "There's only one distance. It's… how far it is."

"Is it? You're not a crow. Let me draw our town. It's a grid, like Chandigarh: straight roads, square blocks."

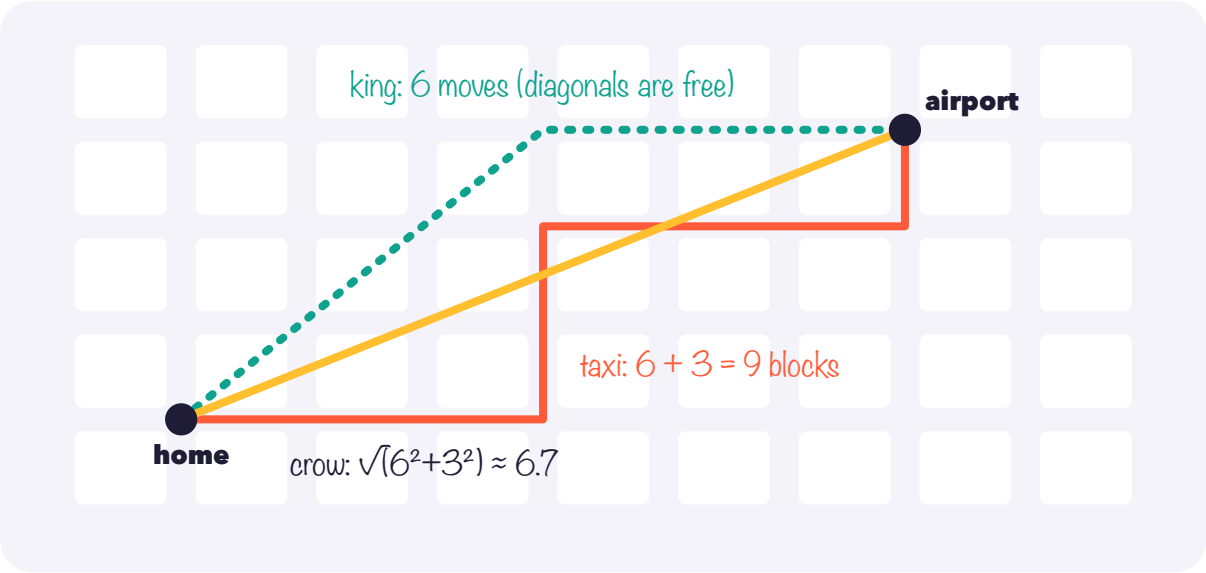


FIGURE 3.1 Three honest answers to "how far?". The crow flies straight. The taxi must follow the roads. A chess king may also cut diagonally through each block. Each one is a different way of measuring distance.

"The crow goes straight: Pythagoras gives about 6.7 blocks. The taxi has to go along the roads, so it's the blocks across plus the blocks up: 9. And if you were a chess king, allowed to cut diagonally through a block in one move, it's 6 moves."

J "The king's cheating."

"The king's just playing by different rules. Let's give these three their own names: `crow(A,B)`, `taxi(A,B)` and `king(A,B)`. Each takes two places and gives back a number. Each is a perfectly good distance, for somebody."

J "Fine. But then is anything a distance? Could I say the distance to the airport is 'blue'?"

"That's exactly the right question: what must something do to deserve the name? Let's find out the way we found the rules for vectors. You try to break it."

J "Right. Distance can't be negative."

"Agreed. What's the distance from your house to your house?"

J "Zero. And if the distance between two places is zero, they're the same place."

"Rule one: never negative, and zero only from a place to itself. Next?"

J "Home to the airport is the same as the airport to home."

"Is it? What about the taxi's forty minutes, and the ride home in rush hour?"

J "...Ninety minutes. Fine, time isn't symmetric. And one-way streets would make the road distance lopsided too."

"Mathematicians insist on it anyway: rule two says the distance from A to B equals the distance from B to A. Anything lopsided, like travel time, is a cousin with its own name, not a true distance. And the third rule is the deepest. Say you stop at a friend's house on the way to the airport."

J "Then it's longer. Or at best the same, if my friend lives right on the way."

"That's rule three: a detour can never be a shortcut. Home to the airport is never longer than home to your friend's plus your friend's to the airport. It's called the triangle inequality, because the three places make a triangle and one side can never be longer than the other two put together."

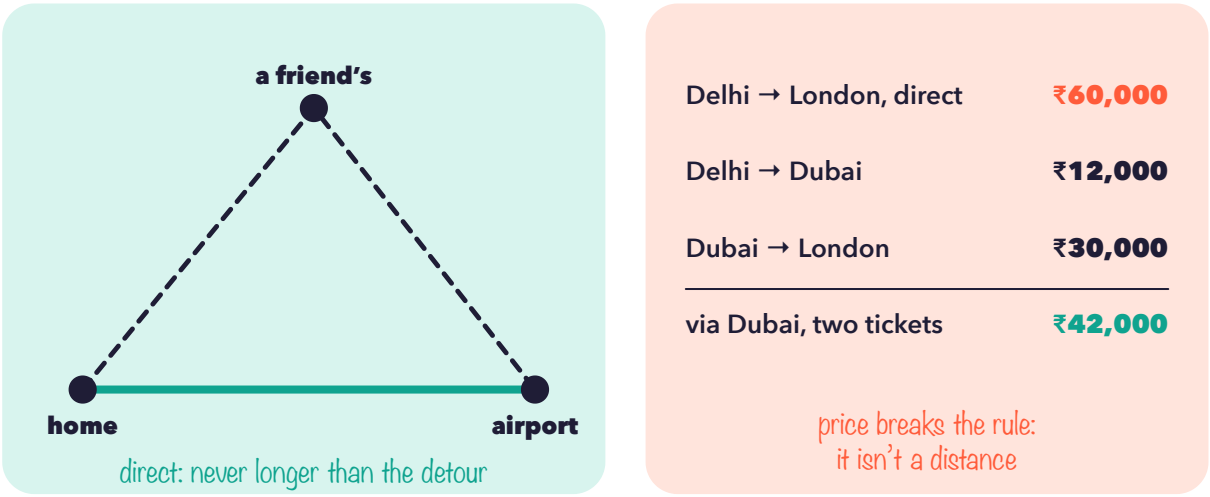


FIGURE 3.2 Left: the triangle inequality, which says the direct route is never longer than a detour. Right: airline prices break it. Two separate tickets via Dubai can cost less than one direct ticket, so ticket price, whatever else it is, isn't a distance.

J "Ha! Flights break it. Delhi to London direct is sixty thousand rupees, but Delhi to Dubai and then Dubai to London on separate tickets comes to forty-two."

"Which is precisely why 'price' isn't a distance, however much it feels like one. Airlines break the triangle inequality all the time, and people have made careers out of exploiting it."

SIDE QUEST: THE WHOLE RULEBOOK

A distance (mathematicians say a metric) is any way of giving two things a number such that: it's never negative, and zero only between a thing and itself; A to B equals B to A; and a detour is never a shortcut. That's all. Anything that obeys these three rules gets to be called a distance, and every theorem about distance then works for it.

J "So crow, taxi and king all pass?"

"All three. Now here's a puzzle for you. What's a circle?"

J "All the points the same distance from the centre."

"Say distance 3. Draw me the taxi's circle: all the corners of the grid that are exactly three blocks away by road."

PAUSE AND PICTURE IT

Try it on squared paper. Start at a dot, and mark every dot you can reach in exactly three blocks of taxi driving. What shape do the marks make?

J "Three across... two across and one up... one across and two up... three up... It's a diamond! A square standing on its point!"

"And the king's circle?"

J "The king reaches anything within three squares in any direction... so it's a square, lying flat."

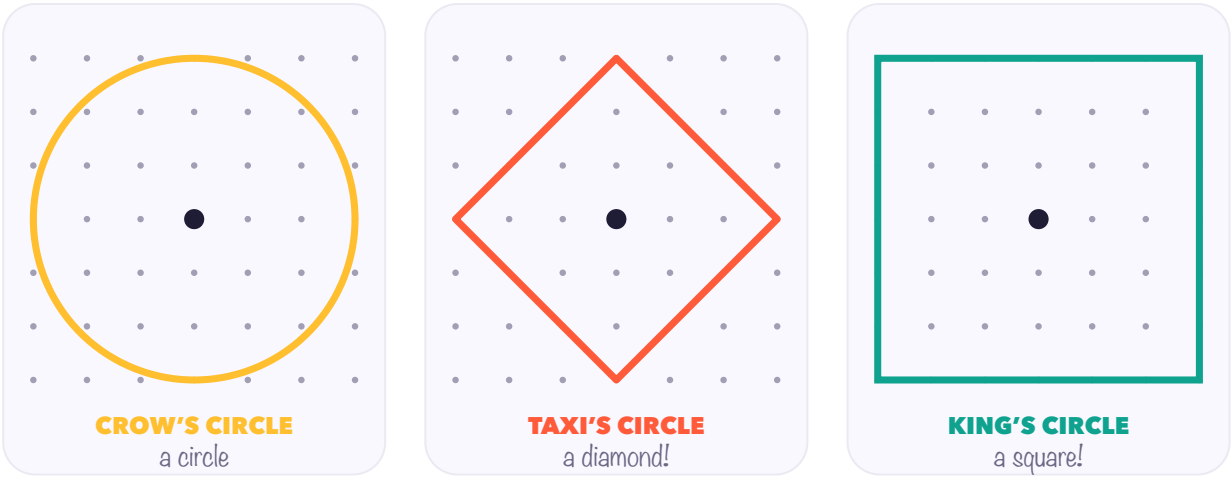


FIGURE 3.3 "All the points at distance 1 from the centre", drawn three ways. The crow's circle is round. The taxi's is a diamond. The king's is a square.

"So what a 'circle' looks like depends entirely on how you measure distance. The crow's circle is only one of many."

J "Then when someone says 'circle', they're secretly assuming a crow."

"Almost always. And that's the point of this chapter: before you ask 'how far?', ask 'far in what sense?' Pick the distance that fits your problem. The taxi's is the right one for a delivery driver, the king's for a chessboard or a warehouse robot that can move both ways at once, the crow's for a helicopter."



J "The robot moves both ways at once. Mine uses the king's distance!"

"Yours called Bruno a wolf, so I'd keep it away from warehouses."

TOOL UNLOCKED

Distance is a choice

A metric is any rule for "how far" that is never negative, symmetric, and obeys the triangle inequality. Different problems need different metrics, and even the shape of a circle depends on which one you pick.

3.2

Distances between things that aren't places

Developing the mathematical architecture for metrics

BY THE END

You'll measure how far apart two words, two songs and two people are, and see why one rule makes search fast.

4 min read · A conversation over the fence

SCENE *Wednesday. Jim is typing on his phone with his thumbs, and losing.*

J "Every time I type 'the', I get 'teh', and my phone fixes it. How does it know I meant 'the' and not 'tea'?"

"It measures how far 'teh' is from every word it knows and picks the closest."

J "How far a word is? Words don't have locations."

"Places aren't the only things with distances. All we need is a rule that obeys the three laws. So invent one for words."

J "Um... how many changes it takes to get from one to the other?"

"Go on. What counts as a change?"

J "Swapping a letter, adding one, or deleting one."

"Then the distance from 'kitten' to 'sitting' is..."



FIGURE 3.4 The edit distance from "kitten" to "sitting" is 3: two swaps and one added letter. No shorter route exists.

J "Swap the k for an s, swap the e for an i, add a g. Three."

"That's called the edit distance. A Soviet mathematician, Vladimir Levenshtein, wrote it down in 1965. Now check it against the three rules."

J "Never negative, and zero only if the words are identical. Symmetric: every change can be undone, so the way back is just as long. And the triangle rule: if I can get from A to B in 3 changes and from B to C in 2, I can certainly get from A to C in 5. Maybe fewer."

"So it's a genuine distance, and every tool built for distances now works on words. Your phone uses it. So do biologists: DNA is a word written in four letters, and the edit distance between two species' DNA tells you roughly how long ago they shared an ancestor."

J "So 'how different are a human and a chimp' is a distance between two very long words."

"Very long: about three billion letters each. Which raises a practical problem. Your phone knows a hundred thousand words. When you type 'cst', does it really measure the distance to every single one?"

J "I'd hope not. That sounds slow."

"This is where the triangle rule stops being a formality and starts saving you time. Suppose the phone already knows that 'cat' is 6 edits from 'elephant'. And it finds that your 'cst' is just 1 edit from 'cat'."

J "Then... going from 'cst' to 'elephant' via 'cat' is at most 7. But that's the wrong way round, isn't it? I want to know it's far."

"Turn the triangle around. Could 'cst' be within 4 of 'elephant'?"

J "If it were, then 'cat' to 'elephant' via 'cst' would be at most 1 + 4 = 5. But it's 6. So no! 'cst' must be at least 5 away from 'elephant'. I don't even need to check."

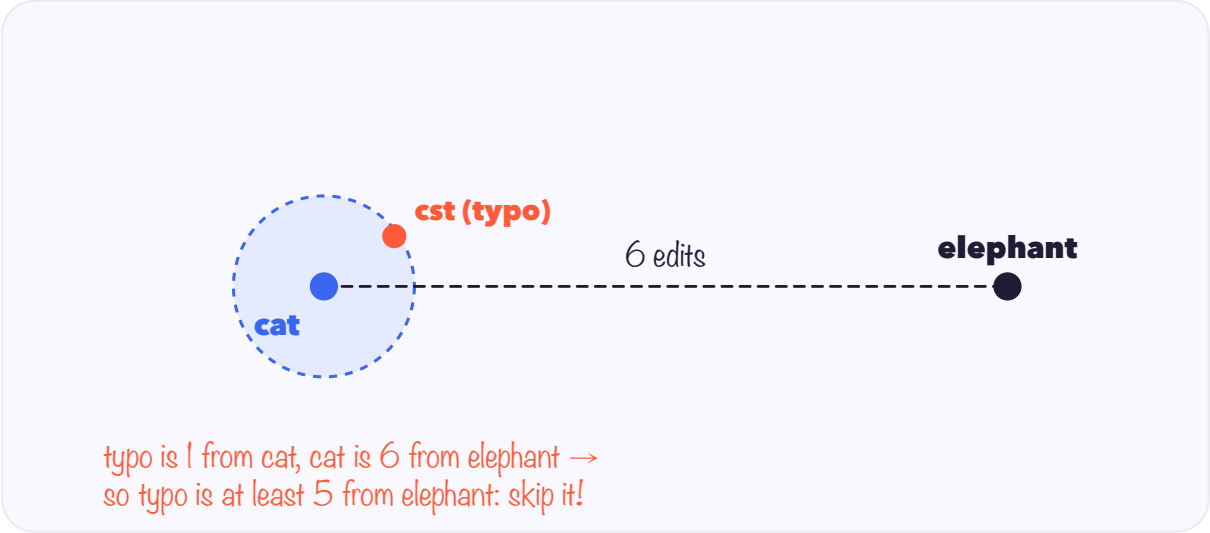


FIGURE 3.5 The triangle inequality as a shortcut. Knowing two distances you've already worked out, you can rule out "elephant" without ever comparing it to the typo. Spell-checkers use this to skip most of the dictionary.

"That's the trick. Arrange the dictionary cleverly, and one comparison rules out thousands of words at once. There are search systems built entirely on this idea. They work for any metric: words, images, songs, fingerprints."

J "Songs have a distance?"

"Anything that's a vector gets one for free. Remember Chapter 2? A song, a recipe, your taste in films: all lists of amounts. For arrows we measured length with Pythagoras, and Pythagoras is just the dot product of an arrow with itself. So the distance between two vectors is the length of the journey between them."

J "So if I rate ten films and you rate the same ten, our ratings are two arrows with ten coordinates..."

"...and the distance between them tells a streaming service how alike our tastes are. 'People near you liked this.'"

J "That explains a lot about my recommendations. What about wiggly lines? How far apart are two songs as wiggly lines?"

"You told me yourself, on the Ducati: 'the distance is the area'. Here are two sensible choices."

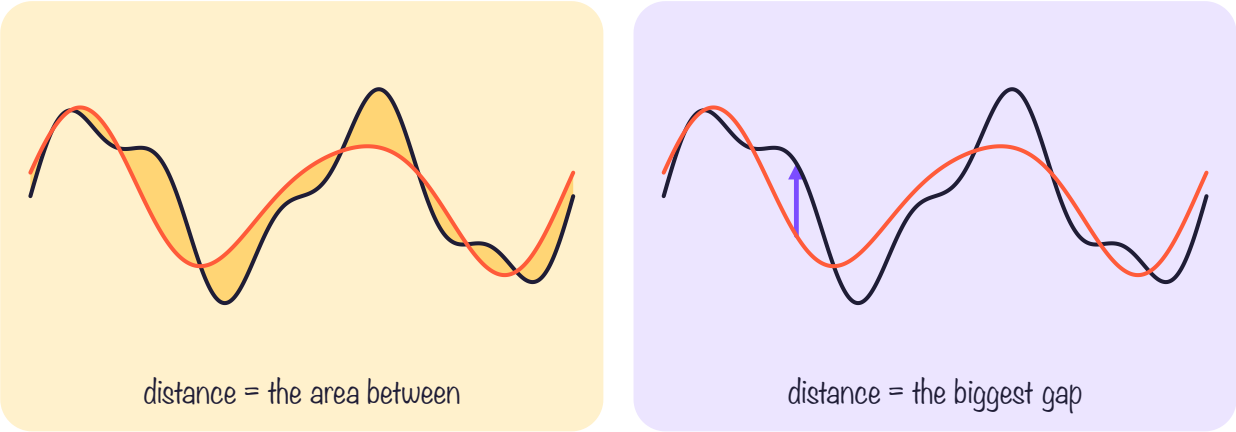


FIGURE 3.6 Two ways to measure the distance between two wiggly lines. Left: the total area between them. Right: the single biggest gap anywhere. Both obey the three rules, and they can disagree.

J "The area between them, or the biggest gap between them."

"Both are genuine distances. The area version asks 'how different are they overall?'. The biggest-gap version asks 'how bad is the worst moment?'. A sound engineer might care about the first, and someone designing a bridge that must never bend too far would care about the second."

J "And do they ever disagree?"

"Spectacularly. And that disagreement is the answer to a question you asked me in Chapter 1, about some little horns."

J "The horns on the square wave!"

"Tomorrow."

HELLO
my name is

Edit distance

How many single-letter changes turn one word into another. A genuine distance between words.

TOOL UNLOCKED

Anything can have a distance

Words, DNA, songs and tastes can all be given distances, as long as the three rules hold. The triangle inequality isn't just a rule: it lets you rule things out without checking them, which is how fast search works.

3.3 Getting closer

Convergence, horns and Stigler's law

BY THE END
You'll learn that "getting closer" depends on how you measure, and finally meet the horns.

4 min read · A conversation over the fence

SCENE *Thursday evening. Jim has drawn a square wave on a napkin, with little horns at every jump.*

J "I've been waiting since Chapter 1. What are the horns?"

"Let's zoom right in on one jump and draw it with 5 hands, then 15, then 60."

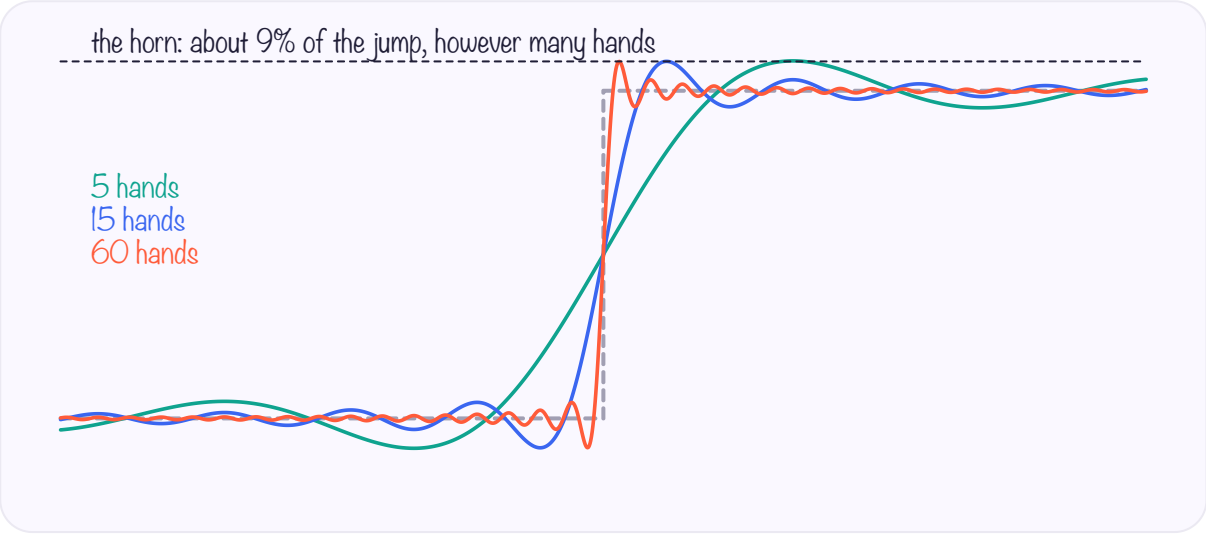


FIGURE 3.7 Zooming in on a jump. With more hands, the horn gets narrower and moves closer to the jump, but its height never drops: it stays at about 9% of the jump.

J "The horn gets thinner, and squashes up against the jump... but it's always the same height."

"About 9% of the jump, forever, no matter how many hands you add. Now, I told you in Chapter 1 that the hands draw the square wave perfectly 'in the end'. Is that true?"

J "It can't be. There's always a horn."

"Then let's measure how far the drawing is from the true square, with both of yesterday's distances."

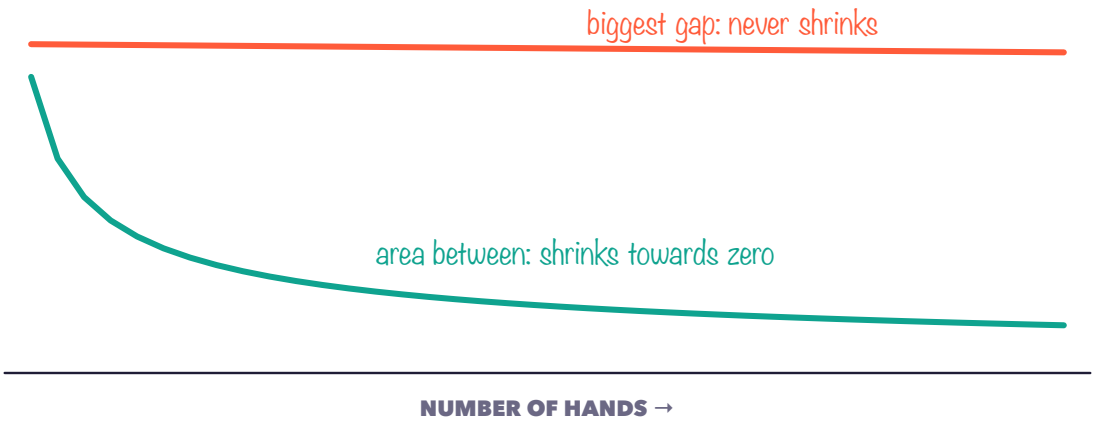


FIGURE 3.8 The same drawings, measured two ways. The total area between the drawing and the square wave shrinks towards zero as you add hands. The single biggest gap never shrinks at all.

J "The area between them shrinks to nothing. But the biggest gap stays put."

"So does the drawing get closer to the square or not?"

J "...Both? It depends which distance you use."

"That's the whole lesson. 'Getting closer' only means something once you've said how you measure distance. By the area distance, the drawing gets as close as you like. By the biggest-gap distance, it never does. Mathematicians call 'getting as close as you like' convergence, and they always have to say which distance they mean."

J "So when you told me the hands draw anything, you meant the area version."

"I did. For sound, the area version is the one your ear cares about: a sliver of a horn carries almost no energy. But if the drawing were the shape of a bridge girder, you'd care about the worst gap."

J "Who found the horns?"

The biggest-gap kind is called uniform convergence. The area kind is convergence in the mean.

"That's a lovely story. In 1898 the American physicist Albert Michelson, the man who measured the speed of light, built a brass machine with 80 springs and levers that added up 80 hands and drew the result. Its drawings of jumpy shapes came out with little horns beside the jumps."

J "He thought his machine was broken!"

"That's how the story is usually told. He certainly wrote to the journal Nature, puzzled about how such a sum could settle down at a jump. The great American physicist Josiah Willard Gibbs replied and explained the horns, and they've been called the Gibbs phenomenon ever since."

J "So Gibbs discovered them."

"Not quite. An English mathematician called Henry Wilbraham had described them exactly, in 1848, fifty years earlier. Nobody noticed."

J "That's unfair."

"It's so common it has its own law. In 1980 the statistician Stephen Stigler proposed Stigler's law of eponymy: no scientific discovery is named after its original discoverer."

J "And who discovered Stigler's law?"

"Stigler himself said the sociologist Robert Merton got there first."

J "That's perfect. That's absolutely perfect."

"And it's our Feynman point from Chapter 1 again: the name tells you nothing about the idea, not even who had it."

SIDE QUEST: WHERE YOU CAN SEE THE HORNS

Whenever something sharp is rebuilt from a limited number of waves, ghost ripples appear beside the edges. MRI scanners build images out of waves, and radiologists have a name for the faint echoes beside sharp boundaries: "Gibbs ringing". JPEG's ripples around text are the same thing.

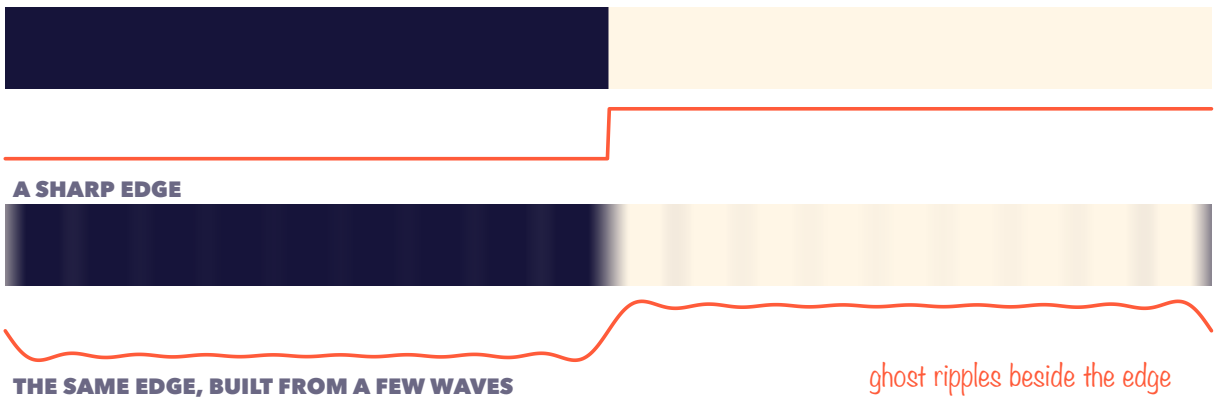


FIGURE 3.9 A sharp edge, and the same edge rebuilt from a few waves. The bright and dark ghost ripples beside it are the Gibbs horns, seen as an image. You'll find them in MRI scans and around text in JPEGs.

J "So the horns are why JPEG text looks smudgy?"

"The horns and their ripples, yes. When you throw away the fast hands, sharp edges get haloes."

J "Let me put this chapter together. A distance is anything that's never negative, the same both ways, and never lets a detour be a shortcut. You can put distances on words, songs, tastes and wiggly lines. And 'getting closer' depends on which distance you pick."

"Beautifully summarised. Next time we're going to do something to vectors instead of just measuring them. Bring pizza dough."

J "Why pizza dough?"

"You'll see."



TOOL UNLOCKED

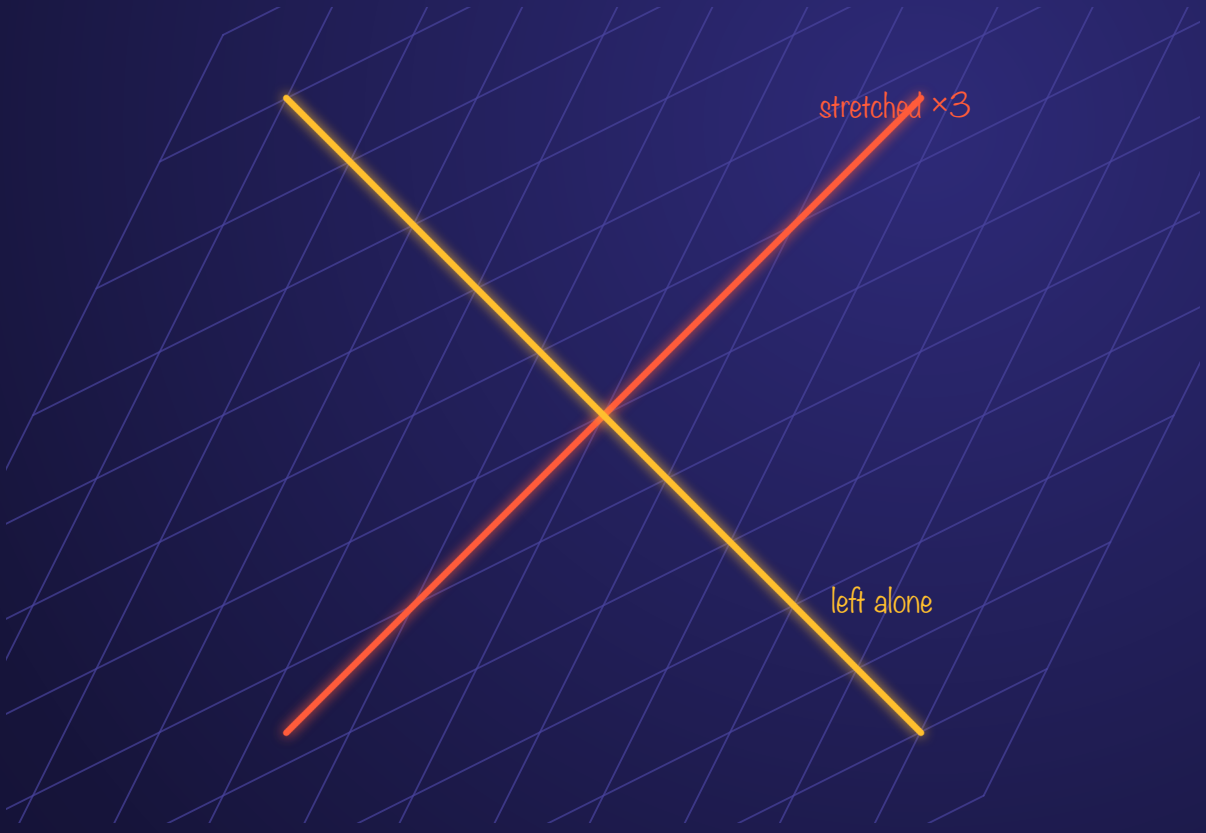
Closeness depends on the ruler

A sequence can converge under one metric and not under another. Fourier's hands draw a square wave perfectly by area, but never by the worst gap: those are the Gibbs horns, about 9% of the jump, in every drawing with a jump.

CHAPTER

4 Eigenvalues and eigenvectors

Every change has a grain



What do pizza dough, Google and a wobbly bridge have in common?

YOUR ROUTE THROUGH THIS CHAPTER

4.1 **The directions that don't turn**
Introduction to eigenvectors and eigenvalues

4.2 **Hunting eigenvectors**
Developing the mathematical architecture for eigenvectors

4.3 **Where the future goes**
Where eigenvectors take us

4.1

The directions that don't turn

Introduction to eigenvectors and eigenvalues

BY THE END
You'll find the hidden "grain" inside any stretch, squash or shove.

4 min read · A conversation over the fence

SCENE *Friday night. Jim's kitchen. There is flour on the ceiling.*

J "I'm making pizza. I drew a smiley face on the dough in tomato sauce, for morale."

"Perfect. Roll it once, only forwards and back, left to right."

J "The smiley's gone fat. Wide eyes, wide grin."

"Now imagine you'd drawn little arrows on the dough first, pointing out from the middle in every direction, like the spokes of a wheel. What happened to them?"

PAUSE AND PICTURE IT
Picture the spokes before and after rolling left to right. Which of them still point the same way afterwards?

J "The ones pointing left and right just got longer. The ones pointing straight up and down... stayed the same. And the slanted ones, the diagonals, got dragged round towards left-right. They turned."

"So almost every arrow turned. Except two directions."

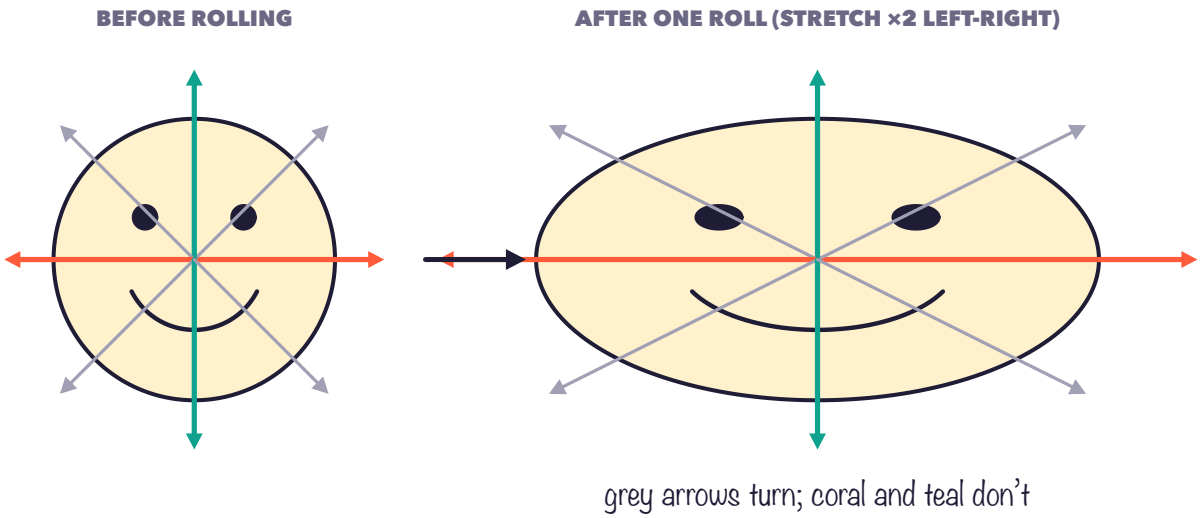


FIGURE 4.1 One roll stretches the dough to twice its width. The grey arrows all turn towards the rolling direction. The coral arrows (along the roll) only get longer, and the teal ones (across it) don't change at all. Those two directions are the dough's own.

J "Along the roll and across it."

"Those are the dough's special directions. Along the roll, arrows get stretched by 2. Across it, stretched by 1, which means left alone. Everything else is a mixture of the two, and that's why it turns: its left-right part doubles, its up-down part doesn't, so it swings towards left-right."

J "So the stretch has a grain, like wood."

"That's a lovely way to put it. Every stretch, squash or shove has a grain: directions it only lengthens or shortens, never turns. They're called eigenvectors, and the amount each one is stretched by is its eigenvalue."

J "Eigen?"

"German, like your robot's manuals. It means 'own'. These are the transformation's own directions. Now, a harder one. Take a deck of cards and push the top sideways, so the stack leans over."

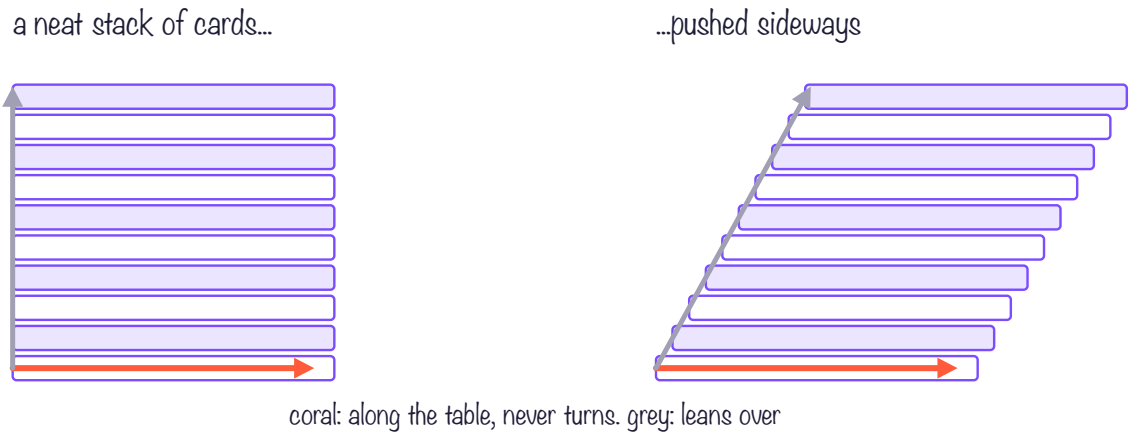


FIGURE 4.2 A stack of cards pushed sideways. An arrow along the table (coral) never turns. An arrow pointing up the stack (grey) leans over. This kind of shove, called a shear, has only one special direction.

J "An arrow flat along the table stays put. An arrow pointing up the stack leans over with it. And the diagonals lean too... so only one special direction?"

"Only one. Some transformations have two, some have one. Now a quarter-turn of my watch face: every arrow rotates by ninety degrees."

J "Then every arrow turns. There are no special directions at all!"

"None that you can draw. Hold that thought, though, because a quarter-turn is the spinning hand from Chapter 1, and it does have special directions if you're willing to use complex numbers. They're exactly the spinning hands."

J "So spinning hands are the grain of spinning."

"Beautifully put. Let me show you why anyone should care. Take your phone and throw it up spinning, in three different ways."

TRY THIS: SPIN A PHONE THREE WAYS

Over a bed, please. Toss your phone spinning end over end around its long axis: it spins cleanly. Spin it flat, like a pizza: also clean. Now spin it around the middle axis, the one that goes across the screen from side to side. It wobbles and flips over, every single time.

J "Oh, that's bizarre! Twice it spins nicely, and around the middle it tumbles."

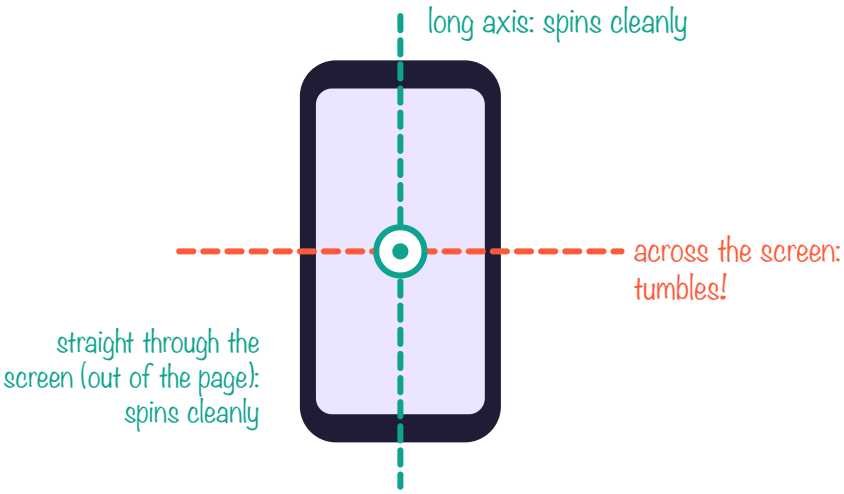


FIGURE 4.3 Every solid object has three special axes, the eigenvectors of how its weight is spread out. It spins cleanly about the longest and the shortest of them, and tumbles about the one in between.

"Every object, however lumpy, has exactly three special axes: the eigenvectors of how its mass is spread out. Spin it about the longest or shortest, and it spins steadily. Spin it about the middle one, and it's unstable. It's called the tennis racket theorem."

J "So the eigenvectors are the directions where a thing behaves simply."

"That's exactly it. In any complicated transformation, look for the directions it doesn't turn. Along those, it's just a stretch, and a stretch is the simplest thing in the world."

In 1985 the cosmonaut Vladimir Dzhanibekov watched a wing nut spinning in zero gravity flip over again and again. It's the same effect.

HELLO
my name is
Eigenvector
A direction a transformation doesn't turn, only stretches. The stretch factor is its eigenvalue.

TOOL UNLOCKED
The grain of a transformation
Most arrows turn when you transform them. The few that only stretch (eigenvectors) reveal the transformation's grain, and the stretch factors (eigenvalues) tell you how strong it is in each direction.

4.2 Hunting eigenvectors

Developing the mathematical architecture for eigenvectors

BY THE END
You'll learn two ways to find a transformation's own directions: a lazy one and a clever one.

4 min read · A conversation over the fence

SCENE *Saturday. The pizza was eaten. The smiley, now a long thin line, was eaten last.*

J "I kept rolling the dough, left to right, over and over. The smiley ended up a single stripe along the roll."

"You've discovered the lazy way to find an eigenvector. Let's make it a bit more interesting than pizza. Here's a transformation I'll call `roll(v)`: it takes an arrow and gives back a new one. The new 'east' is twice the old east plus the old north. The new 'north' is the old east plus twice the old north."

J "That's a sort of slanted stretch."

"Watch what happens to any arrow if you apply `roll(v)` again and again, rescaling each time so it fits on the page."

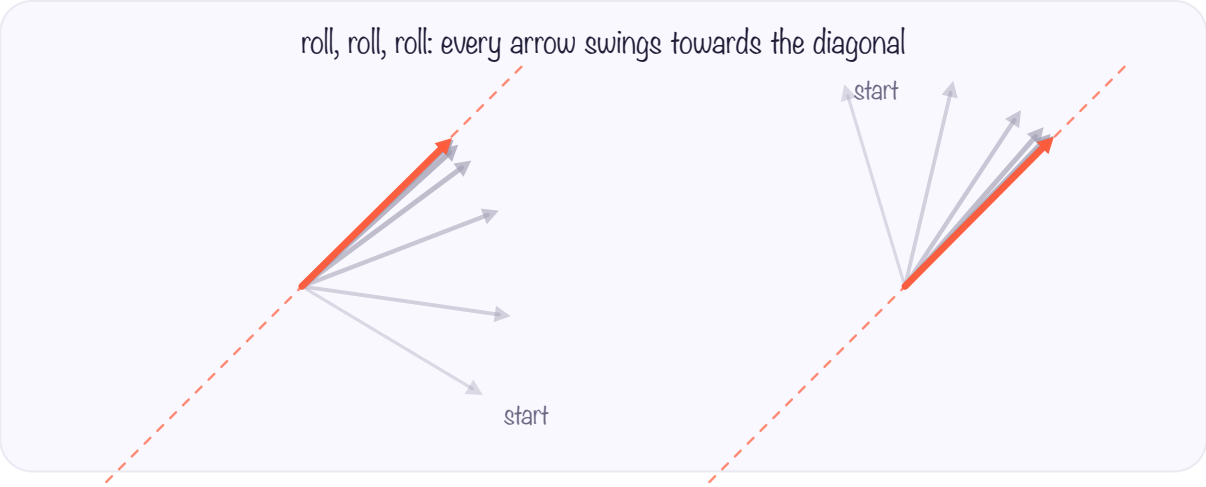


FIGURE 4.4 Apply the same transformation over and over, and every starting arrow swings towards one direction: here, the diagonal. That direction is the transformation's strongest eigenvector.

J "Every arrow swings towards the diagonal. Like my smiley becoming a stripe."

"Because each time you roll, the part along the strongest direction gets stretched more than everything else, so it takes over. That's the lazy method: keep applying it and see where things end up. Mathematicians call it power iteration, and it's more than lazy. As you'll see tomorrow, it's how Google ranked the entire web."

J "But it only finds the strongest one. What about the others?"

"For that we need the clever way. Suppose `v` is an eigenvector with stretch factor λ."

J "λ?"

"The Greek letter 'lambda'. Mathematicians use it for eigenvalues the way you used `ducati(x)` for test waves. So `roll(v)` is just λ times `v`. Now subtract: what does '`roll` minus λ-stretch' do to `v`?"

J "`roll` stretches it by λ, and then I take away λ times it... nothing's left. It sends `v` to zero."

"So '`roll` minus λ' squashes a whole direction down to nothing. Picture what that does to the plane."

J "If one whole direction gets crushed to a point, everything gets flattened. The plane collapses onto a line."

"And what's the area of a flattened square?"

J "Zero!"

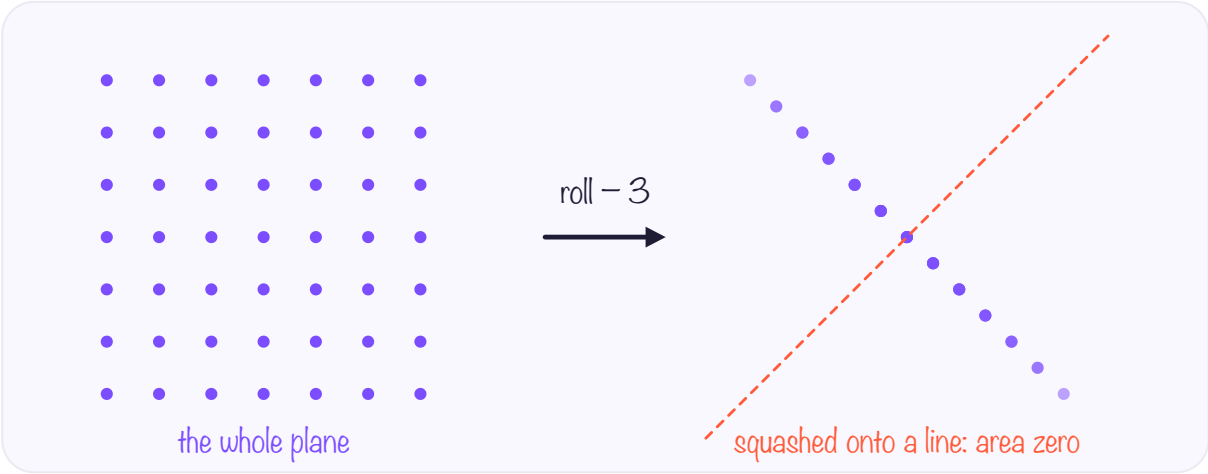


FIGURE 4.5 For the right value of λ (here 3), "roll minus λ" crushes one direction to nothing, and the whole plane collapses onto a line. A flat shape has zero area, and that's how you find λ.

"So here's the clever hunt: look for the values of λ that make 'roll minus λ' squash area to zero. And there's a neat formula for how much a transformation scales area. If it sends east to (a, c) and north to (b, d), areas get multiplied by a times d minus b times c."

J "For ' `roll` minus λ', east goes to (2 − λ, 1) and north to (1, 2 − λ). So the area factor is (2 − λ) times (2 − λ), minus 1."

"Set it to zero."

J "(2 − λ) squared equals 1. So 2 − λ is 1 or −1... λ is 1 or 3!"

"Two eigenvalues. Now find the directions."

J "For λ = 3, the diagonal: (1, 1) becomes (3, 3). Stretched by 3. For λ = 1... (1, −1) becomes (2 − 1, 1 − 2), which is (1, −1). Untouched!"

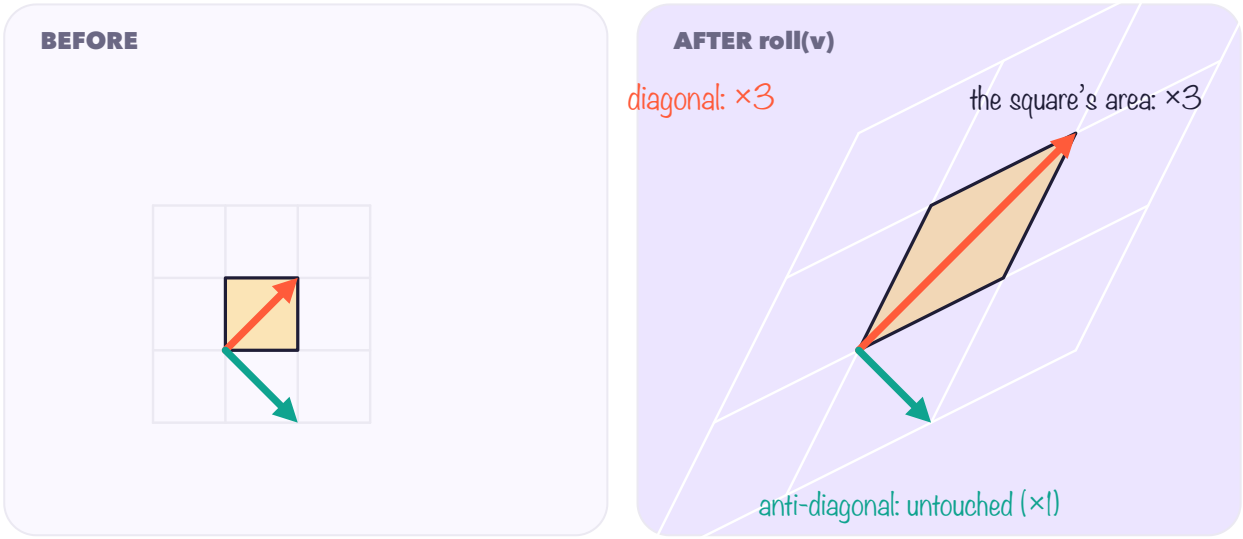


FIGURE 4.6 The grid before and after roll(v). The diagonal (coral) is stretched three times, the anti-diagonal (teal) doesn't move, and the yellow square's area grows three-fold, which is 3 × 1.

"And the area factor of `roll` itself?"

J "2 times 2 minus 1 times 1 is 3. The same as 3 times 1. The area stretch is just the two eigenvalues multiplied together!"

"It always is. The area-scaling number is called the determinant, and the equation you just solved, 'area of roll minus λ equals zero', is the characteristic equation. You've done in five minutes what takes most students a week."

IN TEXTBOOK LANGUAGE (FOR THE CURIOUS)

Textbooks write a transformation as a matrix A and an eigenvector as $Av = \lambda v$. The eigenvalues solve $\det(A - \lambda I) = 0$, where det is the determinant (the area factor) and I is the do-nothing transformation. For our roll, A = [[2, 1], [1, 2]], giving $\lambda = 3$ and $\lambda = 1$.

J "And why bother, if the lazy way works?"

"The lazy way only finds the strongest. The clever way finds them all. And once you have them all, any arrow can be split into eigen-directions, like splitting a song into hands, and then `roll` is trivial: multiply one part by 3, leave the other alone."

J "So eigenvectors are a basis in which the transformation is simple."

"That's the best one-sentence summary of this whole chapter. Chapter 2 said: choose the language in which your problem is simple. For a transformation, that language is its eigenvectors."



TOOL UNLOCKED

Two ways to hunt

Lazy way: keep applying the transformation and see which direction takes over (power iteration). Clever way: find the λ where "transformation minus λ" squashes area to zero (the characteristic equation). In its eigenvector basis, every transformation is just a set of stretches.

4.3

Where the future goes

Where eigenvectors take us

BY THE END

You'll see why rabbits, Google and wobbly bridges are all run by eigenvalues.

4 min read A conversation over the fence

SCENE *Sunday. Jim's nephew has been given two rabbits. Jim is doing sums on the back of an envelope, looking worried.*

J "The pet shop says a pair of rabbits has a new pair every month, starting from their second month. How many will there be by Christmas?"

"That's the oldest puzzle in this business. Leonardo of Pisa, whom we call Fibonacci, posed it in 1202. Count the pairs month by month."

J "1, 1, 2, 3, 5, 8, 13, 21... each month is the last two added together."

"Now divide each month by the month before."

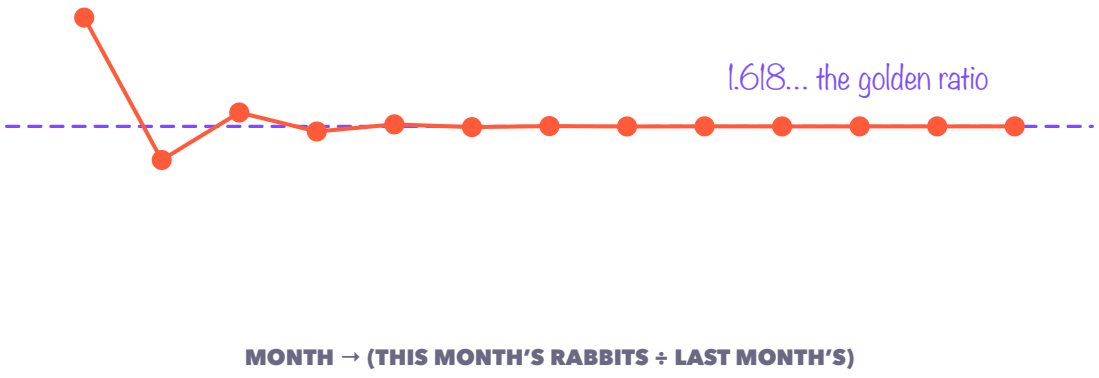


FIGURE 4.7 Divide each month's rabbits by the month before, and the answer quickly settles at 1.618..., the golden ratio. That's the eigenvalue of the rabbit rule.

J "2, 1.5, 1.67, 1.6, 1.625... it's settling down. About 1.618."

“The golden ratio. And here’s why. The rabbit rule is a transformation: it takes this month’s (adults, babies) and gives next month’s. Apply it over and over…”

J “…and, like the pizza, everything swings towards its strongest eigenvector. So in the long run the population just gets multiplied by the biggest eigenvalue every month. That’s the 1.618!”

“You’ve got it. When the same change happens again and again, the future belongs to the biggest eigenvalue. Populations, compound interest, epidemics, the spread of a rumour: in the long run, the biggest eigenvalue says how fast it grows, and its eigenvector says what the mix looks like.”

J “Please tell me the rabbits have a smaller eigenvalue.”

“Sorry, Jim. By Christmas you’ll be well into the hundreds. Now, the most valuable eigenvector in history. How did Google decide which web pages matter?”

J “Counting links? Pages lots of other pages link to?”

“Close, but anyone can make a thousand junk pages all linking to one. Larry Page and Sergey Brin had a better idea. Imagine a bored surfer clicking random links forever. Every so often he gets bored and jumps to a random page. Where does he spend most of his time?”

J “On pages that lots of pages link to… especially pages that important pages link to.”

“And ‘where he spends his time in the long run’ is what?”

J “Something that doesn’t change when he takes one more click… Oh. It’s an eigenvector! The one with eigenvalue 1.”

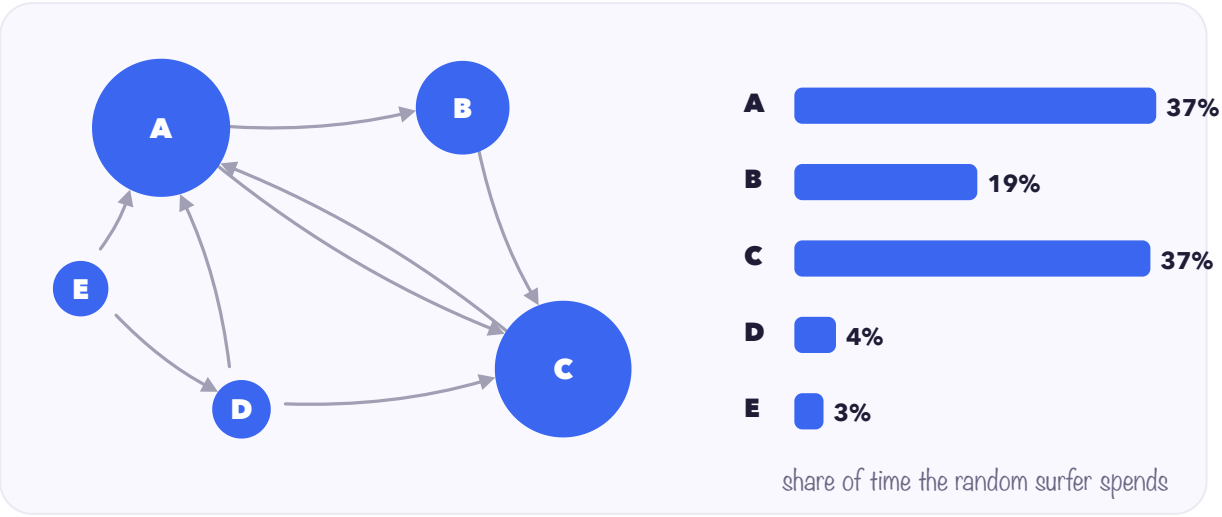


FIGURE 4.8 A tiny web of five pages. The random surfer's long-run share of time on each page is the eigenvector of the "one more click" rule. Page C wins because important pages point to it.

"That's PageRank, the idea Google started with in 1998: rank every page on the web by that eigenvector, found by clicking again and again, which is the lazy method from yesterday. For a while, Google was essentially one enormous eigenvector."

J "A search engine built on the pizza trick."

"One more, and it brings us back to your Ducati. Remember the frame's 'favourite speeds', its natural frequencies?"

J "The buzz at four thousand rpm."

"A frame, a bridge or a guitar string has special shapes it can vibrate in without changing shape, just swinging bigger and smaller. Those shapes are eigenvectors, and their natural frequencies come straight from the eigenvalues."

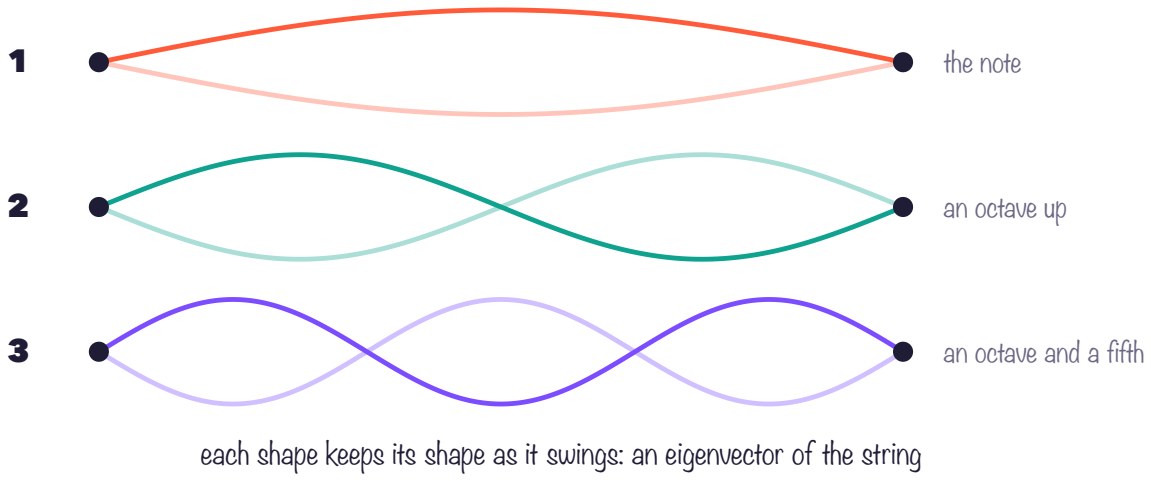


FIGURE 4.9 A guitar string's three simplest ways to vibrate. Each keeps its shape as it swings: an eigenvector of the string. Their frequencies, in the ratio 1 : 2 : 3, are the harmonics from Chapter 1.

J "So the harmonics were eigenvectors all along!"

"The whole of Chapter 1 was sitting on this. Pure waves are the eigenvectors of vibration, and of every echo, filter and loudspeaker too. That's why they're so useful: they pass through such systems without changing shape."

J "What happens if you get the eigenvalues wrong?"

"Ask London. On 10 June 2000, the Millennium Bridge opened across the Thames. Thousands of people walked on to it, and it began to sway sideways. People felt it move and, without meaning to, fell into step with it, which pushed it further. It closed two days later and reopened in 2002, after engineers fitted dampers."

J "Pushing the swing in rhythm. With your feet."

"And the list of a system's eigenvalues has a name you already know: its spectrum. The same word as the rainbow, and as the list of hands in a song."



J "Everything keeps connecting."

"That's how you know you're learning mathematics and not just memorising it. Next week I'll need your mattress."

J "My what?"

TOOL UNLOCKED

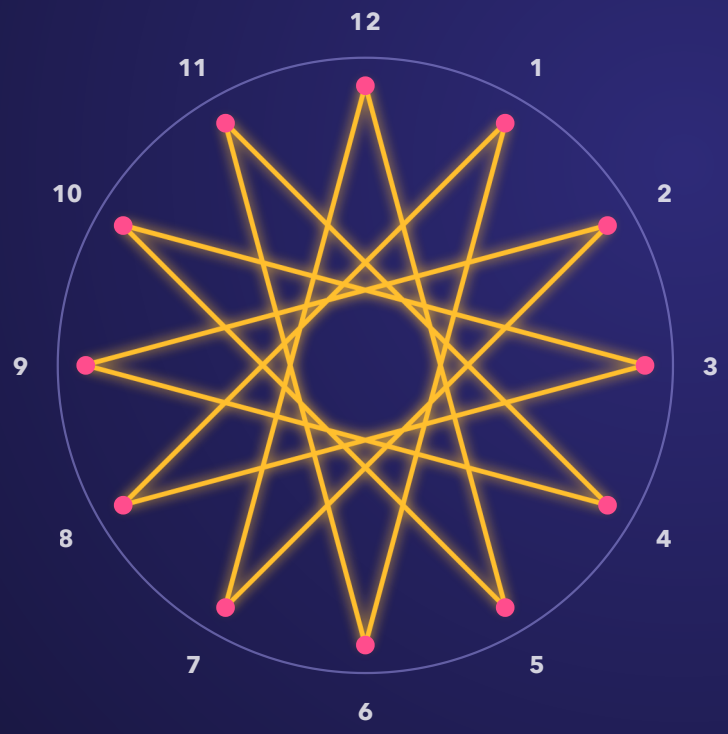
The future belongs to the biggest eigenvalue

Repeat any change and the strongest eigenvector takes over, growing by the biggest eigenvalue each time. That runs population growth, Google's PageRank, and every natural frequency, from guitar strings to wobbling bridges.

CHAPTER

5 Groups

The grammar of moves



Why can't one flip turn your mattress through every position?

YOUR ROUTE THROUGH THIS CHAPTER

5.1 **The mattress problem**
Introduction to groups

5.2 **The grammar of moves**
Developing the mathematical architecture for groups

5.3 **The power of impossibility**
Parity, puzzles and proofs

5.1

The mattress problem

Introduction to groups

BY THE END

You'll discover why no single flip can rotate your mattress through every position, and why your watch can.

4 min read A conversation over the fence

SCENE *A Sunday in spring. Jim is wrestling a mattress through his bedroom door.*

J "The label says 'rotate your mattress regularly so it wears evenly'. Every season I forget which way I turned it last time."

"How many ways can it lie on the bed?"

J "Soft side up or firm side up, and the head end at the top or the bottom. Four."

"And how many moves have you got?"

J "I can spin it end to end, keeping the same side up. I can roll it over the long side. Or I can flop it over the short end."

"Three moves, four positions. So here's your dream: one single move that, done every season, takes the mattress through all four positions and back. Then you'd never need to remember anything."

J "Exactly! Which one?"

PAUSE AND PICTURE IT

Before reading on, try it with a book on the table. Pick any one of the three moves and keep doing it. Does it ever reach all four positions?

J "Spinning... head at the top, head at the bottom, head at the top again. Only two. Rolling... soft, firm, soft. Two. Flopping... two again! Every move, done twice, puts it back where it started."

"So the move you want doesn't exist. Not because you haven't found it, but because it can't. Now look at my watch. One move: 'add an hour'. Keep doing it."

J "One, two, three... all the way round to twelve and back to one. One move reaches everything!"

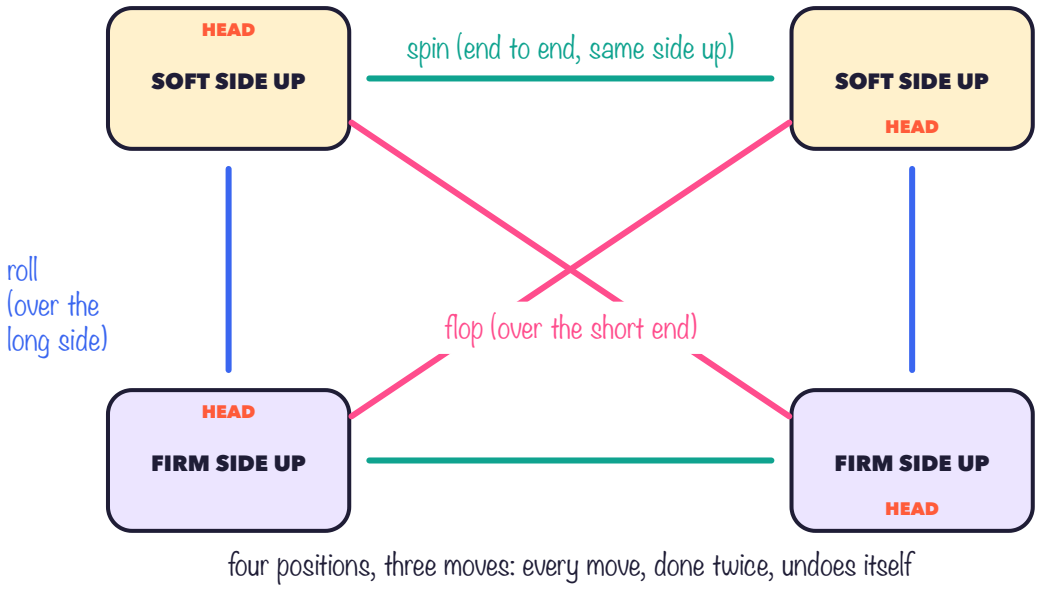


FIGURE 5.1 The mattress's four positions and its three moves. Every move, done twice, brings you back where you started, so no single move can visit all four positions.

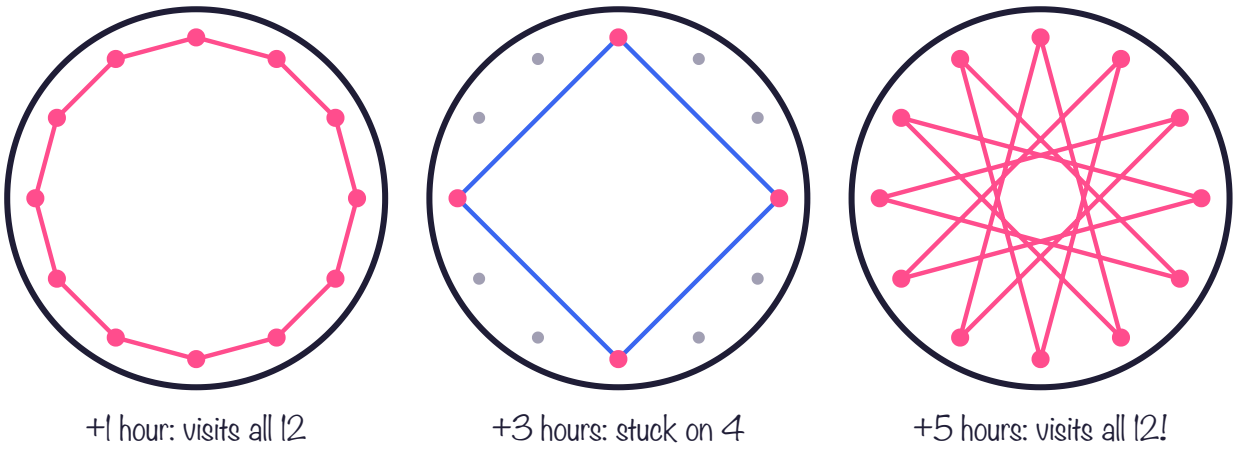


FIGURE 5.2 On a clock, "add 1 hour" reaches every hour. "Add 3 hours" gets stuck on four of them. "Add 5 hours" reaches all twelve again. Steps that share no factor with 12 visit everything.

J "What about 'add 3 hours'?"

"Try it."

J "12, 3, 6, 9, 12... stuck on four hours. And 'add 5'? 5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, 12. All twelve! And 'add 2' only gets the even hours... It works when the step shares no factor with 12!"

"That's a genuine theorem, and you found it in a minute. Now, the big question: what do a mattress and a clock have in common?"

J "They both have... moves. You can do one move after another. You can undo any move. And there's the 'do nothing' move."

"That's it: a collection of moves where you can combine any two, undo any one, and there's a do-nothing move. Mathematicians call it a group. The clock's group is called cyclic, because a single move cycles through everything. The mattress's group is called the Klein four-group, after Felix Klein."



J "So both of them are groups, but they're different shapes of group."

"And the shape tells you what's possible before you ever lift the mattress. That's the power of the idea: you don't need to try every sequence of moves. The structure tells you what can and can't be done."

SIDE QUEST: GROUP THEORY IN THE BEDROOM

The mathematician and writer Brian Hayes wrote an essay with exactly this title in 2005, working out the mattress problem properly. Since no single move works, the practical fix is to alternate two moves: say, spin in spring and autumn and roll in summer and winter. Those two together do visit all four positions.

J "What's the biggest clock?"

"A clock with infinitely many ticks: a circle you can turn by any amount at all. That's the group of the spinning hands from Chapter 1. Every hand, turning steadily, is walking round that infinite clock."

J "So Fourier series are about a group too?"

“Deep down, yes. There’s a whole field that does Fourier analysis on any group you like. But first, let’s learn the grammar of moves.”

TOOL UNLOCKED

Moves have structure

A group is a set of moves you can combine and undo, with a do-nothing move. The clock is cyclic: one move reaches everything. The mattress isn’t: every move undoes itself. The structure decides what’s possible before you try.

5.2

The grammar of moves

Developing the mathematical architecture for groups

BY THE END
You'll find the four rules every group obeys, a hidden Sudoku, and why getting dressed has an order.

4 min read A conversation over the fence

SCENE *Monday morning. Jim is hopping on one foot, trying to put a sock on over his shoe.*

"That's not going to work."

J "I'm in a hurry."

"Then you've just stumbled on the most important rule about undoing things. How do you get dressed?"

J "Socks, then shoes."

"And to undo it?"

J "Shoes off, then socks off. Obviously."

"So undoing reverses the order. The inverse of 'socks then shoes' is 'undo shoes, then undo socks'."



FIGURE 5.3 To undo a sequence of moves, undo them in reverse order. And socks-then-shoes isn't the same as shoes-then-socks: in many groups, order matters.

J "And shoes then socks is a disaster. So order matters."

“Sometimes it does and sometimes it doesn’t. For the mattress, spin then roll is the same as roll then spin. For clothes, and for a Rubik’s cube, it matters enormously. Let’s write down what every group must obey, the way we did with vectors. First: combine any two moves and you get another move in the collection.”

J “Spin then roll is flop. It’s still a mattress move.”

“Second: there’s a do-nothing move. Third: every move can be undone.”

J “And the fourth?”

“The subtlest. If you do three moves, A, B and C, it doesn’t matter whether you think of it as ‘A then (B then C)’ or ‘(A then B) then C’. Same result.”

J “That’s obvious.”

“It is, for moves. But there are ways of combining things where it fails, like dividing numbers, and then the whole theory falls apart. These four rules are the entire definition of a group.”

SIDE QUEST: THE FOUR RULES

Closure: combining two moves gives a move. Identity: there’s a do-nothing move. Inverses: every move can be undone. Associativity: how you bracket a sequence doesn’t matter. Notice what’s not on the list: that order doesn’t matter. Groups where it doesn’t are called abelian, after the Norwegian mathematician Niels Henrik Abel.

“Now let’s write out the mattress’s whole ‘times table’: every move, followed by every move.”

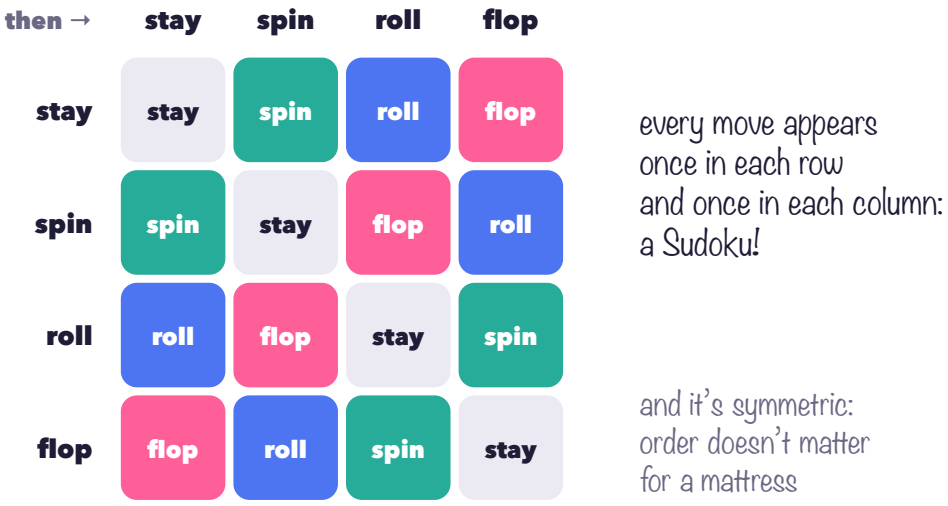


FIGURE 5.4 The mattress's times table. Every move appears exactly once in every row and once in every column, just like a Sudoku.

J "It's a Sudoku! Every move appears exactly once in each row and each column."

"Is that luck?"

PAUSE AND PICTURE IT

Why can't a move appear twice in the same row? Suppose "spin, then X" and "spin, then Y" gave the same result. What could you do to both?

J "Suppose 'spin then X' and 'spin then Y' gave the same result. Then I undo the spin at the start of both... so X and Y must be the same move! No move can appear twice in a row. And since there are only four, each one must appear exactly once."

"That's a real proof. Every group's times table is a Sudoku, and the reason is the undo rule. These tables are called Cayley tables, after Arthur Cayley."

J "And for a group where order matters?"

"Take a triangle cut out of card, with its corners numbered. Two moves: turn it a third of the way round, or flip it over."

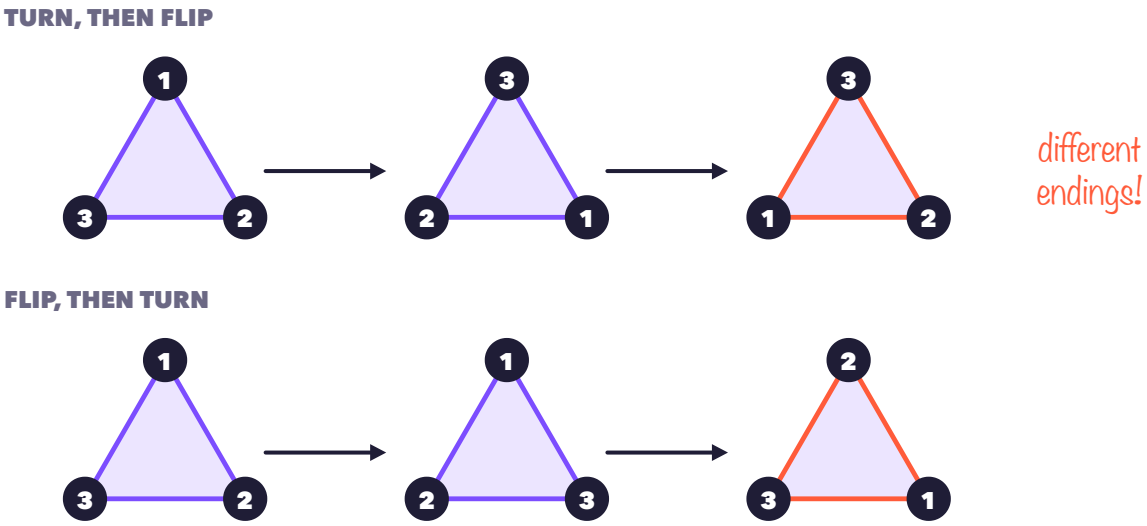


FIGURE 5.5 A triangle turned then flipped ends up different from one flipped then turned. The triangle's group is the smallest one where order matters.

J "Turn then flip gives 3, 2, 1. Flip then turn gives 2, 1, 3. Different!"

"The triangle's six moves make the smallest group where order matters. And that's the key to the Rubik's cube. Solvers use sequences like 'do A, do B, undo A, undo B'. If order didn't matter, that would do nothing at all. Because it does matter, it does something tiny, like swapping just two corners, which is exactly what you need at the end of a solve."

J "So cube solvers are using the fact that the group isn't a Sudoku that reads the same both ways."

"Precisely: they're living in the gap where order matters. Those 'do A, B, undo A, undo B' sequences are called commutators, and they measure exactly how much order matters."



TOOL UNLOCKED

The grammar of moves

Every group obeys four rules: you can combine moves, there's a do-nothing move, everything can be undone, and bracketing doesn't matter. Undoing reverses order, every group's table is a Sudoku, and when order matters, commutators let you make tiny, precise changes.

5.3

The power of impossibility

Parity, puzzles and proofs

BY THE END
You'll prove a famous puzzle can't be solved, without making a single move.

4 min read · A conversation over the fence

SCENE *Tuesday. Jim has found a sliding-tile puzzle at a jumble sale, and has been at it for three hours.*

J "It's the 15 puzzle. Fifteen numbered tiles, one gap. Somebody before me prised two tiles out and put them back swapped, 14 and 15. I can't fix it."

"You can't. Nobody can."

J "How do you know? You haven't even tried."

"That's the point. In 1880 America went mad for this puzzle. It was said that the puzzle-maker Sam Loyd offered a thousand dollars to anyone who could swap just the 14 and 15. Nobody ever collected, and nobody ever will. Let me show you why, without touching a tile."

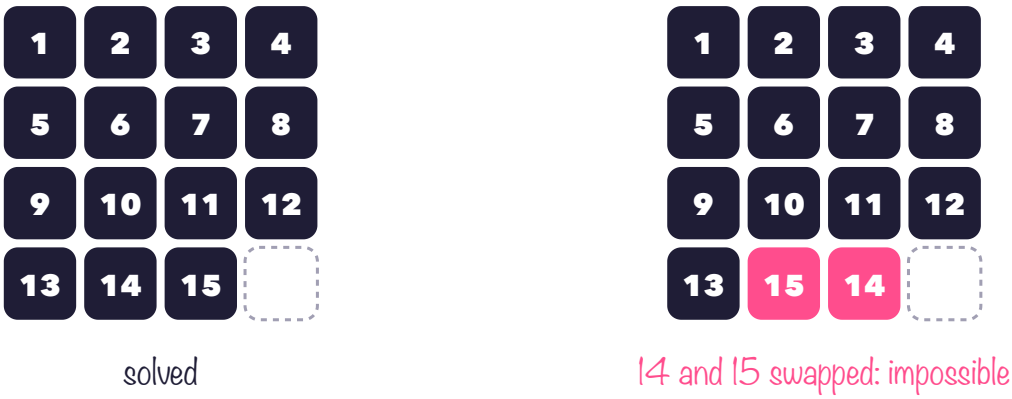


FIGURE 5.6 Left: the solved puzzle. Right: the same puzzle with 14 and 15 swapped. No sequence of slides, however long, gets from one to the other.

"Colour the board like a chessboard. Every slide moves the gap to a neighbouring square. What colour is that square?"

J "The other colour. Always."

"So after one slide the gap is on the other colour, after two slides it's back on its own colour... For the gap to end up back in its home corner, how many slides must you make?"

J "An even number. Otherwise it would be on the wrong colour."

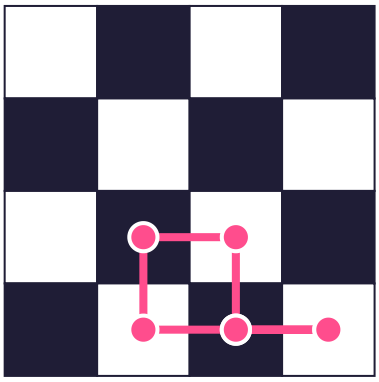


FIGURE 5.7 Every slide moves the gap to the other colour, so it can only come home after an even number of slides. And every slide is a swap.

"Now, every slide is a swap: the gap swaps places with a tile. So any sequence that brings the gap home is an even number of swaps."

J "And swapping 14 and 15 is just one swap. Odd."

"But could an even number of swaps somehow add up to a single swap?"

J "I don't know. Could it?"

"Here's the trick. Write the tiles out in order and count how many pairs are the wrong way round. Every single swap changes that count by an odd number: it either creates or destroys an odd number of wrong-way pairs. So an even number of swaps always changes it by an even amount."

J "And a single swap of 14 and 15 changes it by one, which is odd. So you can't get there with an even number of swaps. It's impossible!"

"That's the proof. Every arrangement is either 'even' or 'odd', and no amount of sliding changes which. Mathematicians call it parity. Half of all the arrangements of the 15 puzzle can never be reached from the solved one, and two Americans, William Johnson and William Story, proved it in 1879."



J "So I've wasted three hours."

"You've learned the best thing group theory teaches: sometimes the most useful thing you can know is that something is impossible. It saves you a lifetime of trying."

SIDE QUEST: A RUBIK'S CUBE IN PIECES

A Rubik's cube has about 43 quintillion positions. But if you take one apart and snap it back together at random, there's only a 1 in 12 chance it can be solved at all. A twisted corner, a flipped edge or a single swapped pair each put it into a part of the group that turning can never reach. Group theory tells you exactly which.

J "What else is impossible?"

"Some of the most famous results in mathematics are impossibilities. The Greeks spent centuries trying to cut any angle into three equal parts with a ruler and compass; it can't be done. There are formulas for solving equations up to x to the fourth power, but none for the fifth. Niels Abel proved that, and a young Frenchman, Évariste Galois, used groups to explain exactly which equations can be solved and why. He was dead in a duel at twenty."

J "At twenty!"

"He spent the night before writing his ideas down. Groups began, more or less, with his letters."

J "So a group is really a way of listing what you can do, and working out from the rules what you can't."

"And the same idea runs chemistry, where the symmetry groups of molecules predict which light they absorb, and physics, where every symmetry of nature comes with something that's conserved. Emmy Noether proved that around 1915: because the laws of physics are the same today as tomorrow, energy is conserved."

J “Symmetry gives you conservation?”

“It’s one of the most beautiful facts in science. But that’s a story for another fence. Next, something more urgent. How sure should you be about anything?”

TOOL UNLOCKED

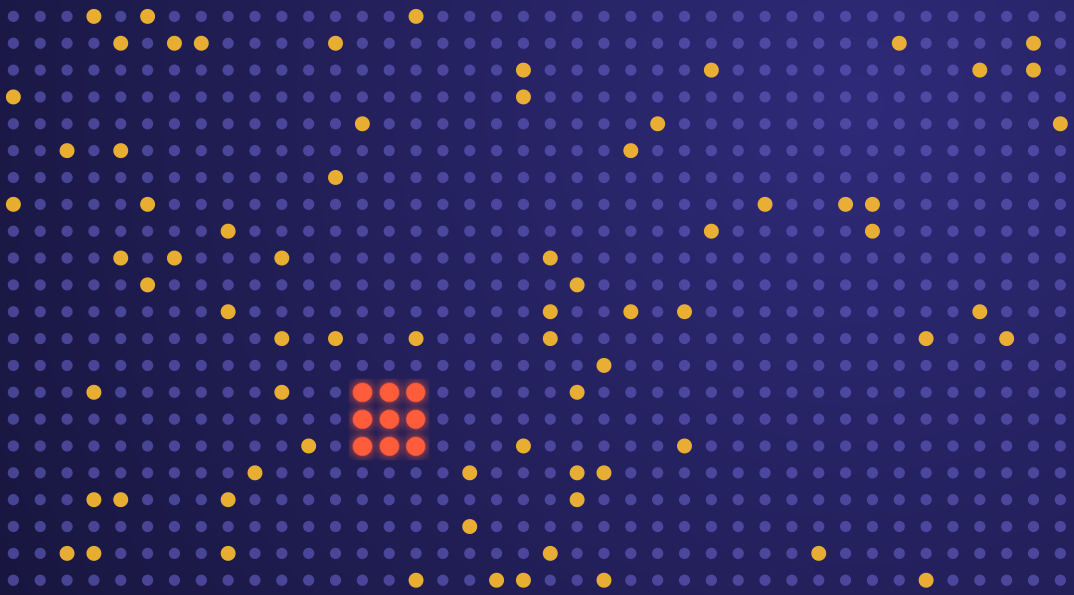
Knowing what’s impossible

Every arrangement has a parity, even or odd, that moves may never change. That proves the swapped 15 puzzle is hopeless and a randomly rebuilt Rubik’s cube is solvable only 1 time in 12. Symmetry is structure, and structure tells you what can never happen.

CHAPTER

6 Bayes' theorem

How much should evidence change your mind?



Why does nobody run to the window when a car alarm goes off?

YOUR ROUTE THROUGH THIS CHAPTER

6.1 Car alarms and rare diseases
Introduction to Bayes' theorem

6.2 Updating, one clue at a time
Developing the mathematical architecture for Bayes' theorem

6.3 The costliest mistake in probability
When "if" gets turned around

6.1
Car alarms and rare diseases

Introduction to Bayes' theorem

BY THE END
You'll learn to tell how much a piece of evidence should really change your mind.

3 min read · A conversation over the fence

SCENE *2 a.m. A car alarm is wailing somewhere down the street. Neither neighbour moves.*

J "(from his window) Is that yours?"

"No idea. Go back to sleep."

J "Shouldn't we check? Somebody might be stealing a car."

"When did you last hear a car alarm that was an actual theft?"

J "...Never. They go off when a cat walks past. Or a lorry."

"So you heard an alarm, which is evidence of a theft, and you didn't budge. Why?"

J "Because thefts are rare and false alarms are common. So an alarm is almost always a false one."

"Congratulations. You've just used Bayes' theorem in your pyjamas, without a single number. Tomorrow I'll show you a case where your instincts get it badly wrong."

> *The next morning.*

"Here's the puzzle. There's a disease that 1 in 100 people have. There's a test for it. If you have the disease, the test catches it 9 times out of 10. If you don't, it wrongly says you do 1 time in 10. You take the test. It's positive. What's the chance you have the disease?"

PAUSE AND PICTURE IT

Make a guess before you read on. Most people, including many doctors, say about 90%.

J "The test is right 9 times out of 10... so about 90%?"

"That's what most people say, including, in surveys, most doctors. The real answer is about 8%."

J "That's impossible."

"Let's not use percentages at all. Picture a crowd of a thousand people."

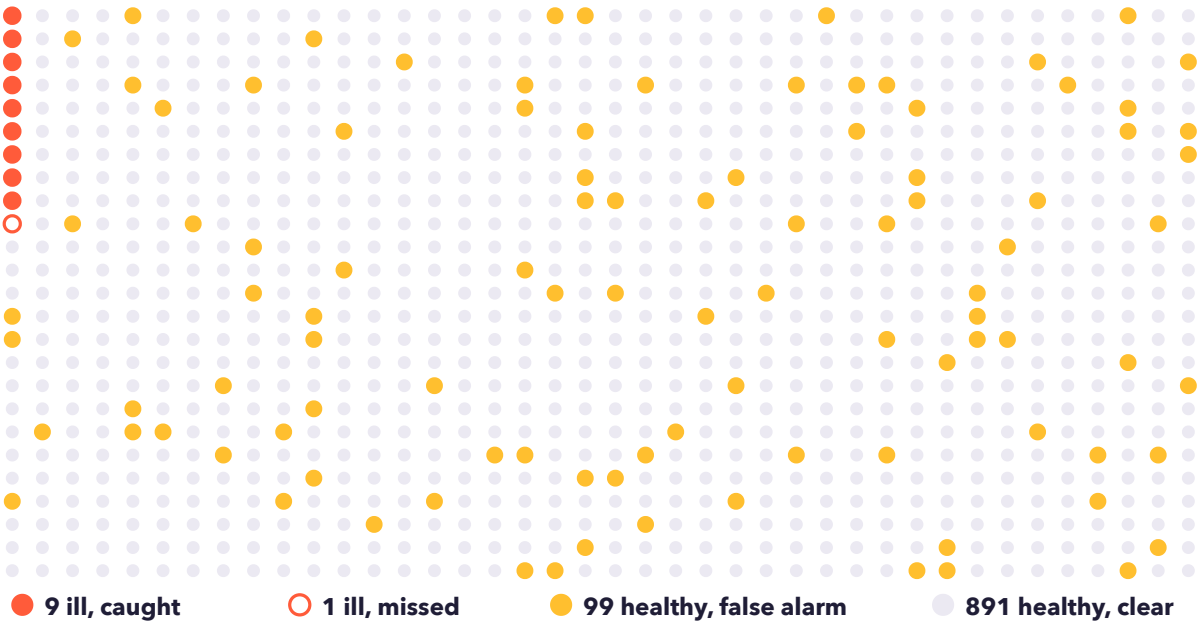


FIGURE 6.1 A thousand people. Ten have the disease (coral): the test catches 9 of them and misses 1. Of the 990 healthy people, the test wrongly flags about 99 (yellow). Count the positives.

"One in a hundred has it, so ten people in the crowd are ill. The test catches nine of them."

J "Nine positives."

"And 990 are healthy. The test wrongly flags one in ten of them."

J "That's 99 false alarms! So 9 plus 99 is 108 positive results, and only 9 of those people are ill."



FIGURE 6.2 Everyone who tested positive, lined up. Only 9 of the 108 are ill. A positive test means about a 1 in 12 chance, not 9 in 10.

"So if you're one of the 108, your chance of being ill is 9 in 108: about 1 in 12, or 8%."

J "So the test is... useless?"

"Not at all. Before the test you had a 1 in 100 chance; now it's 1 in 12, about eight times higher. The test moved you a long way. It just started you from a very low place. It's exactly the car alarm: when the thing is rare, false alarms outnumber real ones."

J "So the rarer the disease, the less a positive result means."

"That's the key insight, and you said it better than most textbooks. The starting rarity is called the base rate. Forgetting it is one of the commonest mistakes people make, in hospitals, courtrooms, airports and newsrooms."

SIDE QUEST: NATURAL FREQUENCIES

The psychologist Gerd Gigerenzer found that when doctors are shown the same problem as counts of people ("10 out of 1,000 are ill...") instead of percentages, far more of them get it right. He called these counts "natural frequencies". Jim wanted to know if they were related to the Ducati's. They're not.

J "Why are percentages so much harder?"

"Because a percentage hides the crowd it came from. '90% accurate' makes you forget that there are 990 healthy people for every 10 ill ones. Counting people puts the crowd back."

HELLO

my name is

Base rate

How common something is before you see any evidence. Always start there.

TOOL UNLOCKED

Picture the crowd

To judge any evidence, imagine a crowd, split it by what's true, and count who gets the evidence. When the thing you're testing for is rare, false alarms can outnumber true ones even for a good test.

6.2

Updating, one clue at a time

Developing the mathematical architecture for Bayes' theorem

BY THE END

You'll turn the crowd into a formula, then into a trick you can do in your head.

4 min read · A conversation over the fence

SCENE *Wednesday. The robot is back from the recycling. Jim has been reprogramming it.*

J "I'm teaching the robot properly this time. With mental models. But I need your crowd trick in a form it can use."

"Then let's write it in words. After a positive test, the chance you're ill is: the number who are ill and test positive, divided by everyone who tests positive."

J "9 divided by 108."

"And if we swap people for fractions of the crowd, that's the whole of Bayes' theorem:"

BAYES' THEOREM, IN WORDS

Chance of being ill, given a positive test = (chance of ill × chance an ill person tests positive) ÷ chance of testing positive at all

J "That's just the crowd count with fractions instead of people."

"That's all it ever is. But there's a version you can do in your head, and it's the one I actually use. Use odds instead of chances."

J "Like betting?"

"Exactly like betting. Before the test, the odds of ill to healthy are 1 to 99. Now: how much more likely is a positive result from an ill person than from a healthy one?"

J "Ill people test positive 9 times in 10, healthy ones 1 time in 10. So nine times more likely."

"That's the strength of the clue. And here's the rule: odds after = odds before × strength of the clue."

J "1 to 99, times 9, is 9 to 99. Which is... 9 ill for every 99 healthy. That's the crowd again!"

"Same answer, no crowd needed. Now the beauty of it. Say you take a second, independent test, and it's also positive."

J "Multiply by 9 again: 81 to 99. Nearly even! And a third: 729 to 99. That's almost 90%."

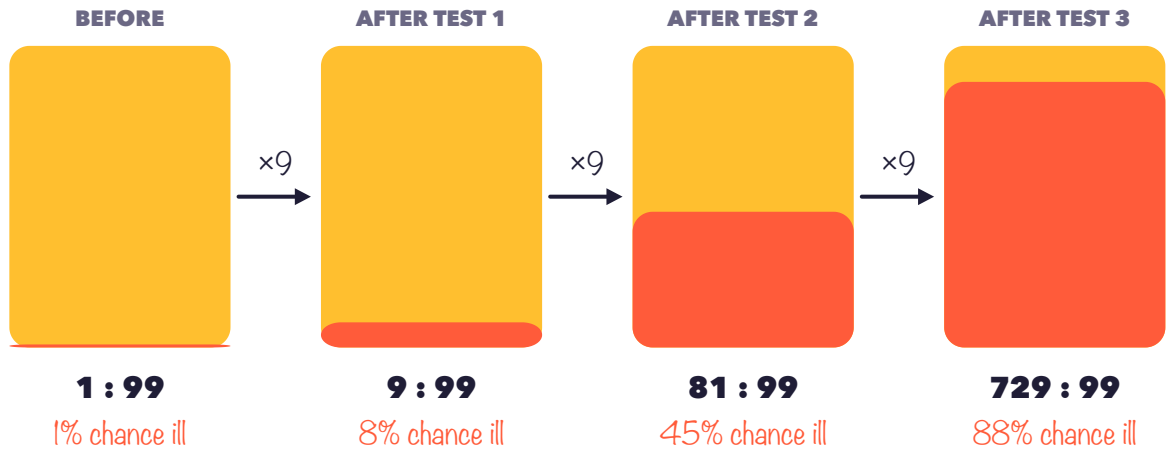


FIGURE 6.3 Each positive test multiplies the odds by 9. One positive takes you from 1% to 8%. Two take you to 45%. Three to 88%. Evidence piles up by multiplying.

"That's why doctors re-test. One clue barely moves a rare suspicion; three independent clues can settle it. But notice the word independent. If the second test is done in the same lab, with the same faulty batch, it's the same mistake twice and not a new clue at all."

J "So 'today's answer is tomorrow's starting point', as long as tomorrow's clue is new."

"Now your robot. How should it decide whether the animal at the gate is a dog or a wolf?"

J "Start with the odds: say, even. Then every clue multiplies them. Pointy ears: dogs and wolves both have them, so maybe ×2 for dog. Barking: wolves hardly ever bark, so ×20 for dog. Lives next door: wolves never do, ×50. Wags its tail at me: ×10."



FIGURE 6.4 The robot, reformed. It starts undecided and multiplies the odds by the strength of each clue. By the fourth clue it's 20,000 to 1 on "dog". Spam filters work exactly this way, clue by clue, word by word.

"And at the end?"

J "2 times 20 times 50 times 10 is 20,000 to 1. It's Bruno."

"And that is, almost word for word, how the first good spam filters worked. Each word in an email is a clue: 'winner' makes spam more likely, your mother's name makes it less. Multiply the strengths together and you've got your verdict."

J "So the robot didn't need five thousand photos. It needed a way to update."

"That's the difference between memorising and understanding, in one sentence."

SIDE QUEST: A MINISTER'S THEOREM

Thomas Bayes was an English Presbyterian minister who worked out the idea in the mid-1700s. He never published it. After his death in 1761 his friend Richard Price found it among his papers, edited it, and sent it to the Royal Society in 1763.

IN TEXTBOOK LANGUAGE (FOR THE CURIOUS)

Textbooks write Bayes' theorem as P(A | B) = P(B | A) × P(A) ÷ P(B): the chance of A given B equals the chance of B given A, times the chance of A, divided by the chance of B. The odds version says: posterior odds = prior odds × likelihood ratio.



TOOL UNLOCKED

Multiply the odds

Odds after = odds before × strength of the clue. Independent clues simply multiply, so weak evidence adds up. It's how doctors re-test, how spam filters sort your inbox, and how a robot should learn.

6.3

The costliest mistake in probability

When "if" gets turned around

BY THE END

You'll learn to spot one reversed sentence that has fooled juries, doctors and everyone who has searched their symptoms online.

4 min read A conversation over the fence

SCENE *Thursday. Jim is pale and holding his phone.*

J "I've had a headache for two days. I looked it up. Serious illnesses cause headaches."

"Most people with a serious brain illness get headaches, yes. So what's your question?"

J "Whether I have one!"

"Then you've just turned a sentence around. 'Most people with the illness have headaches' is one sentence. 'Most people with headaches have the illness' is a completely different one. The first is true. The second is wildly false."

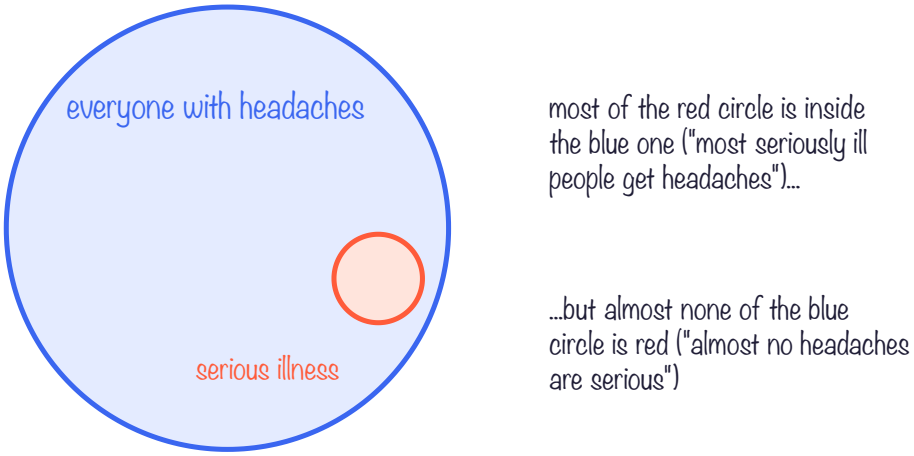


FIGURE 6.5 "Most seriously ill people have headaches" is true: most of the red circle is inside the blue. "Most people with headaches are seriously ill" is false: almost none of the blue circle is red. Same two circles, two very different sentences.

J "Because almost everybody gets headaches, and almost nobody has the illness."

"The base rate again. The chance of the evidence, given the illness, isn't the chance of the illness, given the evidence. Reversing them is called the confusion of the inverse, and it's everywhere. Drink some water and get some sleep."

J "Is it ever more serious than a scare?"

"It has sent people to prison. I'll tell you about Sally Clark, and I'll tell it carefully, because she was real."

> *The joking stops for a while.*

"Sally Clark was a solicitor in England. In 1996 her first baby son died suddenly, and in 1998 her second. She was charged with murdering them. At her trial in 1999, a senior paediatrician, Sir Roy Meadow, told the jury that the chance of two sudden infant deaths in a family like hers was about 1 in 73 million."

J "1 in 73 million. That sounds like proof."

"The number was wrong twice over. First, he got it by taking 1 in 8,543 for one death and squaring it, as if the two deaths were unrelated coin tosses. But families share genes and homes, and a family that has lost one baby this way is at higher risk of losing another."

J "So the two events aren't independent. Like the second test from the same faulty lab."

"Exactly the same mistake. But the deeper error is the one we started with. Even if two sudden deaths really were that rare, the jury had to compare two explanations of a tragedy that had already happened: two natural deaths, or two murders. And a mother murdering two babies is also extraordinarily rare."

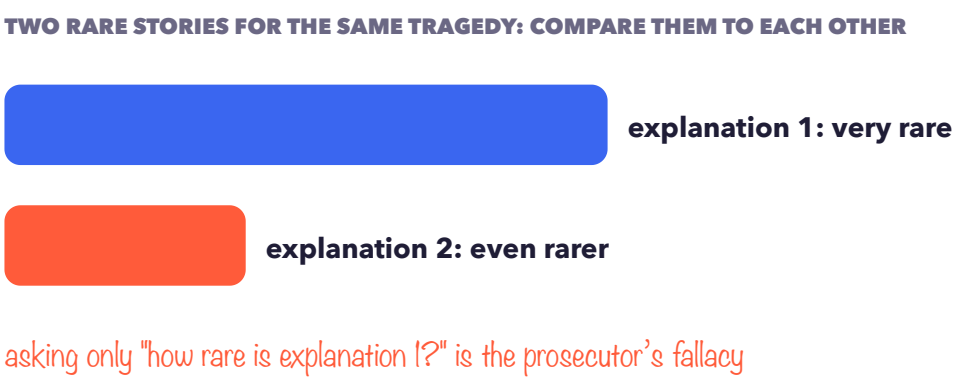


FIGURE 6.6 When something rare has definitely happened, the question isn't "how rare is the innocent explanation?" but "how rare is it compared with the alternative?". Both explanations of the Clark case were very rare.

J "So 'how unlikely is the innocent explanation' is the wrong question. It's 'which explanation is less unlikely?'"

"That's the whole thing. Treating 'the chance of this evidence if she's innocent' as 'the chance she's innocent' is called the prosecutor's fallacy. The Royal Statistical Society wrote publicly to say the figure was wrong. One statistician later estimated that, given the deaths, natural causes were several times more likely than murder."

J "What happened to her?"

"She was convicted in 1999. Her conviction was overturned in 2003, after it emerged that medical evidence that could have helped her had never been disclosed. Other convictions that had leaned on the same kind of evidence were reviewed. She never recovered from the ordeal, and she died in 2007."

Neither of them says anything for a while.

J "Two sentences that sound almost the same."

"'The chance of the evidence if innocent' and 'the chance of innocence given the evidence'. The whole difference is Bayes: the base rate, and the alternatives you compare against. That's why I think this is the most important idea in the whole book, even though it's the easiest."

J "So what's the habit? What do I actually do?"

"Three questions, every time. How common was this before any evidence? How much more likely is this evidence if it's true than if it isn't? And what's the alternative explanation, and how likely is that?"

SIDE QUEST: WHERE THE REVERSED SENTENCE TURNS UP

"Most burglars wear trainers." "Most people who crash were driving close to home." (Most driving happens close to home.) "The screening machine flags 99% of dangerous bags." In every case, ask the reverse: of everything that matches the evidence, how much is actually what you fear?

J "And my headache?"

"Base rate: headaches are common and serious illnesses are rare. Strength of the clue: weak. Alternative: you've been staring at a phone for two days. Water, sleep, and see a doctor if it doesn't go away."



TOOL UNLOCKED

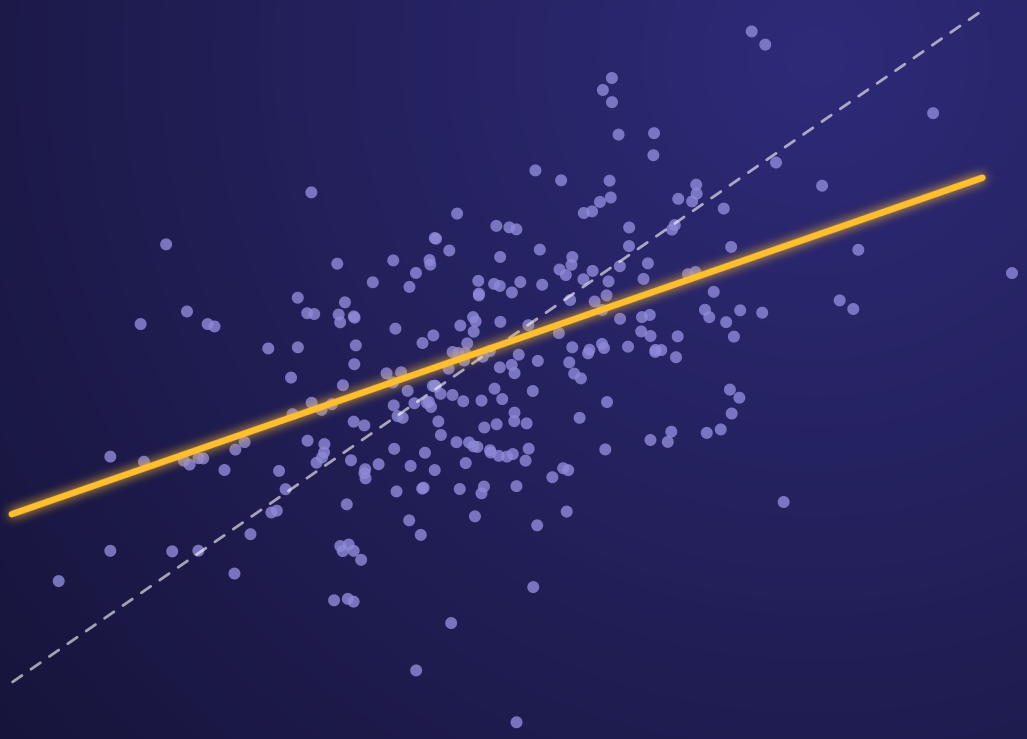
Never flip the "if"

The chance of the evidence given an idea is not the chance of the idea given the evidence. Ask three questions: how common was it before the evidence, how strong is the clue, and what's the alternative?

CHAPTER

7 Regression to the mean

Extremes are partly luck



Why does yelling at a team seem to work?

YOUR ROUTE THROUGH THIS CHAPTER

7.1 **Why the yelling seems to work**
Introduction to regression to the mean

7.2 **How far do things fall back?**
Developing the mathematical architecture for regression

7.1

Why the yelling seems to work

Introduction to regression to the mean

BY THE END

You'll see why praise seems to backfire and criticism seems to work, and why neither is true.

4 min read · A conversation over the fence

SCENE *Saturday. Jim is back from his son's football match, hoarse.*

J "Our coach has a theory. When the boys play badly, he yells at them, and next week they play better. When they play brilliantly, he praises them, and next week they're worse. So he's stopped praising."

"And he's sure it's cause and effect?"

J "He's watched it for years. It happens every time."

"Then here's a question. What if he'd said nothing at all?"

J "Then… I don't know. Nothing would change?"

"I think exactly the same pattern would happen, with no yelling and no praise. Let me show you. Any single performance is two things added together: skill, which carries over from week to week, and luck, which doesn't. The bounce of the ball, the wind, the ref, a good night's sleep."

J "Fair enough."

"Now, when do the boys have their worst game of the season?"

J "When they're off form…"

"And?"

J "…and when they're unlucky. The worst game is the one where bad luck piled on top of everything else."

"So a truly awful game was almost certainly an unlucky one. What happens to that bad luck next week?"

J "It doesn't carry over. Next week their luck is just... average. So they play better. Yelling or no yelling."

"And the best game of the season?"

J "Good form plus good luck. Next week the luck's gone, so they're worse, praise or no praise. The coach is taking credit for the luck running out!"

"That's it. It's called regression to the mean: after an extreme result, the next one is usually closer to average, simply because the extreme was partly luck."

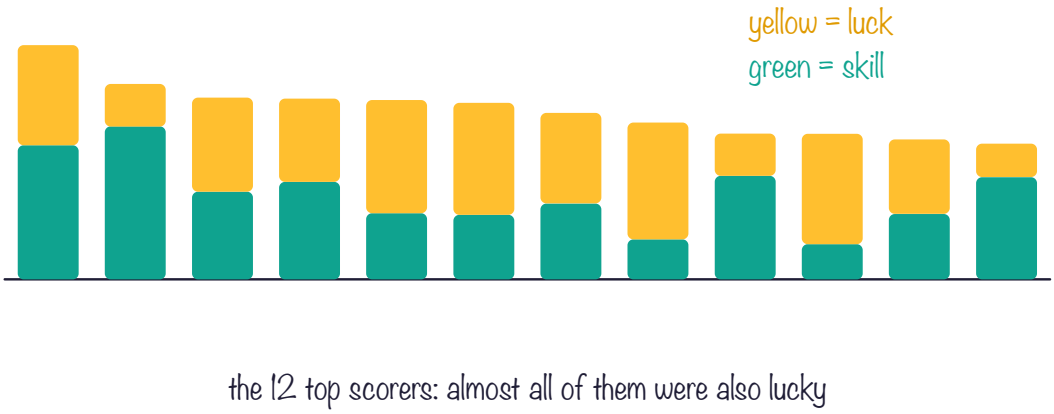


FIGURE 7.1 The twelve top scorers out of four hundred, with each score split into skill (green) and luck (yellow). Almost all of them were lucky as well as good. Next time, the luck won't be there.

J "Can you prove it's not the yelling?"

"Let's try it with a class where there's no skill at all. A hundred people each flip ten coins and count the heads. We take the ten worst 'flippers' and tell them off. We take the ten best and praise them. Then everyone flips again."

PAUSE AND PICTURE IT

What do you expect the ten worst to score in round two? And the ten best?

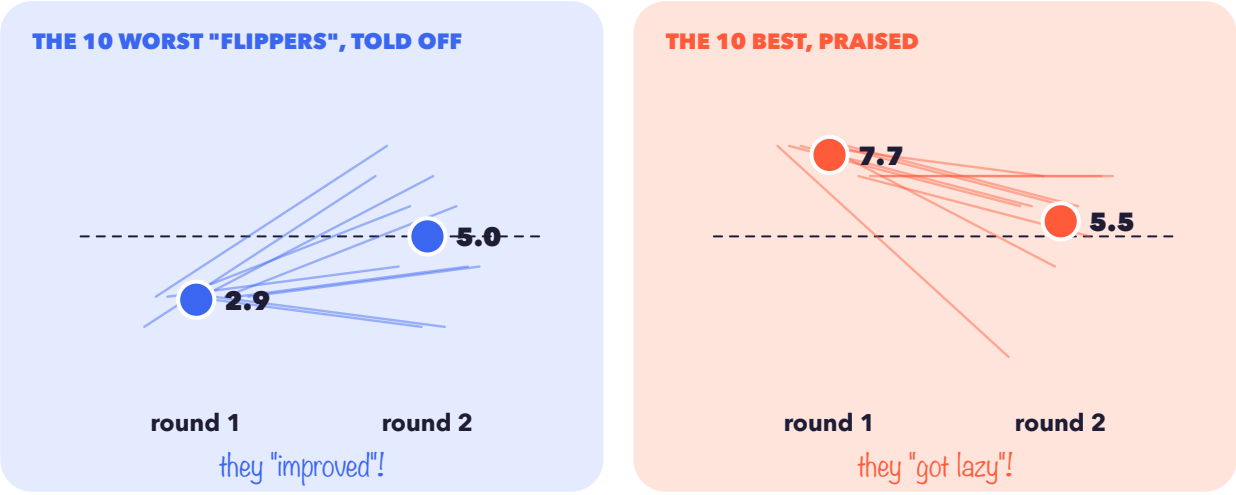


FIGURE 7.2 A hundred people flip ten coins each, twice. The ten worst in round one "improve" to about 5 heads after being told off. The ten best "get lazy" and drop to about 5. It's coins: nobody improved and nobody got lazy.

J "The worst ones jump up to about five, and the best ones drop to about five. But it's coins! There's nothing to improve!"

"Every coach, manager and parent who has ever sworn by shouting has been fooled by this. The psychologist Daniel Kahneman told the story of Israeli air force flight instructors who were sure that praising a good landing made the next one worse. He realised they were watching regression to the mean."

SIDE QUEST: THE COVER JINX

Athletes who appear on the cover of Sports Illustrated often have a worse season afterwards, and fans call it a jinx. But you only get on the cover after an extraordinary run, which is part skill and part luck. The jinx is the luck running out.

J "Where else does this happen?"

"Everywhere somebody picks the extremes. A company sacks its worst-performing sales team, and next quarter the replacements do better. A town puts speed cameras at the junctions with the most crashes last year, and next year there are fewer crashes. A patient tries a remedy on the day their back pain is at its worst, and the next day it's better."

J "Because people go to the doctor when they feel worst, and 'worst' is partly bad luck..."

"...so they'd usually have improved anyway. That's why new medicines are always tested against a group that doesn't get them. You're getting good at this."

"So how do I tell the coach he's wrong without getting yelled at myself?"

"Tomorrow I'll give you the numbers. And the surprising origin of the most-used word in statistics."



TOOL UNLOCKED

Extremes are partly luck

Every result is skill plus luck. Pick out the extremes and you've also picked out extreme luck, which won't repeat. So the worst improve and the best decline, whatever you do to them. Before crediting a cause, ask whether you picked the extremes.

7.2

How far do things fall back?

Developing the mathematical architecture for regression

BY THE END

You'll find out exactly how much of a surprise survives to next time, and where the word "regression" came from.

4 min read · A conversation over the fence

SCENE *Sunday. Jim has brought two years of his son's team scores, written on the back of a programme.*

J "Right. If an extreme result is part skill and part luck, how much falls back?"

"Let's say skill and luck vary by about the same amount from team to team. A team scores 10 goals above the league average this season. How much of that 10 do you expect to be skill?"

J "If they vary by the same amount, then... on average, half? So 5 is skill and 5 is luck."

"And next season?"

J "The 5 of skill stays and the 5 of luck vanishes. So I'd expect them 5 above average."

"So your best guess for next time is:"

YOUR BEST GUESS, IN WORDS

Next result = average + (share that's skill) × (this result − average)

J "And if it were all skill, the share would be 1, and nothing would fall back. All luck, the share is 0, and everyone falls straight back to average."

"That 'share that survives' has a famous name: the correlation. It measures how strongly one result predicts the next. Let's see it in a picture: hundreds

of teams, their first score against their second."

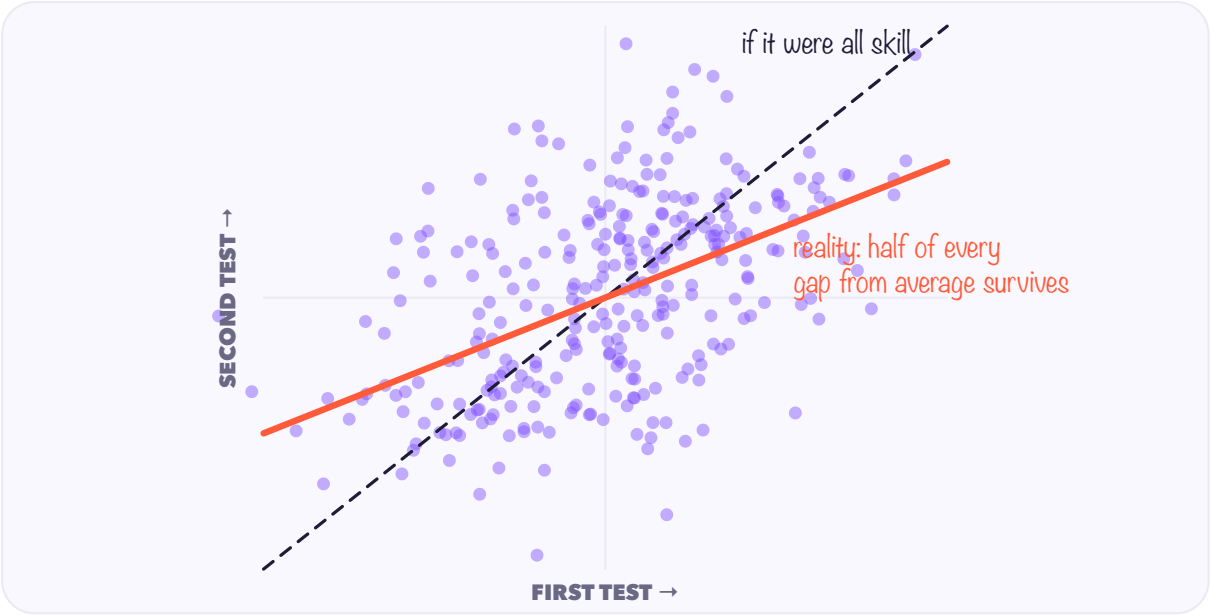


FIGURE 7.3 Each dot is one team: first result across, second result up. If results were all skill they'd sit on the dashed line. In reality the best-guess line (coral) is flatter: here, about half of every gap from average survives to next time.

J "The coral line is flatter than the diagonal. Teams far out on the right come back towards the middle."

"That coral line is the best straight-line guess of the second result from the first. Now for the name. In the 1880s Francis Galton, Charles Darwin's half-cousin, measured the heights of hundreds of fathers and their grown-up sons."

J "Tall fathers have tall sons."

"Taller than average, yes, but less tall than their fathers. Galton called it 're-gression towards mediocrity', and the line he drew through his data was the first regression line."

J "Regression. That's the word in linear regression! The thing every data scientist uses!"

"The most-used tool in statistics is named after this effect, a quirk of skill and luck. Now the twist that proves it's not some force. If you start from tall sons and ask about their fathers..."

J "...the fathers are tall, but less tall than their sons? It goes back-wards in time as well?"

"It does."



FIGURE 7.4 Fathers' and sons' heights. Sons of tall fathers are tall, but less tall (coral). Fathers of tall sons are tall, but less tall (blue). It works in both directions, so it can't be a force acting on the children. It's what happens whenever you pick the extremes.

J "So it can't be a force pushing children towards the average. Nothing can push backwards in time."

"Exactly. It's not a force. It's what happens whenever you select people for being extreme, on anything that's partly luck."

J "So how do I prove to the coach that the yelling does nothing?"

Heights are mostly inherited, so for heights the share that survives is large, but it's still less than 1.

"Compare. Take the weeks the team played badly. On some of them the coach yells; on others, say, he's lost his voice. See whether the teams he yelled at improve more than the ones he didn't."

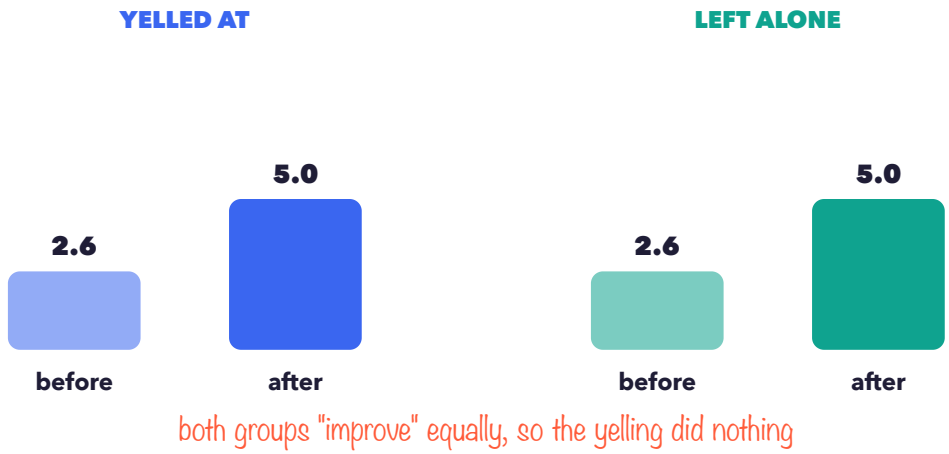


FIGURE 7.5 The only fair test: compare equally bad games, with and without the yelling. If both groups bounce back by the same amount, it was regression to the mean all along.

J "And if both improve by the same amount..."

"...the yelling did nothing. Having a group that doesn't get the treatment is called a control group, and it's one of the greatest inventions in the history of thinking. Nearly every medicine you've ever taken was proved that way."

J "Let me put the whole book together. Everything that repeats is circles. Anything you can add and scale is a vector. Distance is a choice. Every transformation has a grain. Moves have a grammar. Evidence multiplies odds. And extremes are partly luck."

"Seven ideas over a garden fence. None of them needed more than school maths, and every one of them runs a piece of the modern world."

J "And the robot?"

"The robot now updates its beliefs one clue at a time, knows that a dog is a vector of clues, and has stopped calling Bruno a wolf."

J "Bruno's forgiven it."

"Bruno forgives everyone. That's his mental model."



TOOL UNLOCKED

Before you credit the cause

Best guess for next time = average + correlation × (this time − average). Extremes fall back by exactly the share that was luck. The effect runs backwards in time too, so it isn't a force. To tell real causes from regression, compare against a control group.

YOUR TOOLKIT

Seven mental models

1. **Every repeating thing is circles.** Find the hands with a resonance test.
2. **A vector is anything you can add and scale.** Pick the language where it's simple.
3. **Distance is a choice.** Even "getting closer" depends on the ruler.
4. **Every change has a grain.** The future belongs to the biggest eigenvalue.
5. **Moves have a grammar.** Structure tells you what's impossible.
6. **Evidence multiplies odds.** Never flip the "if".
7. **Extremes are partly luck.** Check for a control group before crediting a cause.

"Bruno forgives everyone. That's his mental model."

Guhan Purushothaman

M.SC. MATHEMATICS · INDIAN INSTITUTE OF TECHNOLOGY MADRAS

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